



Mansoura University
Faculty of Computers and Information
Department of Information System



[IS422P] Data Mining

Grade:4 Th

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AGENDA

Textbook and Course Coverage

Lab work, Assignments, and Exams

Course Project and Evaluation

Introduction

- Why Data Mining?
- What is Data Mining?
- Knowledge Discovery Process
- What Kinds of Data Can Be Mined?
- What Kinds of Patterns Can Be Mined?
- What Kinds of Technologies Are Used?
- What Kinds of Applications are Targeted?
- Major Issues in Data Mining



TEXTBOOK & COURSE COVERAGE

- “Data Mining: Concepts and Techniques, 3rd Edition”
 - by J. Han, M. Kamber, and J. Pei – 2012
 - See the book's home page (<http://www.cs.illinois.edu/~hanj/bk3/>) for reference slides and other materials
- We will cover the following:
- Introduction
- Getting to Know Your Data
- Data Preprocessing
- Data Warehouse
- Mining Frequent Patterns & Association: Basic Concepts
- Classification: Basic Concepts
- Cluster Analysis: Basic Concepts and Methods
- Outlier Analysis
- Data Mining Trends



LAB WORK, ASSIGNMENTS, PROJECT

- Python → Written a program for data mining algorithm
- Written Assignments



EVALUATION

- Final grades:
- Written Assignments/ lab activity: 10%
- Practical Exam: 10%
- Oral exam : 10%.
- Midterm exam: 10%
- Final exam: 60%
- Quiz: ++



INTRODUCTION

- Why Data Mining?
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WHY DATA MINING

WHAT IS DATA MINING



WHY DATA MINING?

- The era of Explosive Growth of Data: in the petabytes!
 - Automated data collection and availability: tools, database systems, Web, computerized society
 - Major sources of abundant data
 - Business: Web, transactions, stocks, ...
 - Science: Remote sensing, bioinformatics, ...
 - Society and everyone: news, digital cameras, social feeds
 - The ability to economically store and manage petabytes of data online
 - The Internet and computing Grid that makes all these archives universally accessible
 - Linear growth of data management tasks with data volumes
- Massive data volumes, but still little insight!
- Solution! Data mining—The automated analysis of massive data sets

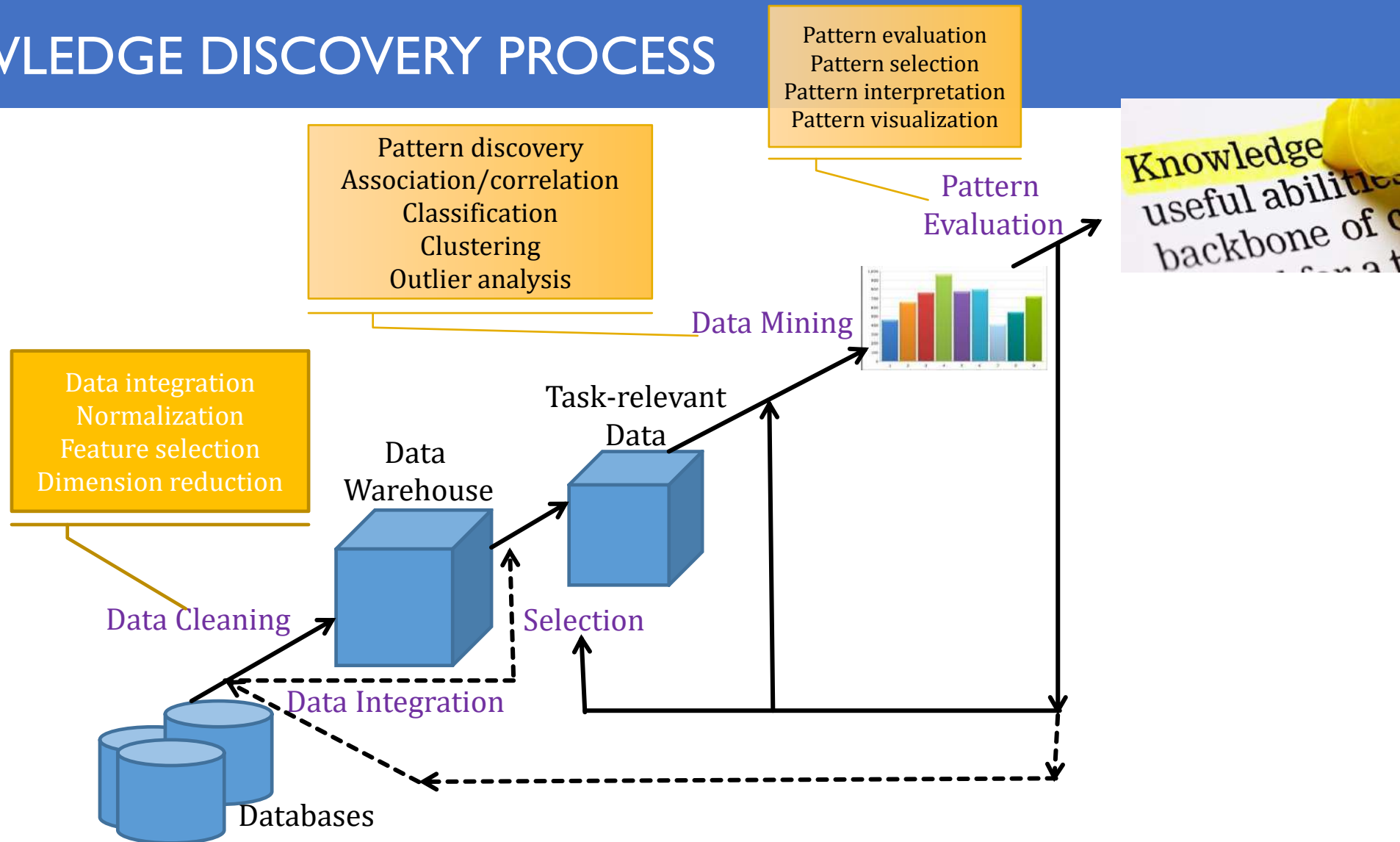


WHAT IS DATA MINING?

- Data mining (knowledge discovery from data)
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
 - Data mining: a misnomer?
- Alternative names
 - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, information harvesting, business intelligence, etc.
- Is everything “data mining”?
 - Simple search and query processing
 - (Deductive) expert systems



KNOWLEDGE DISCOVERY PROCESS



WHAT KINDS OF DATA CAN BE MINED?

- Database-oriented data sets and applications
 - Relational database, data warehouse, transactional database
- Advanced data sets and advanced applications
 - Data streams and sensor data
 - Time-series data, temporal data, sequence data (incl. bio-sequences)
 - Structure data, graphs, social networks and multi-linked data
 - Object-relational databases
 - Heterogeneous databases and legacy databases
 - Spatial data and spatiotemporal data
 - Multimedia database
 - Text databases
 - The World-Wide Web



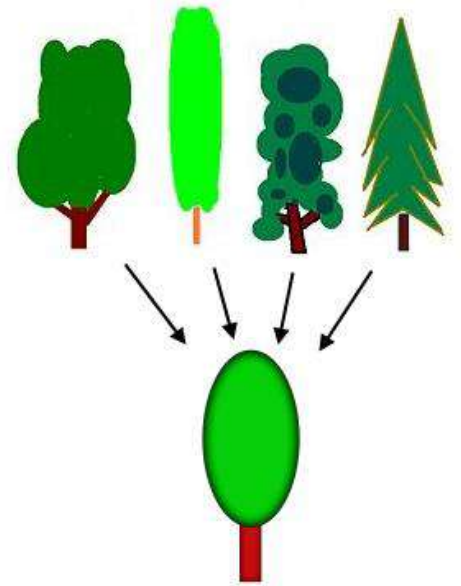
WHAT KINDS OF PATTERNS CAN BE MINED?

- Generalization
- Association / Correlation analysis
- Classification
- Cluster analysis
- Outlier analysis



GENERALIZATION

- Information integration and data warehouse construction
 - Data cleaning, transformation, integration, and multidimensional data model
- Data cube technology
 - Scalable methods for computing, multidimensional aggregates
 - OLAP (online analytical processing)
- Multidimensional concept description: Characterization and discrimination
 - Generalize, summarize, and contrast data characteristics → classes, features, and attributes are used



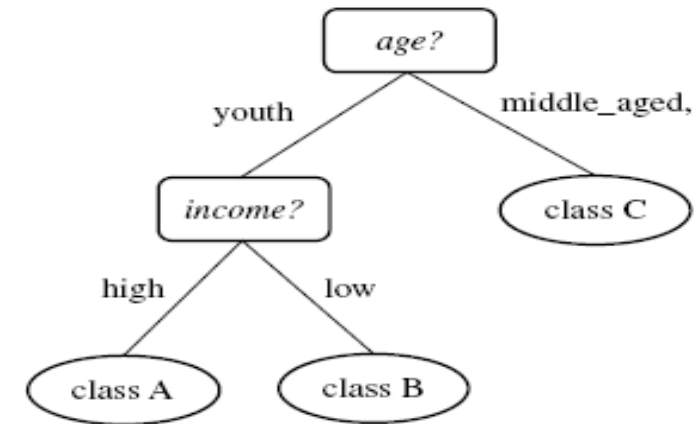
ASSOCIATION / CORRELATION ANALYSIS

- Frequent patterns (or frequent itemsets)
 - What items are frequently purchased together in your supermarket basket?
- Association, correlation vs. causality
 - A typical association rule
 - Milk $\rightarrow$ Bread[0.5%, 75%] (support, confidence)
 - Are strongly associated items also strongly correlated?
- How to mine such patterns and rules efficiently in large datasets?
- How to use such patterns:
 - classification,
 - clustering,
 - and other applications?



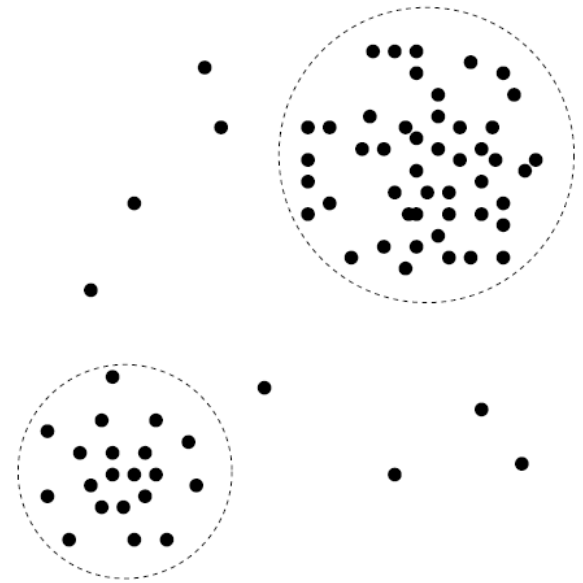
CLASSIFICATION

- Classification and label prediction
 - Construct models (functions) based on some training examples
 - Describe and distinguish classes or concepts for future prediction
 - e.g., classify countries based on (climate), or classify cars based on (gas mileage)
 - Predict some unknown class labels
- Typical methods
 - Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, ...
- Typical applications
 - Credit card fraud detection, direct marketing, diseases, web-pages, ...



CLUSTER ANALYSIS

- Unsupervised learning (i.e., Class label is unknown)
- Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- Principle: Maximizing intra-class similarity & minimizing interclass similarity
- Many methods and applications



TIME AND ORDERING

- Sequence, trend and evolution analysis
 - Trend, time-series, and deviation analysis: e.g., regression and value prediction
 - Sequential pattern mining
 - e.g., first buy digital camera, then buy large SD memory cards
 - Periodicity analysis
 - Similarity-based analysis
- Mining data streams
 - Ordered, time-varying, potentially infinite, data streams

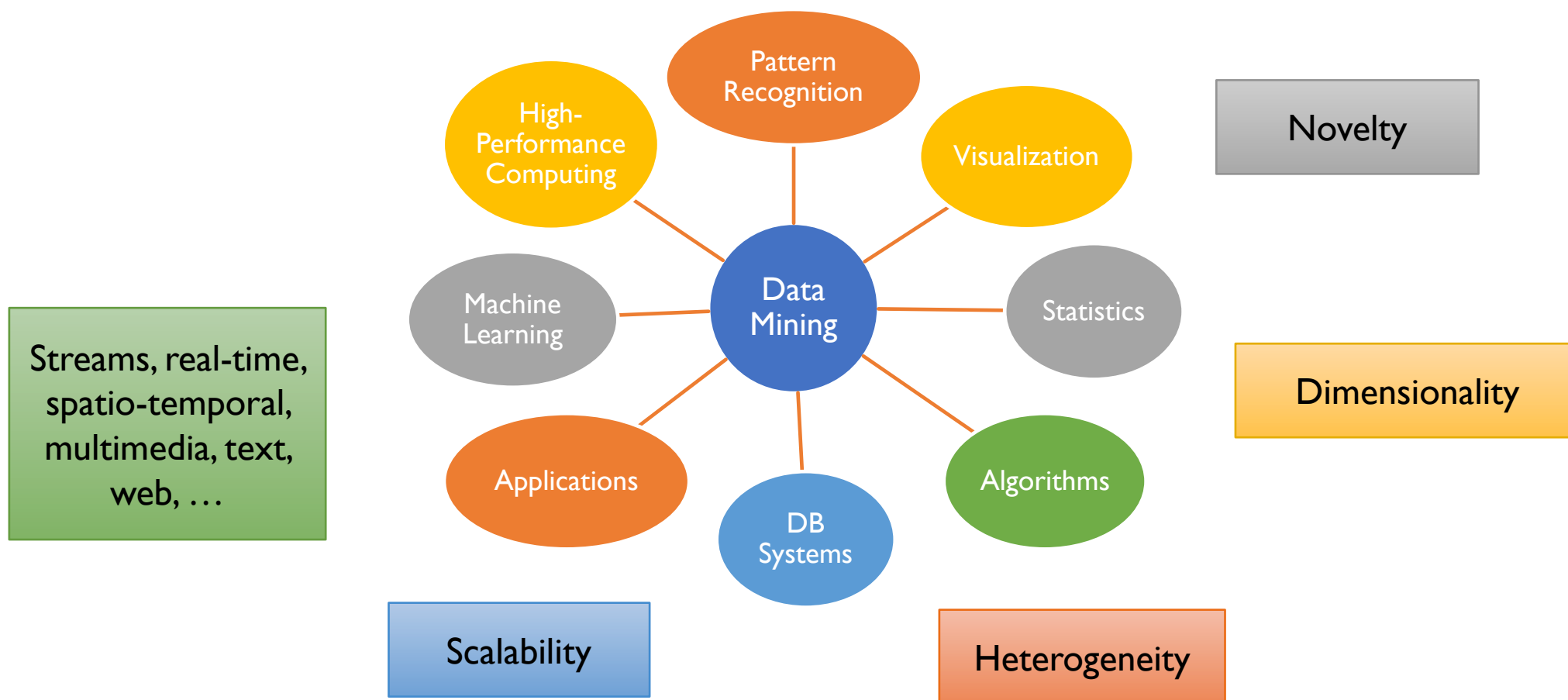


EVALUATION OF KNOWLEDGE

- Are all mined knowledge interesting?
 - One can mine tremendous amount of “patterns” and knowledge
 - Some may fit only certain dimension space (time, location, ...)
 - Some may not be representative, may be transient, ...
- Evaluation of mined knowledge → directly mine only interesting knowledge?
 - Descriptive vs. predictive
 - Coverage
 - Typicality vs. novelty
 - Accuracy
 - Timeliness
 - ...



WHAT KINDS OF TECHNOLOGIES ARE USED?



WHAT KINDS OF APPLICATIONS ARE TARGETED?

- Web page analysis: from web page classification, clustering to PageRank & HITS algorithms
- Collaborative analysis & recommender systems
- Basket data analysis to targeted marketing
- Biological and medical data analysis: classification, cluster analysis (microarray data analysis), biological sequence analysis, biological network analysis
- Data mining and software engineering
- From major dedicated data mining systems/tools (e.g., SAS, MS SQL-Server Analysis Manager, Oracle Data Mining Tools) to invisible data mining



MAJOR ISSUES IN DATA MINING

■ Mining Methodology

- Mining various and new kinds of knowledge
- Mining knowledge in multi-dimensional space
- Boosting the power of discovery in a networked environment
- Handling noise, uncertainty, and incompleteness of data
- Pattern evaluation and pattern- or constraint-guided mining

■ User Interaction

- Interactive mining
- Incorporation of background knowledge
- Presentation and visualization of data mining results



MAJOR ISSUES IN DATA MINING (2)

- Efficiency and Scalability
 - Efficiency and scalability of data mining algorithms
 - Parallel, distributed, stream, and incremental mining methods
- Diversity of data types
 - Handling complex types of data
 - Mining dynamic, networked, and global data repositories
- Data mining and society
 - Social impacts of data mining
 - Privacy-preserving data mining
 - Invisible data mining



THANK YOU



IS422P - DATA MINING

2- KNOW YOUR DATA



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AGENDA

Data Objects & Attribute Types

- What is an Object?
- What is an Attribute?
- Attribute Types
- Continuous vs. Discrete

Basic Statistical Descriptions of Data

- Measuring central tendency
- Measuring Data dispersion
- Basic Graphic displays

Measuring Data similarity & Dissimilarity

- Data Matrix & Dissimilarity Matrix
- Proximity Measures for
 - Nominal Attributes
 - Binary Attributes
- Dissimilarity Measures for Numerical Attributes



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OBJECTS AND ATTRIBUTES

- A data **object** represents an entity
 - Also sample, example, instance, data point, or object (in a DB : Data Tuple)
 - e.g. customers, students, patients, books
- An **attribute** is a data field, representing a characteristic or feature of a data object
 - Also noun attribute, dimension, feature, and variable (DB and DM, DWs, ML, Statistics)
 - e.g. name, age, salary, gender, grade, ...
 - Attribute (feature) vector → A set of attributes that describe an object



ATTRIBUTE TYPES

NOMINAL ATTRIBUTES

- Symbol or names of things
- Each value represents category, code, or state
 - also referred to as categorical
 - e.g. hair color, marital status, customer ID
 - Possible to be represented as numbers (coding)
- Qualitative



ATTRIBUTE TYPES

BINARY ATTRIBUTES

- Nominal with only two values representing two states or categories: 0 or 1 (absent or present)
- Also Boolean (true or false)
- Qualitative
- **Symmetric**: both states are equally valuable and have the same weight
 - e.g. gender
- **Asymmetric**: states are not equally important
 - e.g. medical test outcomes



ATTRIBUTE TYPES

ORDINAL ATTRIBUTES

- Qualitative
- Values have a meaningful order or ranking, but magnitude between successive values is not known
 - e.g. professional rank, grade, customer satisfaction
 - Useful for data reduction of numerical attributes (Chapter 3)



ATTRIBUTE TYPES

NUMERIC ATTRIBUTES

- Quantitative
- **Interval-scaled:** measured on a scale of equal-size units
 - e.g. temperature, year
 - Do not have a true zero point
 - Not possible to be expressed as multiples
- **Ratio-scaled:** have a true zero point
 - A value can be expressed as a multiple of another
 - e.g. years of experience, weight, salary



DISCRETE VS. CONTINUOUS ATTRIBUTES

- **Discrete Attribute:** has a finite or countably infinite set of values, integers or otherwise
 - e.g. hair color, smoker, medical test, Customer_ID
 - Customer_ID is countably infinite → infinite values but one-to-one correspondence with natural numbers
- If an attribute is not discrete, it's continuous
 - e.g. height, weight, age



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MEASURING CENTRAL TENDENCY

For N observations of numerical variable X: $x_1, x_2, \dots, x_N$

- **Mean**: or average of values

- $$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

- **Weighted Average** : a weight is associated with each value

- $$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{N} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_N x_N}{N}$$

- **Problem**: sensitivity to outlier values

- e.g. mean salary, mean student score

- **Trimmed mean** → chop off extreme values at both ends



MEASURING CENTRAL TENDENCY

- **Median:** middle value in set of ordered values
 - N is odd $\rightarrow$ median is middle value of ordered set
 - N is even $\rightarrow$ median is not unique $\rightarrow$ average of two middlemost values
 - Expensive to compute for large # of observations
- **Mode:** value that occurs most frequently in the attribute values
 - Works for both qualitative and quantitative attributes
 - Data can be unimodal, bimodal, or trimodal – no mode?



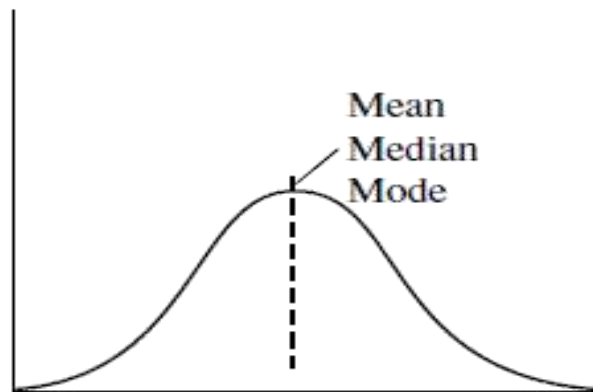
MEASURING CENTRAL TENDENCY

- **Example**
- Salary (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110
 - Mean = ?
 - Median = ?
 - Mode = ?

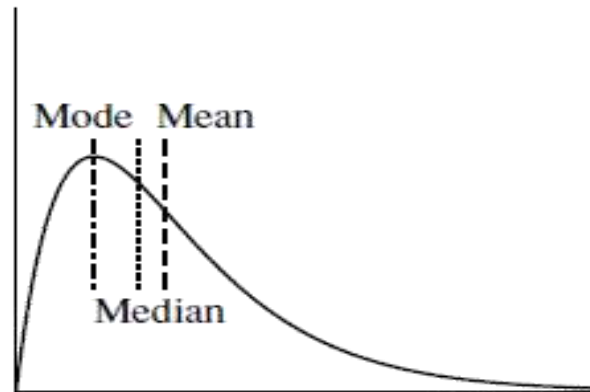


MEASURING CENTRAL TENDENCY

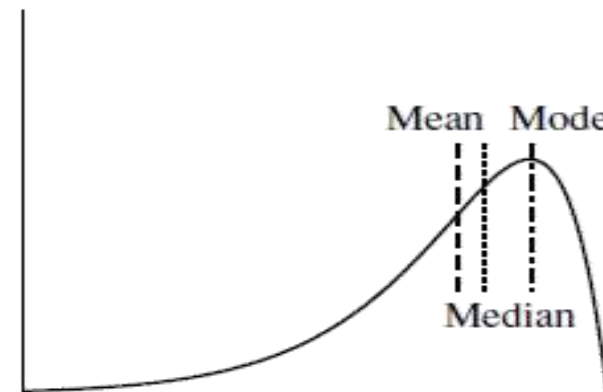
- Example
- Salary (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110
 - Mean = 58,000
 - Median = 54,000
 - Mode = 52,000 and 70,000 – bimodal



(a) Symmetric data



(b) Positively skewed data



(c) Negatively skewed data



MEASURING DISPERSION OF DATA

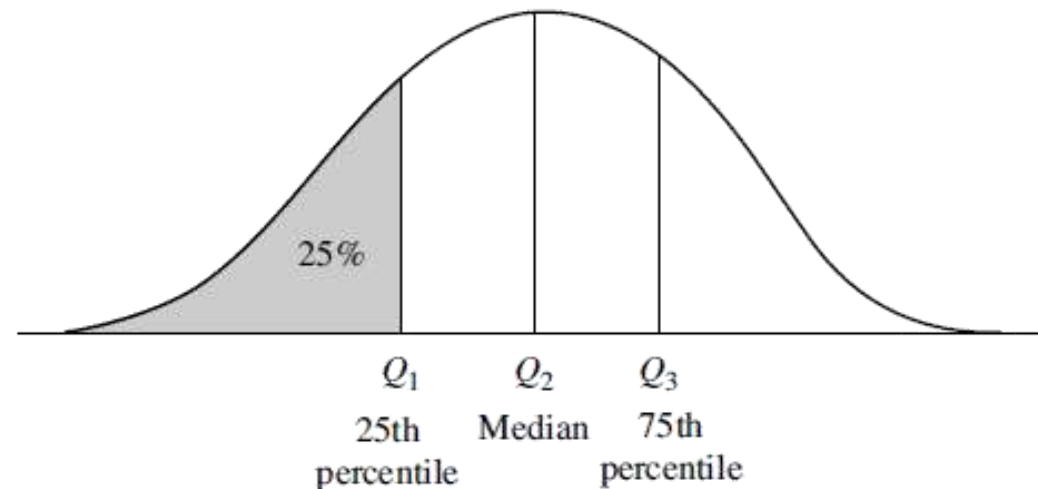
- **Variance & SD**: indicate how **spread out** a data distributed is
- **Low SD** → data observations tend to be very close to the mean
- **High SD** → data is spread out over a large range of values
- $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2$
- SD = σ



MEASURING DISPERSION OF DATA

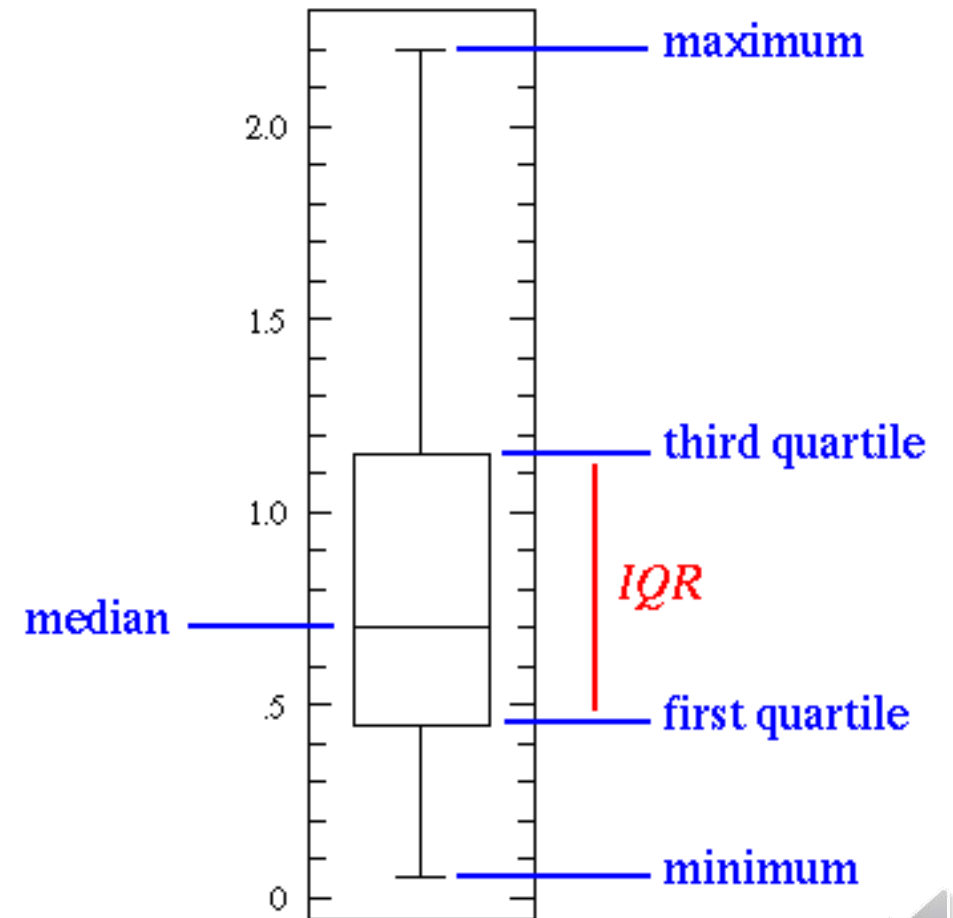
For N observations of numerical variable X : $x_1, x_2, \dots, x_N$

- **Rang**: difference between the largest and smallest values
- **Quantiles**: points taken at **regular intervals** of a data distribution, dividing it into (almost) equal-size consecutive sets
 - Most famous $\rightarrow$ percentile
 - 100 equal-sized sets
 - **Quartiles** $\rightarrow$ 4 Quantiles
- **Interquartile Range** := $Q_3 - Q_1$



MEASURING DISPERSION OF DATA

- **Five-Number Summary:**
 - Median (Q2), quartiles Q1 and Q3, & smallest and largest individual observations – in order
- **Boxplots:** visualization technique for the five-number summary
 - **Whiskers** terminate at min & max OR the most extreme observations within
 - $1.5 \times \text{IQR}$ of the quartiles – with remainder points (outliers) plotted individually



EXAMPLE

Age	23	23	27	27	39	41	47	49	50	52	54	54	56	57	58	58	60	61
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results:

- Draw the boxplots for age and %fat.



SOLUTION

Age	23	23	27	27	39	41	47	49	50	52	54	54	56	57	58	58	60	61
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

%fat

7.8

9.5

17.8

25.9

26.5

27.2

27.4

28.8

30.2

31.2

31.4

32.9

33.4

34.1

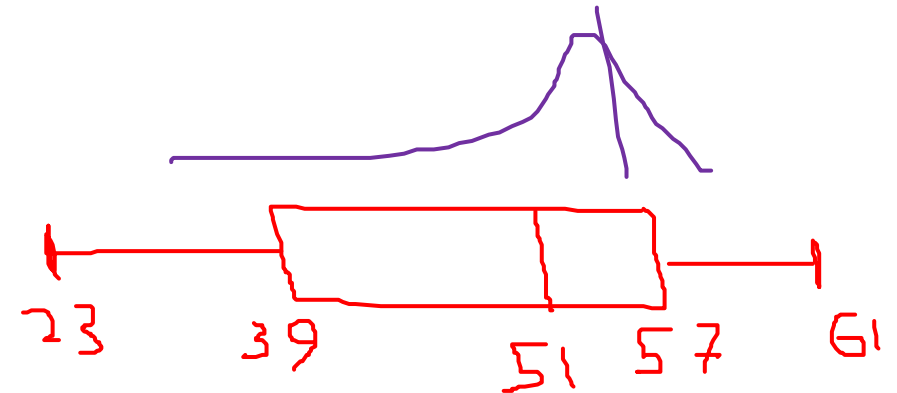
34.6

35.7

41.2

42.5

- Draw the boxplots for age and %fat.
- For Age
 - $Q1=39$, median= 51, $Q3=57$, min=23, max=61
 - $IQR=57-39=18$, $1.5 IQR=27$
 - newMin= $39-27=12$, newMax= $57+27=84$
- For Fat
 - $Q1=26.5$, median= 30.7, $Q3=34.1$, min=7.8, max=42.5
 - $IQR=34.1-26.5=7.6$, $1.5 IQR=11.4$
 - newMin= $26.5-11.4=15.1$
 - newMax= $34.1+11.4=45.5$



* *
7.8 9.5 15.1

DESCRIPTIVE STATISTICS WITH PANDAS

- `pandas.DataFrame.describe`
- `DataFrame.describe(percentiles=None, include=None, exclude=None)`
- *Percentiles*: list-like of numbers, optional
- *Include*: 'all', list-like of dtypes or None (default), optional → `data.describe(include=['O'])`
- *Exclude*: list-like of dtypes or None (default), optional,

DESCRIPTIVE STATISTICS WITH PANDAS

```
df = pd.DataFrame({
    'categorical': pd.Categorical(['d', 'e', 'f']),
    'numeric': [1, 2, 3],
    'object': ['a', 'b', 'c']
})
```

```
df.describe()
   numeric
count    3.0
mean     2.0
std      1.0
min      1.0
25%     1.5
50%     2.0
75%     2.5
max      3.0
```

```
df.describe(include='all')
   categorical  numeric  object
count         3      3.0      3
unique         3      NaN      3
top           f      NaN      a
freq          l      NaN      l
mean         NaN      2.0     NaN
std          NaN      1.0     NaN
min          NaN      1.0     NaN
25%          NaN      1.5     NaN
50%          NaN      2.0     NaN
75%          NaN      2.5     NaN
max          NaN      3.0     NaN
```

DESCRIPTIVE STATISTICS WITH PANDAS

`data.describe()`

`data[“Attr”]. describe()`

- count 32561.000000
mean 40.437456
std 12.347429
min 1.000000
25% 40.000000
50% 40.000000
75% 45.000000
max 99.000000
dtype: float64

- ***For numeric attributes***

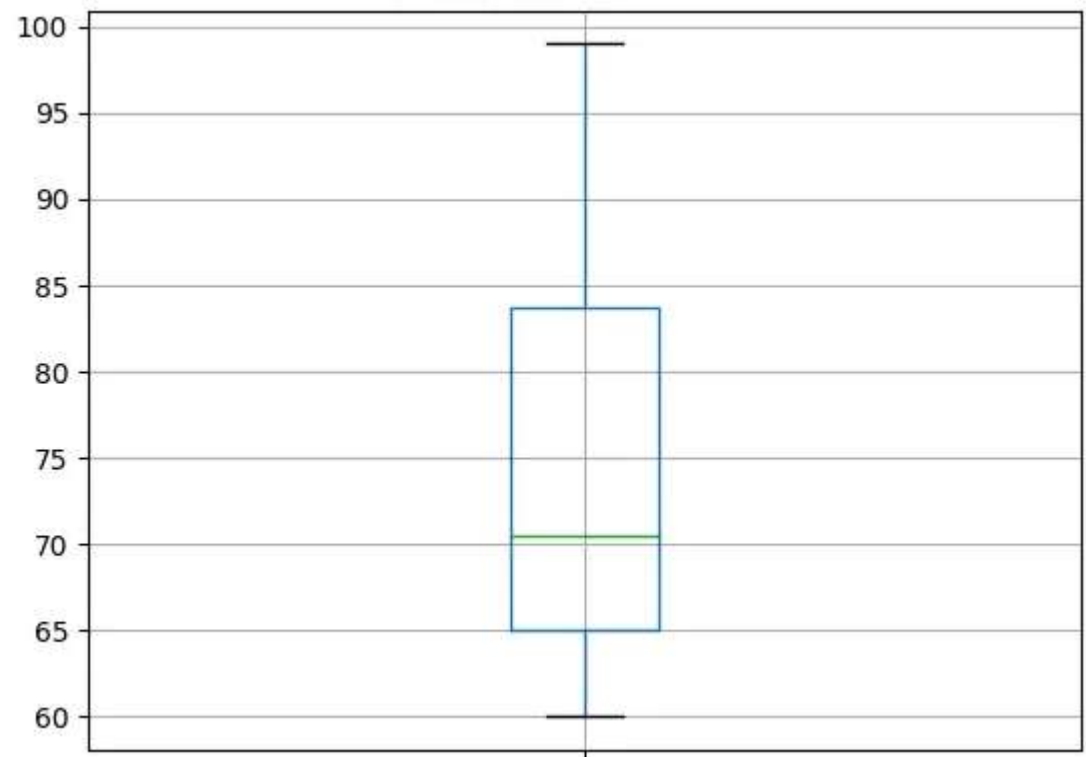
`data.describe(include=['O'])`

- count 200
- unique 3
- top bird
- freq 65
- dtype: object

Data.count() : Count number of non-NA/null observations.
Data.max() : Maximum of the values in the object.
Data.min () : Minimum of the values in the object.
Data.mean() : Mean of the values.
Data.std() : Standard deviation of the observations.

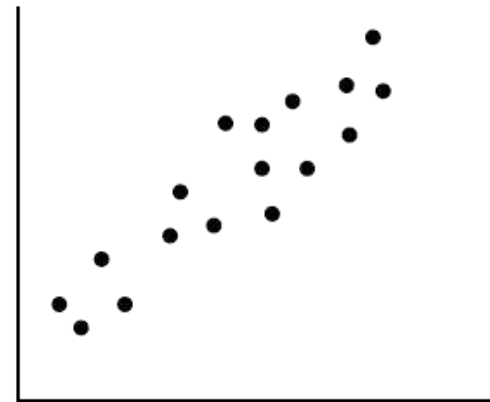
PANDAS.PLOTTING.BOXPLOT

- `pandas.plotting.boxplot(data, column=None, by=None, ax=None, fontsize=None, rot=0, grid=True, figsize=None, layout=None, return_type=None, **kwargs)`
- `Data.boxplot(column= 'Grade')`

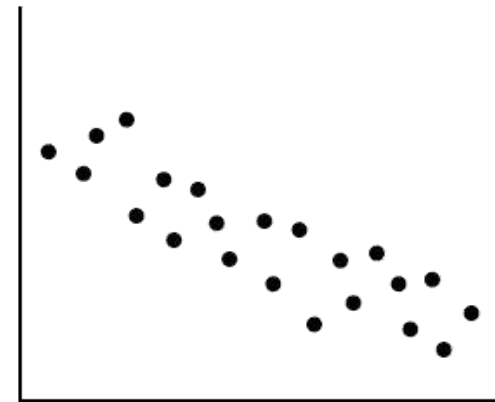


VISUAL REPRESENTATIONS OF DATA DISTRIBUTIONS

- **Histograms**
- **Scatter Plots:** each pair of values is treated as a pair of coordinates and plotted as points in plane
 - X and Y are correlated if one attribute implies the other
 - positive, negative, or null (uncorrelated)
 - For more attributes, we use a scatter plot matrix



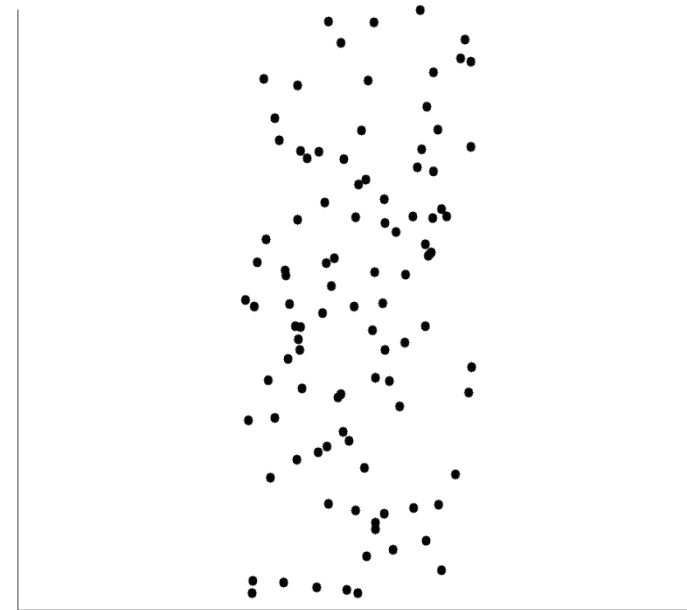
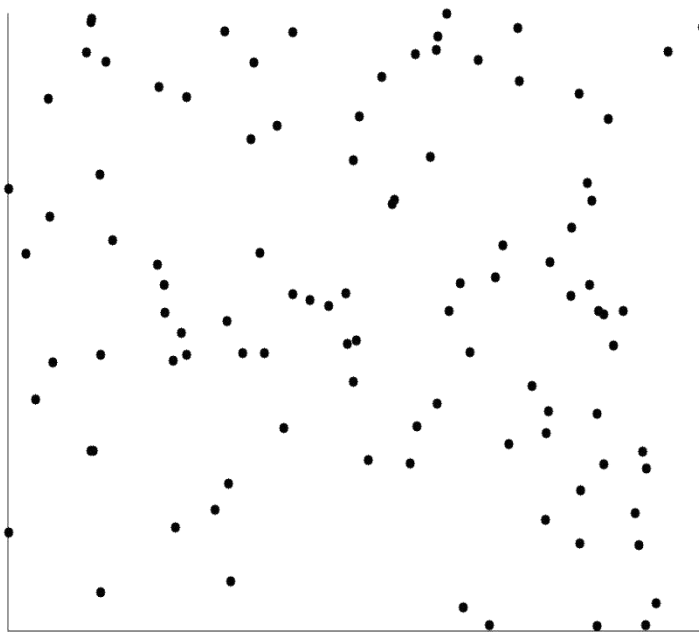
X increase as Y
increase



X increase as Y
decrease



VISUAL REPRESENTATIONS OF DATA DISTRIBUTIONS



Uncorrelated data



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- Statistical descriptions are about **Attributes**
 - e.g. age, weight, *color*, grade, product name, ...
- Similarity/Dissimilarity is about **Objects**
 - e.g. customers, students, ...
- Similarity/Dissimilarity measures objects *proximity*
- Similarity of i and $j \rightarrow 0$ if totally unlike, larger means more alike
- Dissimilarity (distance) of i and $j \rightarrow 0$ if totally alike, larger means less alike



DATA MATRIX & DISSIMILARITY MATRIX

- Data matrix (*object-by-attribute structure*)

- Stores n data *objects* that have p *attributes* as an n -by- p matrix

Also called
*feature
vectors*

p attributes

	a_1	...	a_j	...	a_p
x_1	x_{11}	...	x_{1j}	...	x_{1p}
...	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_i	x_{i1}	...	x_{ij}	...	x_{ip}
...	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	x_{n1}	...	x_{nj}	...	x_{np}

n objects



DATA MATRIX & DISSIMILARITY MATRIX

- Dissimilarity matrix (*object-by-object structure*)
 - stores a *collection of proximities* for all pairs of n objects as an n -by- n matrix

$d(i, j) \rightarrow$ difference between objects i and j
 $d(i, i) = 0$
 $d(i, j) = d(j, i)$

	n objects					
	x_1	x_2	...	x_i	...	x_n
x_1	0					
x_2	$d(2,1)$	0				
...	...	...	...			
x_i	$d(i,1)$	$d(i,2)$	...	0		
...	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
x_n	$d(n,1)$	$d(n,2)$	...	$d(n,i)$	...	0

n objects

$sim(i, j) = 1 - d(i, j)$



PROXIMITY MEASURES FOR NOMINAL ATTRIBUTES

$$d(i, j) = \frac{p - m}{p}$$

m is # matches
 p is # attributes

$$\text{sim}(i, j) = 1 - \frac{p - m}{p} = \frac{m}{p}$$

Example

ID	Test-1	Test-2
1	Code A	Excellent
2	Code B	Good
3	Code C	Good
4	Code A	Excellent

$$\begin{matrix} \rightarrow & \begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ d(3,1) & d(3,2) & 0 & \\ d(4,1) & d(4,2) & d(4,3) & 0 \end{bmatrix} & \rightarrow & \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 0.5 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$



PROXIMITY MEASURES FOR BINARY ATTRIBUTES

Symmetric $\rightarrow d(i, j) = \frac{r+s}{q+r+s+t}$

Asymmetric $\rightarrow d(i, j) = \frac{r+s}{q+r+s}$

Jaccard coefficient $\rightarrow$

$$sim(i, j) = 1 - d(i, j) = \frac{q}{q + r + s}$$

q is # attributes that equal 1 for i & j
 r is # attributes that equal 1 for i but 0 for j
 s is # attributes that equal 0 for i but 1 for j
 t is # attributes that equal 0 for i & j

		Object j	
		1	0
Object i	1	q	r
	0	s	t

When both symmetric and asymmetric binary attributes occur in the same data set, they are treated as mixed attributes



PROXIMITY MEASURES

BINARY ATTRIBUTES - EXAMPLE

name	gender	fever	cough	Test-1	Test-2	Test-3	Test-4
Ali	M	Y	N	P	N	N	N
Adam	M	Y	Y	N	N	N	N
Selma	F	Y	N	P	N	P	N
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Symmetric

Asymmetric

Y and P → 1

N and N → 0

Assuming we use only
asymmetric attributes



$$d(\text{Ali}, \text{Adam}) = \frac{1 + 1}{1 + 1 + 1} = 0.67$$

$$d(\text{Ali}, \text{Selma}) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

$$d(\text{Adam}, \text{Selma}) = \frac{1 + 2}{1 + 1 + 2} = 0.75$$



DISSIMILARITY OF NUMERIC DATA

- Normalize → give all attributes an *equal weight*
 - If object “patient” is described as having height (in meters) and weight (in g), what data ranges can we have for both attributes? Which will be more dominant?

ID	Height	Weight
Patient 1	1.5	150000
Patient 2	1.8	90000

- Map values for each attribute to interval → $[-1, 1]$ or $[0, 1]$
- More on that in Chapter 3

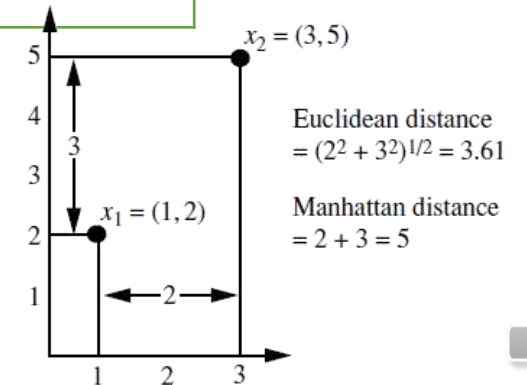


DISSIMILARITY OF NUMERIC DATA

MINKOWSKI DISTANCE

Euclidean Distance	$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2}$
Manhattan (City Block) Distance	$d(i, j) = x_{i1} - x_{j1} + x_{i2} - x_{j2} + \cdots + x_{ip} - x_{jp} $
Minkowski Distance	$d(i, j) = \sqrt[h]{(x_{i1} - x_{j1})^h + (x_{i2} - x_{j2})^h + \cdots + (x_{ip} - x_{jp})^h}$

- Distance is a non-negative number
- Distance of an object to itself is 0
- Distance is a symmetric function
- $d(i, j) \leq d(i, k) + d(k, j)$



DISSIMILARITY OF NUMERIC DATA

MINKOWSKI DISTANCE

point	x	y
p_1	1	2
p_2	3	5
p_3	2	0
p_4	4	5

Simple in-lecture Example

Find dissimilarity matrices using the Manhattan distance and the Euclidean distance



SUMMARY

- Attributes describe Objects
- Attributes can be nominal, binary, ordinal, or numeric
- To measure the central tendency of attribute observations, use mean, SD, median, and mode for numeric attributes, and only median and mode for the rest
- To measure objects similarity/dissimilarity, use:
 - matching percent for objects with nominal attributes
 - symmetric/asymmetric binary dissimilarity or the Jaccard coefficient for objects with binary attributes
 - Minkowski distance for objects with numerical attributes



TASK I

- **HOW CAN YOU CONSTRUCT A DISTANCE OR DISSIMILARITY MATRIX WITH PANDAS?**

THANK YOU





Mansoura University
Faculty of Computers and Information
Department of Information System



[IS422P] Data Mining

Grade:4 Th

Dr. Amira Rezk



3- DATA PRE-PROCESSING



REVIEW

- Draw the Box-Plot for the following dataset
- 4.3, 5.1, 3.9, 4.5, 4.4, 4.9, 5.0, 4.7, 4.1, 4.6, 4.4, 4.3, 4.8, 4.4, 4.2, 4.5, 4.4

REVIEW

- Draw a box-plot for the following data set:

4.3, 5.1, 3.9, 4.5, 4.4, 4.9, 5.0, 4.7, 4.1, 4.6, 4.4, 4.3, 4.8, 4.4, 4.2, 4.5, 4.4

1- order

3.9, 4.1, 4.2, 4.3, 4.3, 4.4, 4.4, 4.4, 4.4, 4.5, 4.5, 4.6, 4.7, 4.8, 4.9, 5.0, 5.1

2- Median

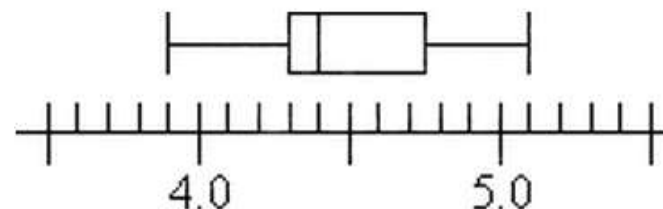
3.9, 4.1, 4.2, 4.3, 4.3, 4.4, 4.4, 4.4, **4.4**, 4.5, 4.5, 4.6, 4.7, 4.8, 4.9, 5.0, 5.1

3- Q1 & Q3

3.9, 4.1, 4.2, 4.3, **4.3**, 4.3, 4.4, 4.4, 4.4, **4.4**, 4.5, 4.5, 4.6, 4.7, **4.75**, 4.8, 4.9, 5.0, 5.1

4- **$IQR = 4.75 - 4.3 = 0.45 \rightarrow 1.5 IQR = 0.675 \cong 0.7$**

5- **Whisker:** **$\min = 4.3 - 0.7 = 3.6$** ; **$\max = 4.7 + 0.7 = 5.4$**



AGENDA



Overview

Why pre-process data?
Major tasks



Data Cleaning

Missing values
Noisy data



Data Integration

Entity Identification Problem
Redundancy and correlation analysis
Tuple duplication
Data value conflict detection



Data Reduction

Strategies
Wavelet transforms
PCA
Attribute subset selection
Regression
Histograms
Clustering
Sampling



Data Transformation

Strategies
Normalization
Binning
Histogram analysis
Cluster/Decision trees/Correlation analyses
Concept hierarchy

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OVERVIEW

- Databases are highly susceptible to noisy, missing, and inconsistent data
- Low-quality data will lead to low-quality mining results
- “How can the data be preprocessed in order to help improve the quality of the data and, consequently, of the mining results?”
- How can the data be preprocessed so as to improve the efficiency and ease of the mining process?”



WHY PREPROCESS DATA?

- To satisfy the requirements of the intended use
- Factors of data quality:
 - **Accuracy** → lack of due to faulty instruments, errors caused by human/computer/transmission, deliberate errors ...
 - **Completeness** → lack of due to different design phases, optional attributes
 - **Consistency** → lack of due to semantics, data types, field formats ...
 - **Timeliness**
 - **Believability** → how much the data are trusted by users
 - **Interpretability** → how easy the data are understood

MAJOR PREPROCESSING TASKS THAT IMPROVE QUALITY OF DATA

- **Data cleaning** → filling in missing values, smoothing noisy data, identifying or removing outliers, and resolving inconsistencies
- **Data integration** → include data from multiple sources in your analysis, map semantic concepts, infer attributes ...
- **Data reduction** → obtain a reduced representation of the data set that is much smaller in volume, while producing almost the same analytical results
- **Discretization** → raw data values for attributes are replaced by ranges or higher conceptual levels
- **Data transformation** → normalization

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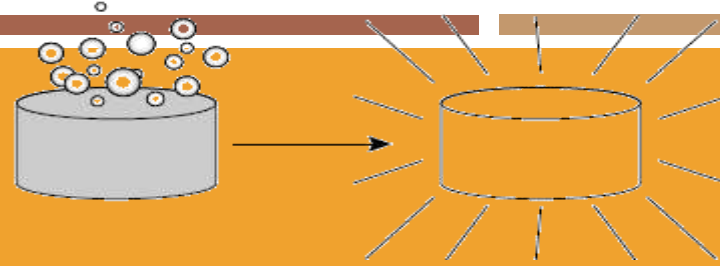
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DATA CLEANING

- Data in the Real World Is Dirty!
 - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., Occupation="" (missing data)
 - noisy: containing noise, errors, or outliers
 - e.g., Salary="-10" (an error)
 - inconsistent: containing discrepancies in codes or names, e.g.,
 - Age="42", Birthday="03/07/2010"
 - Was rating "1, 2, 3", now rating "A, B, C"
 - discrepancy between duplicate records
 - Intentional → Jan. 1 as everyone's birthday?

DATA CLEANING



- ... fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data
- A missing value may not imply an error in the data!
 - e.g. driver's license number



DATA CLEANING

MISSING VALUES

- Ignore the tuple → not very effective, unless the tuple contains several attributes with missing values
- Fill in the missing value manually → time consuming, not feasible for large data sets
- Use a global constant → replace all missing attribute values by same value (e.g. unknown)
 - may mistakenly think that “unknown” is an interesting concept

DATA CLEANING

MISSING VALUES

- Use mean or median → For normal (symmetric) data distributions, the mean is used, while skewed data distribution should employ the median
- Use mean or median for all samples belonging to the same class as the given tuple → e.g. mean or median of customers in a certain age group
- Use the most probable value → using regression, inference-based tools such as Bayesian formula or decision tree
 - Most popular

DATA CLEANING

NOISY DATA

- Noise is a random error or variance in a measured variable
- Data smoothing techniques:
 - Binning
 - Regression
 - Outlier Analysis

DATA CLEANING

NOISY DATA

- Binning → smooth a sorted data value by consulting its “neighborhood”
 - sorted values are partitioned into a # of “buckets,” or bins → local smoothing
 - equal-frequency bins → each bin has same # of values
 - equal-width bins → interval range of values per bin is constant
- Smoothing by bin means → each bin value is replaced by the bin mean
- Smoothing by bin medians → each bin value is replaced by the bin median
- Smoothing by bin boundaries → each bin value is replaced by the closest boundary value (min & max in a bin are bin boundaries)

DATA CLEANING

NOISY DATA

Example:

Sorted data for *price* (in dollars):
4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

Smoothing by bin means

Bin 1: 9, 9, 9

Bin 2: 22, 22, 22

Bin 3: 29, 29, 29

Smoothing by bin boundaries

Bin 1: 4, 4, 15

Bin 2: 21, 21, 24

Bin 3: 25, 25, 34

DATA CLEANING NOISY DATA

Example:

Sorted data for *price* (in dollars):

4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-width) bins

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24, 25, 28

Bin 3: 34

Smoothing by bin means

Bin 1: 9, 9, 9

Bin 2: 24, 24, 24, 24, 24

Bin 3: 34

Smoothing by bin boundaries

Bin 1: 4, 4, 15

Bin 2: 21, 21, 21, 28, 28

Bin 3: 34

DATA CLEANING

NOISY DATA

- Regression → Conform data values to a function
 - Linear regression → find “best” line to fit two attributes so that one attribute can be used to predict the other
- Outlier Analysis
- Potter's Wheel → Automated interactive data cleaning tool

Questions?





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DATA PRE-PROCESSING

(PART2)



AGENDA



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DATA INTEGRATION

- Merging data from multiple data stores
- Helps reduce and avoid redundancies and inconsistencies in the resulting data set
- Challenges:
 - Semantic heterogeneity → entity identification problem
 - Structure of data → functional dependencies and referential constraints
 - Redundancy

DATA INTEGRATION

ENTITY IDENTIFICATION PROBLEM

- Schema integration and object matching
- Metadata → name, meaning, data type, and range of values permitted, null rules for handling blank, zero, or null values
 - can help avoid errors in schema integration and data transformation

DATA INTEGRATION

REDUNDANCY AND CORRELATION ANALYSIS

- An Attribute is redundant if it can be “derived” from another attribute or set of attributes
- Redundancy can be detected by **correlation analysis** → measure how strongly one attribute implies the other, based on the available data
 - Chi-square (χ^2) test for nominal data
 - Correlation coefficient and covariance for numeric attributes

DATA INTEGRATION

REDUNDANCY AND CORRELATION ANALYSIS

- Chi-square (χ^2) test for nominal data
- Example: Are Gender and preferred reading correlated?
 - Contingency table → summary of observed values

		gender		
		male	female	Total
Preferred reading	Fiction	250	200	450
	Non-fiction	50	1000	1050
	Total	300	1200	1500

DATA INTEGRATION

REDUNDANCY AND CORRELATION ANALYSIS

- Hypothesis: the two attributes are **independent**
- Expected frequencies $\rightarrow e_{ij} = \frac{\text{count}(A=a_i) \times \text{count}(B=b_i)}{n}$
- e.g. $e_{11} = \frac{\text{count}(\text{male}) \times \text{count}(\text{fiction})}{n}$

$$= \frac{300 \times 450}{1500} = 90$$

		gender		
		male	female	Total
Preferred reading	Fiction	250 (90)	200 (360)	450
	Non-fiction	50 (210)	1000 (840)	1050
	Total	300	1200	1500

DATA INTEGRATION

REDUNDANCY AND CORRELATION ANALYSIS

$$\begin{aligned} \blacksquare \chi^2 &= \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}} = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} \\ &= 507.93 \end{aligned}$$

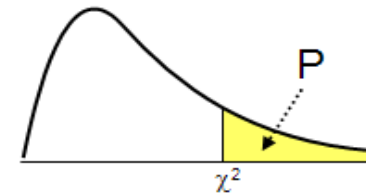
➤ o_{ij} → observed frequency

➤ e_{ij} → expected frequency

		gender		
		male	female	Total
Preferred reading	Fiction	250 (90)	200 (360)	450
	Non-fiction	50 (210)	1000 (840)	1050
	Total	300	1200	1500

DATA INTEGRATION REDUNDANCY AND CORRELATION ANALYSIS

- For one degree of freedom at a 0.001 significance level, the χ^2 value needed to reject the hypothesis is 10.828
- Degree of freedom:
 - If $r > 1$ and $c > 1$, then $df = (r-1)(c-1)$
 - If $r = 1$ and $c > 1$, then $df = c-1$ or if $r > 1$ and $c = 1$ then $df = r-1$
 - $r=c=1$ is not allowed
 - $507.93 > 10.828 \rightarrow$ reject hypothesis that preferred reading and gender are independent $\rightarrow$ they are strongly correlated



	P										
DF	0.995	0.975	0.20	0.10	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	0.0000393	0.000982	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.550	10.828
2	0.0100	0.0506	3.219	4.605	5.991	7.378	7.824	9.210	10.597	12.429	13.816
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	0.207	0.484	5.989	7.779	9.488	11.143	11.668	13.277	14.860	16.924	18.467
5	0.412	0.831	7.289	9.236	11.070	12.833	13.388	15.086	16.750	18.907	20.515
6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458
7	0.989	1.690	9.803	12.017	14.067	16.013	16.622	18.475	20.278	22.601	24.322
8	1.344	2.180	11.030	13.362	15.507	17.535	18.168	20.090	21.955	24.352	26.124
9	1.735	2.700	12.242	14.684	16.919	19.023	19.679	21.666	23.589	26.056	27.877
10	2.156	3.247	13.442	15.987	18.307	20.483	21.161	23.209	25.188	27.722	29.588
11	2.603	3.816	14.631	17.275	19.675	21.920	22.618	24.725	26.757	29.354	31.264
12	3.074	4.404	15.812	18.549	21.026	23.337	24.054	26.217	28.300	30.957	32.909
13	3.565	5.009	16.985	19.812	22.362	24.736	25.472	27.688	29.819	32.535	34.528
14	4.075	5.629	18.151	21.064	23.685	26.119	26.873	29.141	31.319	34.091	36.123
15	4.601	6.262	19.311	22.307	24.996	27.488	28.259	30.578	32.801	35.628	37.697
16	5.142	6.908	20.465	23.542	26.296	28.845	29.633	32.000	34.267	37.146	39.252
17	5.697	7.564	21.615	24.769	27.587	30.191	30.995	33.409	35.718	38.648	40.790
18	6.265	8.231	22.760	25.989	28.869	31.526	32.346	34.805	37.156	40.136	42.312
19	6.844	8.907	23.900	27.204	30.144	32.852	33.687	36.191	38.582	41.610	43.820
20	7.434	9.591	25.038	28.412	31.410	34.170	35.020	37.566	39.997	43.072	45.315
21	8.034	10.283	26.171	29.615	32.671	35.479	36.343	38.932	41.401	44.522	46.797
22	8.643	10.982	27.301	30.813	33.924	36.781	37.659	40.289	42.796	45.962	48.268
23	9.260	11.689	28.429	32.007	35.172	38.076	38.968	41.638	44.181	47.391	49.728

DATA INTEGRATION

REDUNDANCY AND CORRELATION ANALYSIS

- **Correlation coefficient for numeric attributes**

- $$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - (n\bar{A}\bar{B})}{n\sigma_A\sigma_B}$$

- $-1 \leq r_{A,B} \leq +1$

- If $r_{A,B}$ is greater than 0, then A and B are **positively correlated**

- The higher the value, the stronger the correlation

- If $r_{A,B} = 0$ then A and B are **independent**

- Correlation does not imply causality!

DATA INTEGRATION REDUNDANCY AND CORRELATION ANALYSIS

Covariance for numeric attributes

- $E(A) = \bar{A} = \frac{\sum_{i=1}^n a_i}{n}, E(B) = \bar{B} = \frac{\sum_{i=1}^n b_i}{n}$
- $Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$
- It follows that $r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B} \rightarrow Cov(A, B) = E(A \cdot B) - \bar{A}\bar{B}$

Covariance is *+ve* $\rightarrow$ A & B change together, and $A > \bar{A}$ then $B > \bar{B}$

Covariance is *-ve* $\rightarrow$ one is above its mean and one is below

If A and B are independent $\rightarrow$ Covariance = 0

DATA INTEGRATION

REDUNDANCY AND CORRELATION ANALYSIS

- Example: Stock price for two companies

- $E(\text{AllElectronics}) = 20/5 = \4

- $E(\text{HighTech}) = 54/5 = \10.80

- $\text{Cov}(\text{AllElectronics}, \text{HighTech})$

$$= \frac{6 \times 20 + 5 \times 10 + 4 \times 14 + 3 \times 5 + 2 \times 5}{5} - 4 \times 10.80$$

$$= 50.2 - 43.2 = 7$$

Time point	AllElectronics	HighTech
T1	6	20
T2	5	10
T3	4	14
T4	3	5
T5	2	5

Covariance is +ve → Stock prices of the two companies rise together

DATA INTEGRATION MORE ISSUES

- **Tuple duplication**
- The use of denormalized tables (often done to improve performance by avoiding joins) is another source of data redundancy.
 - e.g. purchaser name and address, and purchases
- **Data value conflict**
 - e.g. grading system in two different institutes → A, B, ... versus 90%, 80% ...

QUIZ

- Calculate the correlation coefficient. Are these two attributes positively or negatively correlated? Compute their covariance.
 - (Hint: $n = 18$)
 - SD for Age and fat are 12.85 and 8.99 respectively
 - Mean for Age and fat are 46.44 and 28.78 respectively
 - $E(\text{age} * \text{fat}) = 1431.29$

Age	23	23	27	27	39	41	47	49	50	52	54	54	56	57	58	58	60	61
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

QUIZ .. SOL.

- $r_{A,B} = \frac{\sum_{i=1}^n (a_i b_i) - (n \bar{A} \bar{B})}{n \sigma_A \sigma_B}$
- $r_{A,B} = \frac{25763.2 - 18 \times 46.44 \times 28.78}{18 \times 12.85 \times 8.99} = 0.82 \rightarrow$ The two attributes are positively correlated

- $Cov(A, B) = E(A.B) - \bar{A}\bar{B}$

$$Cov(A, B) = 1431.29 - 46.44 \times 28.78 = 94.64$$

$$r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B} = \frac{94.64}{12.85 \times 8.99} = 0.82$$



AGENDA



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DATA REDUCTION STRATEGIES

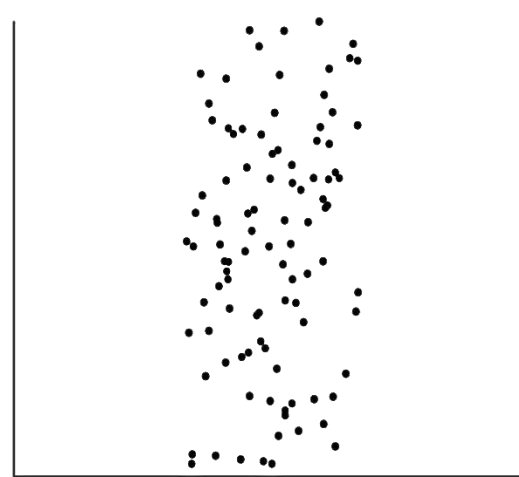
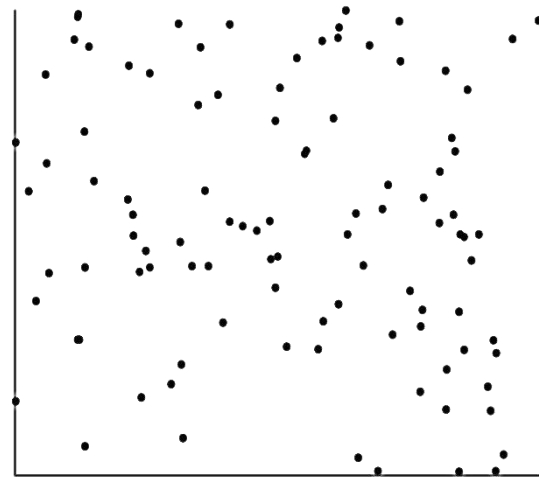
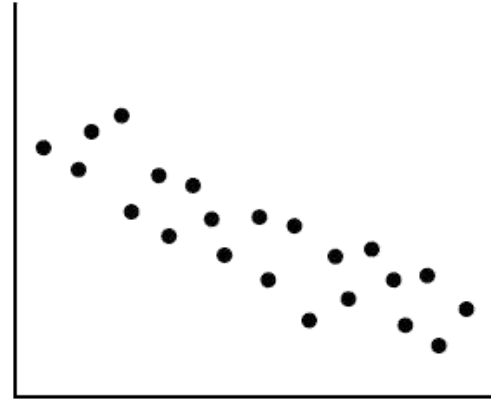
- **Dimensionality reduction** → reduce number of attributes
 - Wavelet transforms, PCA, Attribute subset selection
- **Numerosity reduction** → replace original data volume by smaller data representation
 - Parametric → a model is used to estimate the data - only the data parameters are stored
 - Regression
 - Nonparametric → store reduced representations of the data
 - Histograms, clustering, sampling
- **Compression** → transformations applied to obtain a “compressed” representation of original data
 - Lossless, Lossy

DATA REDUCTION

ATTRIBUTE SUBSET SELECTION

- Find a min set of attributes such that the resulting probability distribution of data is as close as possible to the original distribution using all attributes
- An exhaustive search can be prohibitively expensive
- Heuristic (Greedy) search
 - Stepwise forward selection: start with empty set of attributes as reduced set. The best of the attributes is determined and added to the reduced set. At each subsequent iteration, the best of the remaining attributes is added to the set
 - Stepwise backward elimination: start with the full set of attributes. At each step, remove the worst attribute remaining in the set
 - Combination of forward selection and backward elimination
 - Decision tree induction
- **Attribute construction** → e.g. area attribute based on height and width attributes

DATA REDUCTION REGRESSION



DATA REDUCTION

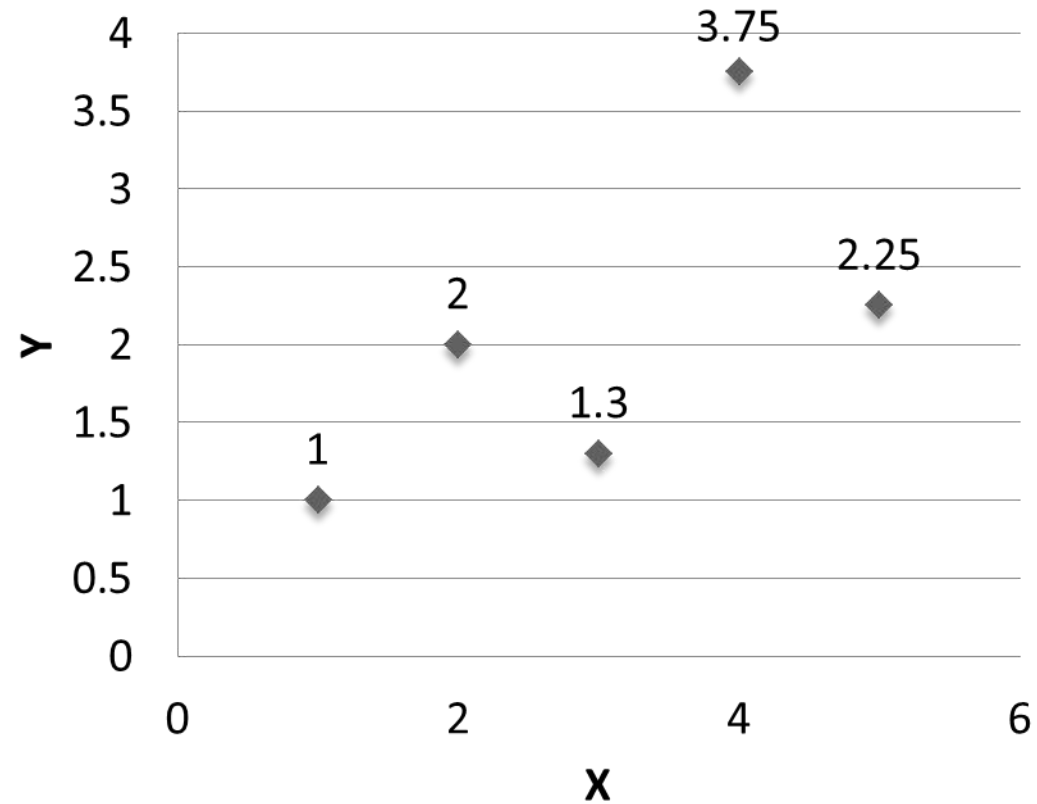
REGRESSION

- Data is modeled to fit a straight line
- A random variable y (response variable), can be modeled as a linear function of another random variable x (predictor variable)
- Regression line equation $\rightarrow y = wx + b$
- w and b are regression coefficients $\rightarrow$ they specify the slope of the line and y -intercept
- Solved for by the method of least squares $\rightarrow$ minimize error between actual line separating data and estimate of the line (best-fitting line)

DATA REDUCTION REGRESSION

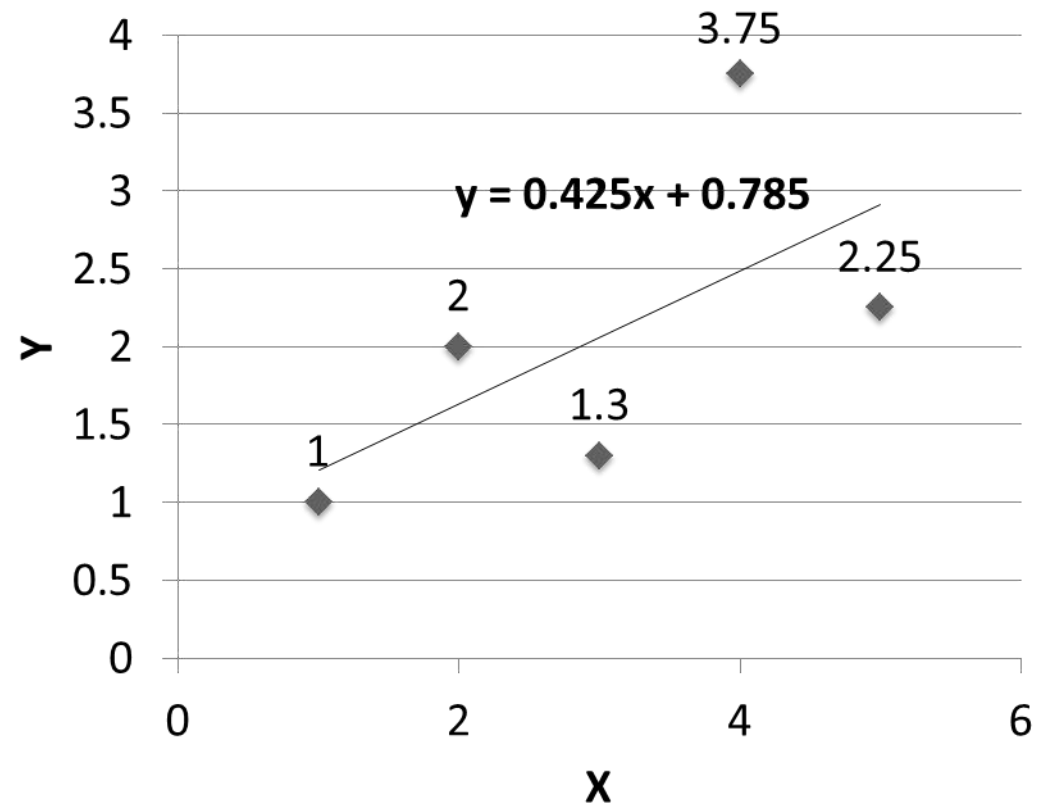
■ Example:

X	Y
1.00	1.00
2.00	2.00
3.00	1.30
4.00	3.75
5.00	2.25



DATA REDUCTION REGRESSION

X	Y
1.00	1.00
2.00	2.00
3.00	1.30
4.00	3.75
5.00	2.25



DATA REDUCTION

HISTOGRAMS

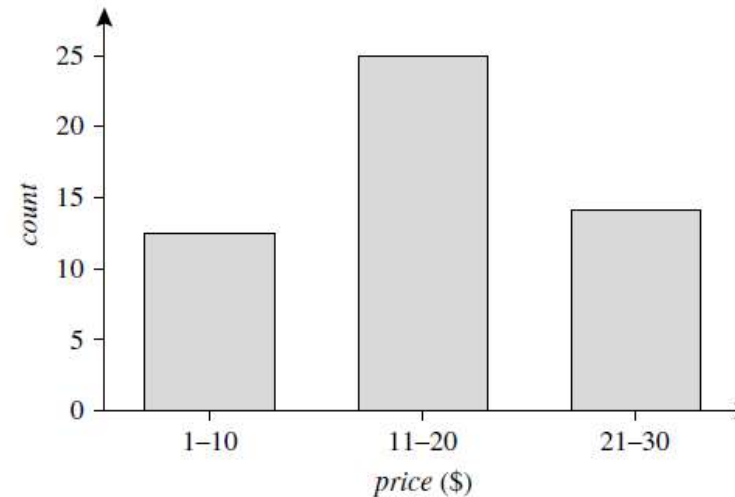
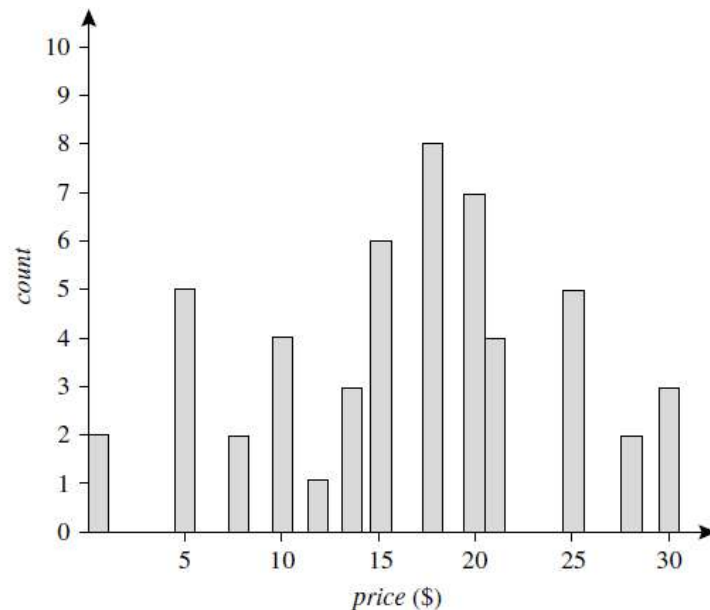
- A **histogram** for an attribute, A , partitions the data distribution of A into disjoint subsets, referred to as buckets or bins.
 - a single attribute–value/frequency pair $\rightarrow$ singleton buckets
 - Often, buckets represent continuous ranges for the given attribute.
- Equal-width: the width of each bucket range is uniform (e.g., the width of \$10 for the buckets).
- Equal-frequency (or equal-depth): roughly, the frequency of each bucket is constant (i.e., each bucket contains roughly the same number of contiguous data samples).

DATA REDUCTION

HISTOGRAMS

- The following data are a list of AllElectronics prices for commonly sold items (rounded to the nearest dollar). The numbers have been sorted:

1, 1, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30.

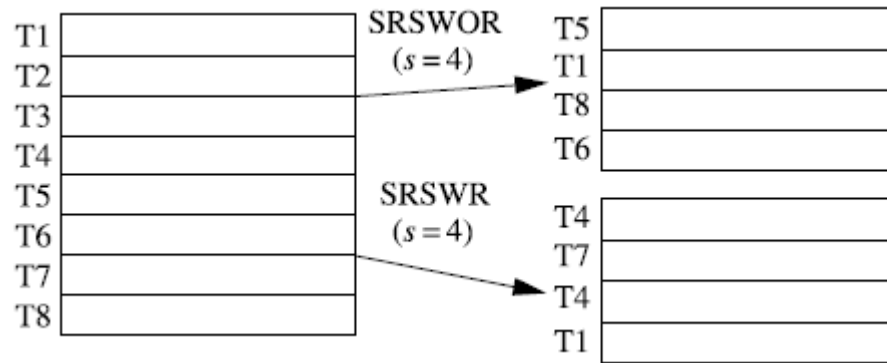


DATA REDUCTION

SAMPLING

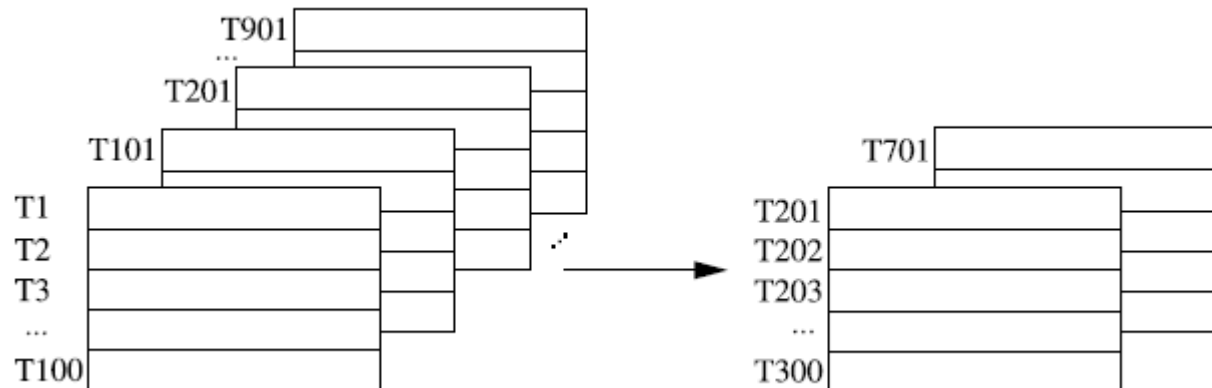
- A large data set represented by a smaller random data sample
- Simple random sample without replacement (**SRSWOR**) of size $s \rightarrow$ draw s of the N tuples ($s < N$)
 - all tuples are equally likely to be sampled
- Simple random sample with replacement (**SRSWR**) of size $s \rightarrow$ similar to SRSWOR, but each time a tuple is drawn, it's recorded then placed back so it may be drawn again
- **Cluster sample** $\rightarrow$ If tuples are grouped into M “clusters,” an SRS of s clusters can be obtained
- **Stratified sample** $\rightarrow$ If tuples are divided into strata, a stratified sample is generated by obtaining an SRS at each stratum
 - e.g. stratum is created for each customer age group

DATA REDUCTION SAMPLING



T38	youth
T256	youth
T307	youth
T391	youth
T96	middle_aged
T117	middle_aged
T138	middle_aged
T263	middle_aged
T290	middle_aged
T308	middle_aged
T326	middle_aged
T387	middle_aged
T69	senior
T284	senior

T38	youth
T391	youth
T117	middle_aged
T138	middle_aged
T290	middle_aged
T326	middle_aged
T69	senior



AGENDA



Overview

Why pre-process data?
Major tasks



Data Cleaning

Missing values
Noisy data



Data Integration

Entity Identification Problem
Redundancy and correlation analysis
Tuple duplication
Data value conflict detection



Data Reduction

Strategies
Wavelet transforms
PCA
Attribute subset selection
Regression
Histograms
Clustering
Sampling



Data Transformation

Strategies
Normalization
Binning
Histogram analysis
Cluster/Decision trees/Correlation analyses
Concept hierarchy

TRANSFORMATION AND DISCRETIZATION

TRANSFORMATION STRATEGIES

- Smoothing → binning, regression
- Attribute construction
- Aggregation
- Normalization → raw values of a numeric attribute (e.g. age) replaced by interval labels (e.g. 0–10, 11–20) or conceptual labels (e.g., youth, adult, senior)
- Concept hierarchy → e.g. street generalized to higher-level concepts (city or country)

TRANSFORMATION AND DISCRETIZATION

TRANSFORMATION BY NORMALIZATION

- To help avoid dependence on the choice of measurement units
- Give all attributes equal weight
- Methods:
 - min-max normalization
 - z-score normalization

TRANSFORMATION AND DISCRETIZATION

TRANSFORMATION BY NORMALIZATION

- **min-max normalization** → linear transformation
- For value v_i
- $$\hat{v}_i = \frac{v_i - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$
- Example: income *min* and *max* are \$12000 and \$98000. Mapping to [0.0, 1.0], a value of \$73600 for income is transformed to

$$\frac{73600 - 12000}{98000 - 12000} (1.0 - 0) + 0 = 0.716$$

TRANSFORMATION AND DISCRETIZATION

TRANSFORMATION BY NORMALIZATION

- **Z-score normalization** → attribute value normalized based on mean and SD
- For value v_i
- $$v'_i = \frac{v_i - \bar{A}}{\sigma_A}$$
- Example: income *mean and SD* are \$54000 and \$16000, a z-score for a value of \$73600 for income is

$$\frac{73600 - 54000}{16000} = 1.225$$

TRANSFORMATION AND DISCRETIZATION

CONCEPT HIERARCHY

- **Concept hierarchy** organizes concepts (i.e., attribute values) hierarchically
- Concept hierarchies facilitate drilling and rolling to view data in multiple granularity
- Concept hierarchy formation: Recursively reduce data by collecting and replacing low level concepts (e.g. age values) by higher level concepts (e.g. age groups: youth, adult, or senior)
- Concept hierarchies can be explicitly specified by domain experts
- Concept hierarchy can be automatically formed for both numeric and nominal data
→ discretization

TRANSFORMATION AND DISCRETIZATION

CONCEPT HIERARCHY

- For nominal data:
- Specification of a partial ordering of attributes explicitly at the schema level by users or experts
 - street, city, province or state, country $\rightarrow$ street $<$ city $<$ province or state $<$ country
- Specification of a set of attributes, but not of their partial ordering $\rightarrow$ order automatically generated by system
 - e.g. Location $\rightarrow$ country contains a smaller #distinct values than street
 - automatically generate concept hierarchy based on # distinct values per attribute in the given attribute set
 - Not for all concepts! Time $\rightarrow$ year (20), month (12), day of week (7)

SUMMARY

Cleaning	Integration	Reduction	Transformation/Discretization
Binning			Binning
Regression		Regression	Regression
	Correlation analysis		Correlation
		Histograms	Histogram analysis
		Clustering	Clustering
		Attribute construction	Attribute construction
			Aggregation
			Normalization
Outlier analysis			
		Wavelet transforms	
		PCA	
		Attribute subset selection	
		Sampling	
			Concept hierarchy

QUIZ

- You have this data for the attribute age: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.
- Use smoothing by bin means to smooth these data, using a bin depth of 3
- Use min-max normalization to transform the value 35 for age onto the range [0.0, 1.0].
- Use z-score normalization to transform the value 35 for age, where the standard deviation of age is 12.94 years.

Questions?





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[IS422P] Data Mining

Grade:4 Th

Dr. Amira Rezk



MINING FREQUENT PATTERNS, ASSOCIATIONS, & CORRELATIONS



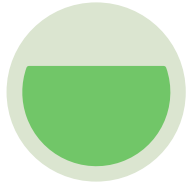


The Basics

Market Basket
Analysis

Frequent Itemsets

Association Rules

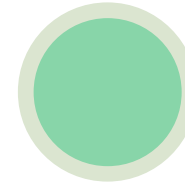


Frequent Itemset Mining Methods

Apriori Algorithm

Generating
Association Rules
from Frequent
Itemsets

FP-Growth



Pattern Evaluation Methods



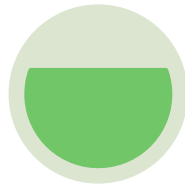


The Basics

Market Basket
Analysis

Frequent Itemsets

Association Rules

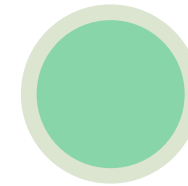


Frequent Itemset
Mining Methods

Apriori Algorithm

Generating
Association Rules
from Frequent
Itemsets

FP-Growth



Pattern Evaluation
Methods



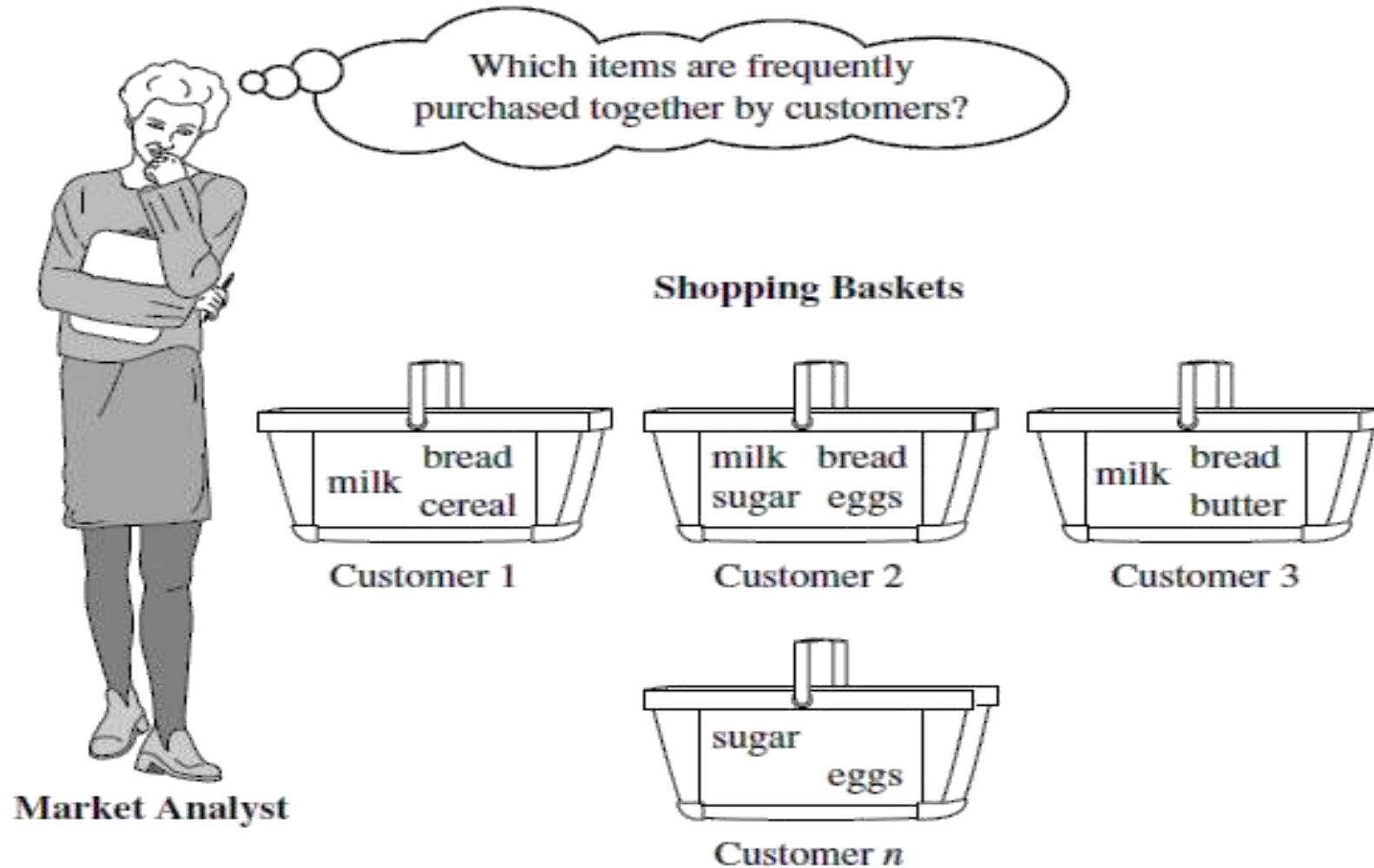
THE BASICS

WHAT IS FREQUENT PATTERN ANALYSIS?

- Frequent pattern: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [AIS93] in the context of frequent itemsets and association rule mining



THE BASICS



THE BASICS

- **Motivation:** Finding inherent regularities in data
 - What products were often purchased together?— Beer and diapers?!
 - What are the subsequent purchases after buying a PC?
 - What kinds of DNA are sensitive to this new drug?
 - Can we automatically classify web documents?
- **Applications**
 - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis



THE BASICS

- Frequent Pattern are itemsets that appear frequently in a data set (e.g. Transaction record)
- Items that are frequently associated (e.g purchased) together can be represented as association rules

Computer → antivirus_SW [Support = 2% , Confidence =60%]

- **Support** and **Confidence** are measures of rule interestingness
- 2% Support means 2% of Transactions Show that computers and antivirus_SW are bought Together
- 60% Confidence means 60 % of customers who bought a computer also bought antivirus_SW



THE BASICS

FREQUENT ITEM-SETS

- Itemset $X = \{x_1, \dots, x_k\}$ ex: $X = \{A, B, C, D, E, F\}$
- Find all the rules $X \rightarrow Y$ with minimum support and confidence
 - support, s , probability that a transaction contains $X \cup Y$
 - confidence, c , conditional probability that a transaction having X also contains Y

$$\text{support } X \rightarrow Y = P(X \cup Y) = \frac{n(X \cup Y)}{N}$$

$$\text{confidence } (X \rightarrow Y) = P(Y|X) = \frac{n(X \cup Y)}{n(X)}$$



THE BASICS

ASSOCIATION RULES

Ex: Let $\text{min_Sup.} = 50\%$, $\text{min_conf.} = 50\%$

Frequent Patterns:

$\{A:3, B:3, D:4, E:3, AD:3\}$

Association rules:

$A \rightarrow D$ (60%, 100%)

$D \rightarrow A$ (60%, 75%)

$$\text{conf}(A \rightarrow D) = \frac{3}{3} = 100\%$$

$$\text{conf}(D \rightarrow A) = \frac{3}{4} = 75\%$$

Transaction-id	Items bought
10	A, B, D
20	A, C, D
30	A, D, E
40	B, E, F
50	B, C, D, E, F

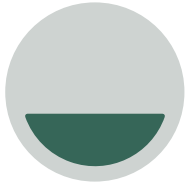


THE BASICS

ASSOCIATION RULES

- If frequency of itemset I satisfies min_support count then I is a frequent itemset
- If a rule satisfies min_support and min_confidence thresholds, it is said to be strong
 - problem of mining association rules reduced to mining frequent itemsets
- Association rules mining becomes a two-step process:
 - Find all frequent itemsets that occur at least as frequently as a predetermined min_support count
 - Generate strong association rules from the frequent itemsets that satisfy min_support and min_confidence



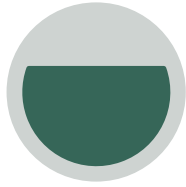


The Basics

Market Basket
Analysis

Frequent Itemsets

Association Rules

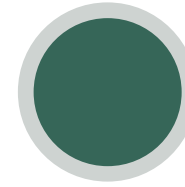


Frequent Itemset Mining Methods

Apriori Algorithm

Generating
Association Rules
from Frequent
Itemsets

FP-Growth



Pattern Evaluation Methods



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

- Goes as follows:
 - Find frequent 1-itemsets $\rightarrow L_1$
 - Use L_1 to find frequent 2-itemsets $\rightarrow L_2$
 - ... until no more frequent k-itemsets can be found
- Each L_k itemset requires a full dataset scan
- To improve efficiency, use the Apriori property:
 - “All nonempty subsets of a frequent itemset must also be frequent” – if a set cannot pass a test, all of its supersets will fail the same test as well – if $P(I) < \text{min_support}$ then $P(I \cup A) < \text{min_support}$



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

Transactional data example
N=10, min_supp count=2

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

Scan dataset for
count of each
candidate

C_1

Itemset	Support count
{I1}	7
{I2}	8
{I3}	6
{I4}	2
{I5}	2

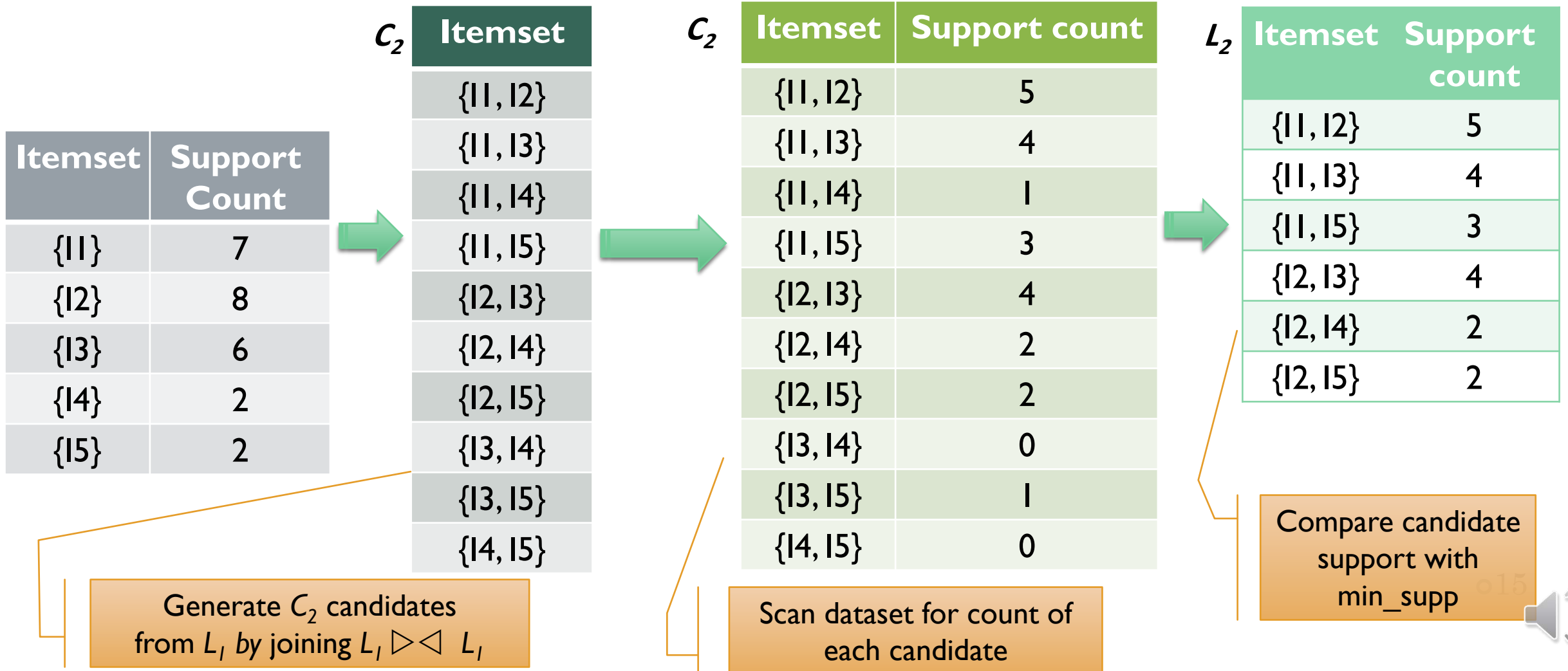
Compare candidate
support with
min_support

L_1

Itemset	Support Count
{I1}	7
{I2}	8
{I3}	6
{I4}	2
{I5}	2

MINING FREQUENT ITEMSETS

APRIORI ALGORITHM



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

$C_3 = L_2 \triangleright \triangleleft L_2 = \{\{I1, I2, I3\}, \{I1, I2, I5\},$
 ~~$\{I1, I3, I5\}, \{I2, I3, I4\}, \{I2, I3, I5\}, \{I2, I4, I5\}\}$~~

Not all subsets are frequent $\rightarrow$ **Prune** (Apriori property)

Itemset	Support count
{I1, I2}	5
{I1, I3}	4
{I1, I5}	3
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2



Itemset
{I1, I2, I3}
{I1, I2, I5}



C_3

Itemset	Support count
{I1, I2, I3}	2
{I1, I2, I5}	2

Scan dataset for count of each candidate



L_3

Itemset	Support count
{I1, I2, I3}	2
{I1, I2, I5}	2

Compare candidate support with min_supp

Generate C_3 candidates from L_2 by **joining** $L_2 \triangleright \triangleleft L_2$



Two joining (lexicographically ordered) k-itemsets must share first k-1 items $\rightarrow \{I1, I2\}$ is not joined with $\{I2, I4\}$



MINING FREQUENT ITEMSETS

APRIORI ALGORITHM

Itemset	Support count
{11, 12, 13}	2
{11, 12, 15}	2



Itemset
{11, 12, 13, 15}



Not all subsets are frequent → **Pruning**

$C_4 = \phi \rightarrow \text{Terminate}$



APRIORI ALGORITHM

Algorithm: Apriori. Find frequent itemsets using an iterative level-wise approach based on candidate generation.

Input:

- D , a database of transactions;
- min_sup , the minimum support count threshold.

Output: L , frequent itemsets in D .

Method:

```
(1)  $L_1 = \text{find\_frequent\_1-itemsets}(D)$ ;  
(2) for ( $k = 2; L_{k-1} \neq \phi; k++$ ) {  
(3)    $C_k = \text{apriori\_gen}(L_{k-1})$ ;  
(4)   for each transaction  $t \in D$  { // scan  $D$  for counts  
(5)      $C_t = \text{subset}(C_k, t)$ ; // get the subsets of  $t$  that are candidates  
(6)     for each candidate  $c \in C_t$   
(7)        $c.\text{count}++$ ;  
(8)   }  
(9)    $L_k = \{c \in C_k | c.\text{count} \geq min\_sup\}$   
(10) }  
(11) return  $L = \cup_k L_k$ ;
```

Generate C_k using L_{k-1} to find L_k

procedure $\text{apriori_gen}(L_{k-1}:\text{frequent } (k-1)\text{-itemsets})$

```
(1) for each itemset  $l_1 \in L_{k-1}$   
(2)   for each itemset  $l_2 \in L_{k-1}$   
(3)     if ( $(l_1[1] = l_2[1]) \wedge (l_1[2] = l_2[2])$   
        $\wedge \dots \wedge (l_1[k-2] = l_2[k-2]) \wedge (l_1[k-1] < l_2[k-1])$ ) then {  
(4)        $c = l_1 \bowtie l_2$ ; // join step: generate candidates  
(5)       if  $\text{has\_infrequent\_subset}(c, L_{k-1})$  then  
(6)         delete  $c$ ; // prune step: remove unfruitful candidate  
(7)       else add  $c$  to  $C_k$ ;  
(8)     }  
(9) return  $C_k$ ;
```

Join

Prune

procedure $\text{has_infrequent_subset}(c:\text{candidate } k\text{-itemset};$

```
   $L_{k-1}:\text{frequent } (k-1)\text{-itemsets})$ ; // use prior knowledge  
(1) for each  $(k-1)$ -subset  $s$  of  $c$   
(2)   if  $s \notin L_{k-1}$  then  
(3)     return TRUE;  
(4) return FALSE;
```



MINING FREQUENT ITEMSETS

GENERATING ASSOCIATION RULES FROM FREQUENT ITEMSETS

- Association rules can be generated using the confidence equation, as follows”

$$\text{confidence}(A \Rightarrow B) = P(B|A) = \frac{\text{support_count}(A \cup B)}{\text{support_count}(A)}$$

- For each frequent itemset L, generate all nonempty subset of L
- For every nonempty subset S of L, output rule: $S \rightarrow L-S$

- $\frac{\text{Supportcount}(L)}{\text{Supportcount}(S)} \geq \text{min_conf},$

where min_conf is the minimum confidence threshold.



MINING FREQUENT ITEMSETS

GENERATING ASSOCIATION RULES FROM FREQUENT ITEMSETS

Itemset	Support count
{11, 12, 13}	2
{11, 12, 15}	2

Nonempty subsets
{11, 12}
{11, 15}
{12, 15}
{11}
{12}
{15}

Association Rules
{11, 12} → 15
{11, 15} → 12
{12, 15} → 11
11 → {12, 15}
12 → {11, 15}
15 → {11, 12}

Confidence
$2/5 = 40\%$
$2/2 = 100\%$
$2/2 = 100\%$
$2/7 = 28\%$
$2/8 = 25\%$
$2/2 = 100\%$

For a *min_confidence* = 70%



MINING FREQUENT ITEMSETS

FP-GROWTH

- To avoid costly candidate generation
- Divide-and-conquer strategy:
- Compress database representing frequent items into a frequent pattern tree (FP-tree) – 2 passes over dataset
- Divide compressed database (FP-tree) into conditional databases, then mine each for frequent itemsets – traverse through the FP-tree



MINING FREQUENT ITEMSETS

FP-GROWTH

Transactional data example
N=10, min_supp count=2

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

Scan dataset for count
of each candidate

C_1

Itemset	Support count
{I1}	7
{I2}	8
{I3}	6
{I4}	2
{I5}	2

Compare candidate
support with min_supp

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2



MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2

FP-tree



TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
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T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

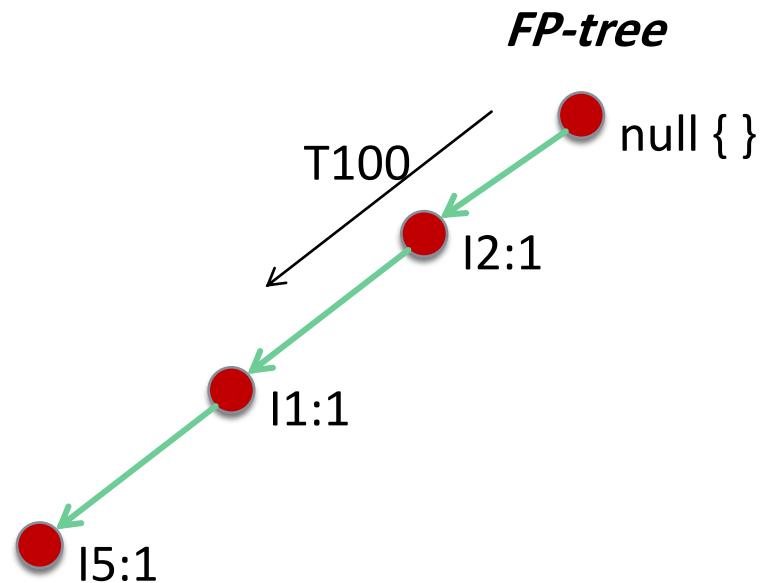


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2



Order of items is kept throughout path construction, with common prefixes shared whenever applicable

TID	List of items
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T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
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T700	I1, I3
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T1000	I1, I2

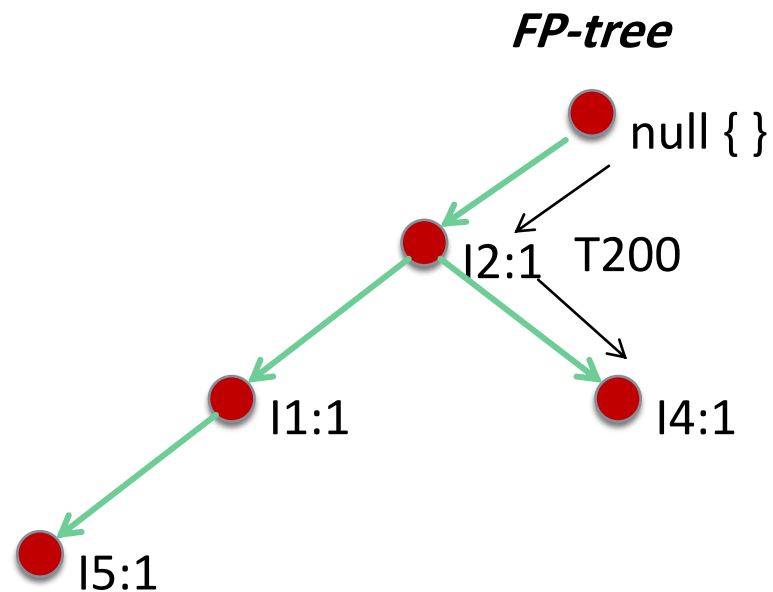


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

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T1000	I1, I2

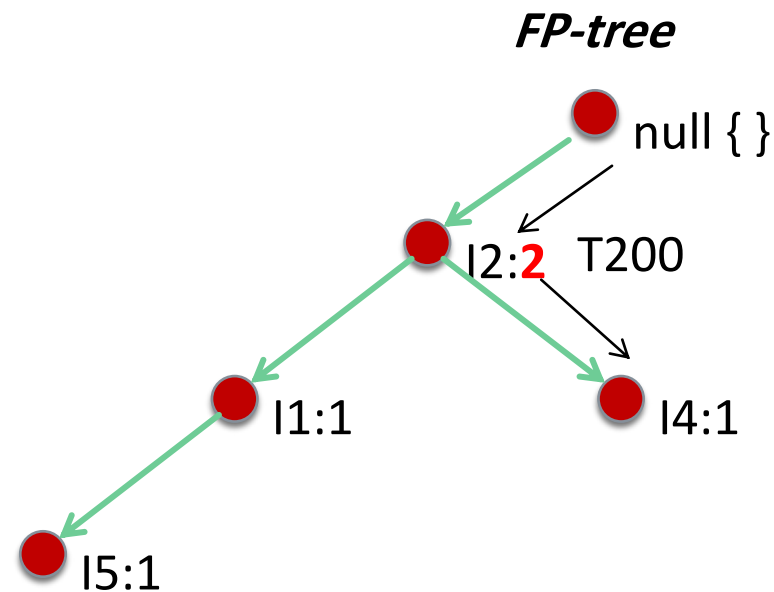


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

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T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

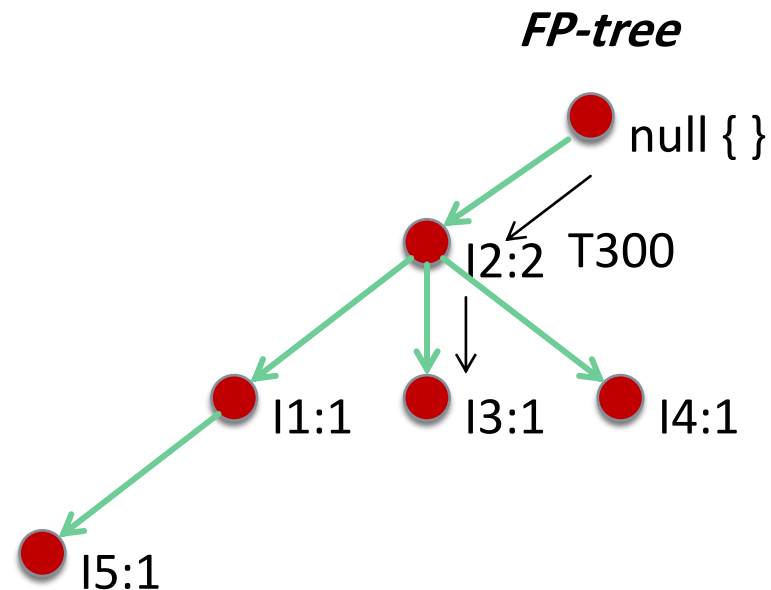


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

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T1000	I1, I2

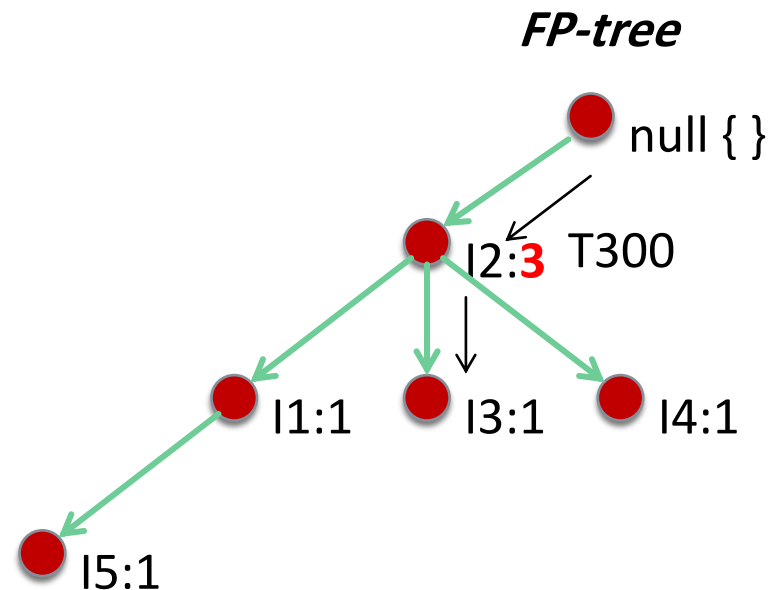


MINING FREQUENT ITEMSETS

FP-GROWTH – FP-TREE CONSTRUCTION

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T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

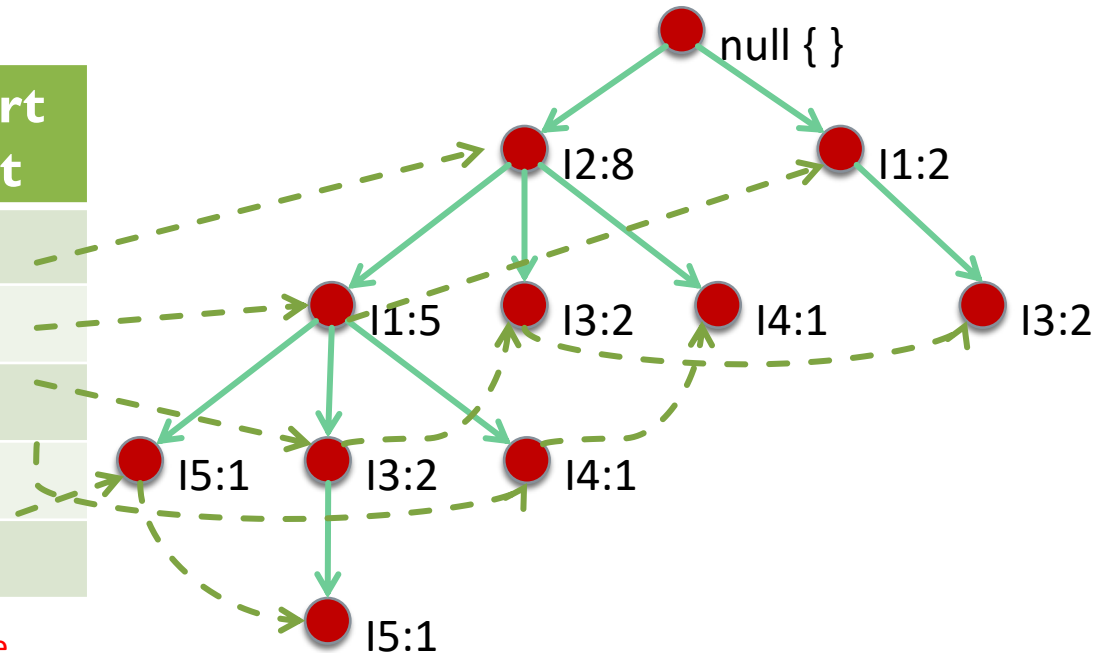
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{I3}	6
{I4}	2
{I5}	2

For Tree Traversal

FP-tree



TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



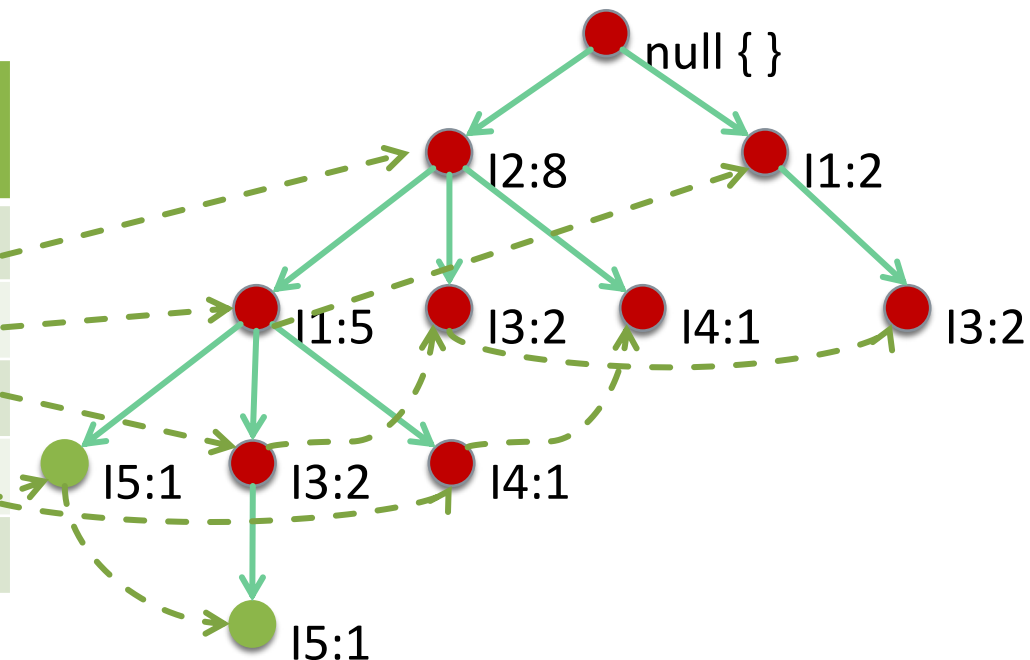
MINING FREQUENT ITEMSETS

FP-GROWTH – FREQUENT PATTERNS MINING

L_1 - Reordered

Itemset	Support count
{I2}	8
{I1}	7
{I3}	6
{I4}	2
{I5}	2

FP-tree



TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2

Bottom-up algorithm – start from leaves and go up to root



MINING FREQUENT ITEMSETS

FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I5

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree

 null { }

Eliminate I5

Eliminate transactions
not including I5

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

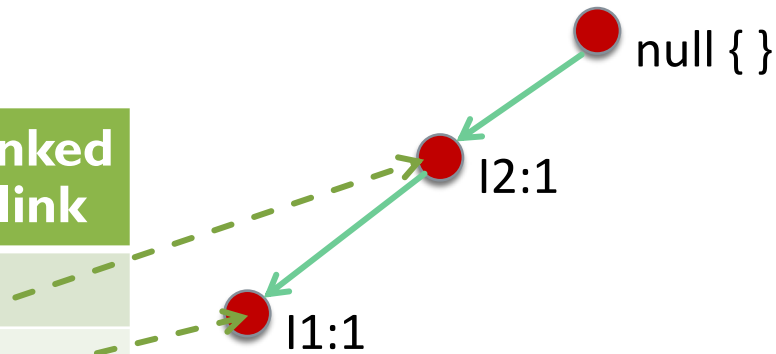
FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I5

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree



Eliminate I5

Eliminate transactions
not including I5

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



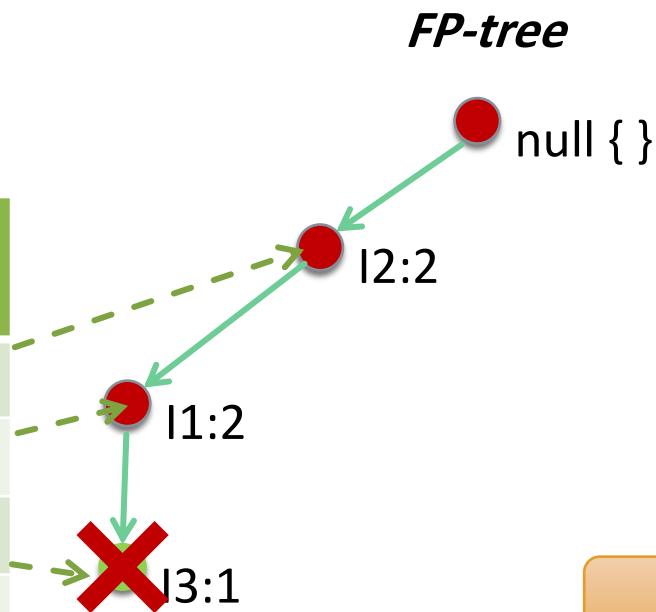
MINING FREQUENT ITEMSETS

FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I5

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	



Eliminate I5

Eliminate transactions
not including I5

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



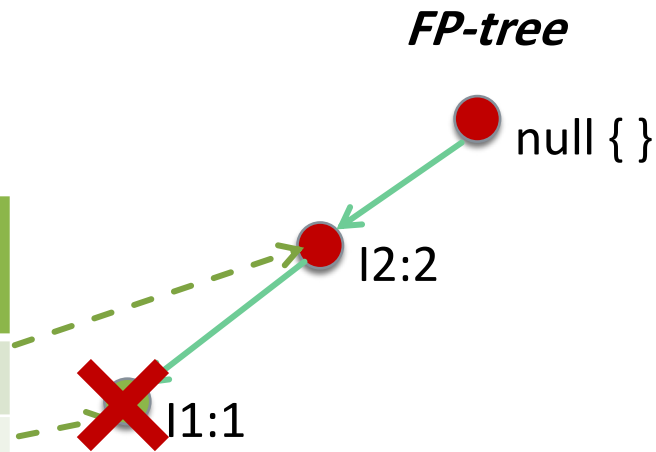
MINING FREQUENT ITEMSETS

FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I4

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	



Eliminate I4

Eliminate transactions
not including I4

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

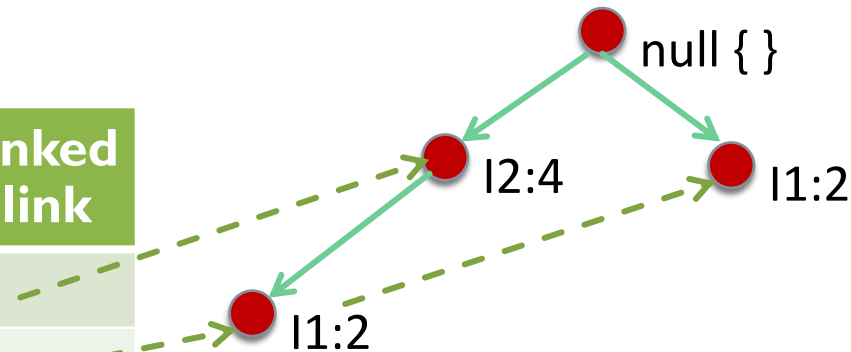
FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I3

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree



Eliminate I3

Eliminate transactions
not including I3

TID	List of items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3 , I5
T900	I1, I2, I3
T1000	I1, I2



MINING FREQUENT ITEMSETS

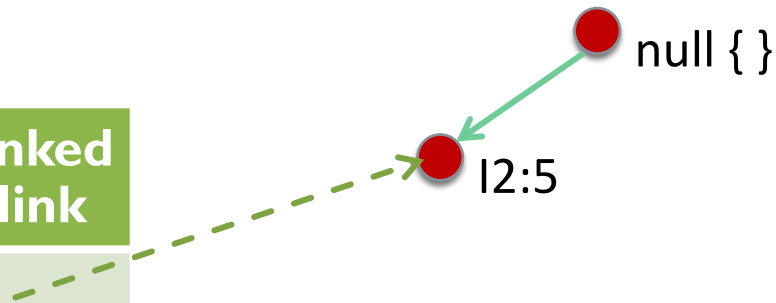
FP-GROWTH – CONDITIONAL FP-TREE CONSTRUCTION

For I1

L_1 - Reordered

Itemset	Support count	Linked link
{I2}	8	
{I1}	7	
{I3}	6	
{I4}	2	
{I5}	2	

FP-tree



Eliminate I1

Eliminate transactions
not including I1

TID	List of items
T100	I1 , I2, I5
T200	I2, I4
T300	I2, I3
T400	I1 , I2, I4
T500	I1 , I3
T600	I2, I3
T700	I1 , I3
T800	I1 , I2, I3, I5
T900	I1 , I2, I3
T1000	I1 , I2

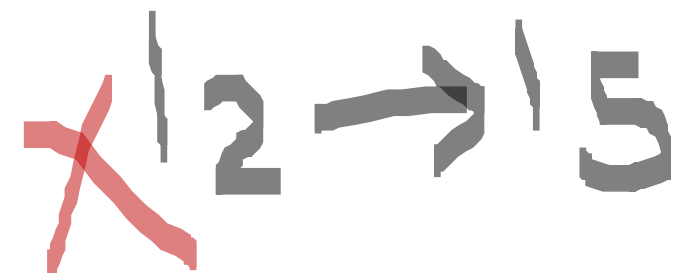
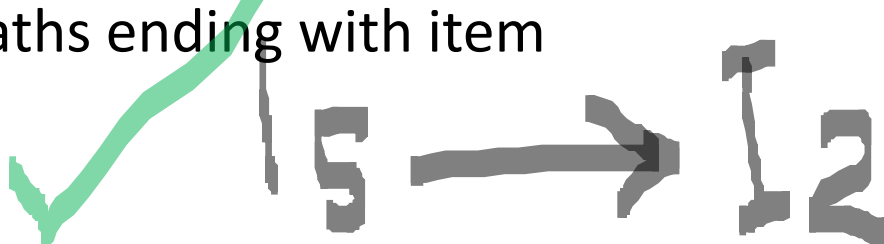


MINING FREQUENT ITEMSETS

FP-GROWTH

Item	Conditional Pattern Base	Conditional FP-tree	Frequent Patterns Generated
15	$\{\{I2, I1: 1\}, \{I2, I1, I3: 1\}\}$	$\langle I2:2, I1:2 \rangle$	$\{I2, I5: 2\}, \{I1, I5: 2\}, \{I2, I1, I5: 2\}$
14	$\{\{I2, I1: 1\}, \{I2: 1\}\}$	$\langle I2:2 \rangle$	$\{I2, I4: 2\}$
13	$\{\{I2, I1: 2\}, \{I2: 2\}, \{I1: 2\}\}$	$\langle I2:4, I1:2 \rangle, \langle I1:2 \rangle$	$\{I2, I3: 4\}, \{I1, I3: 4\}, \{I2, I1, I3: 2\}$
11	$\{\{I2: 5\}\}$	$\langle I2:5 \rangle$	$\{I2, I1: 5\}$

Paths ending with item



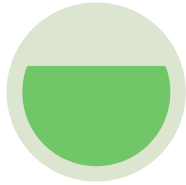


The Basics

Market Basket
Analysis

Frequent Itemsets

Association Rules

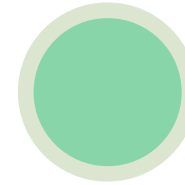


Frequent Itemset Mining Methods

Apriori Algorithm

Generating
Association Rules
from Frequent
Itemsets

FP-Growth



Pattern Evaluation Methods



PATTERN EVALUATION METHODS

- Not all association rules are interesting
 - $\text{Buys}(X, \text{"Computer games"}) \rightarrow \text{buys}(X, \text{"Videos"})$ [40%, 66%]
 - $P(\text{"videos"})$ is 75% > 66%
 - The two items are negatively associated means buying one decreases the likelihood of buying the other
 - We need to measure “real strength” of rule
- **Correlation analysis**
 - $A \rightarrow B$ [support , confidence , correlation]



PATTERN EVALUATION METHODS

1. **Lift** = $\frac{P(A \cup B)}{P(A)P(B)}$
 - A and B are independent if $P(A \cup B) = P(A)P(B)$
 - Otherwise, **dependent and correlated** occurrence
 - If lift < 1, A is **Negatively correlated** with B
 - If lift > 1, A is **Positively correlated** with B A's occurrence "lifts" the occurrence of B
2. χ^2 → already discussed in previous lecture



QUESTIONS?

NEXT ...





Mansoura University
Faculty of Computers and Information
Department of Information System



[IS422P] Data Mining

Grade:4 Th

Dr. Amira Rezk



CLASSIFICATION (PART I)



AGENDA



The Basics

What is Classification?
General Approach



Decision Tree Induction

The Algorithm
Attribute Selection Measures
Tree Pruning
Extracting Rules from Decision Trees



Bayes Classification

Bayes' Theorem
Naïve Bayesian Classification



Lazy Learners

K-Nearest Neighbor Classifiers



Regression analysis

Linear regression



Model Evaluation

Metrics for Evaluating Classifiers Performance
Cross-Validation
Bootstrap



Improving Classification Accuracy

Bagging
Boost and AdaBoost



THE BASICS

WHAT IS CLASSIFICATION

- Motivation: Prediction
 - Is a bank loan applicant “safe” or “risky”?
 - Which treatment is better for patient, “treatmentX” or “treatmentY”?
- Classification is a data analysis task where a model is constructed to predict class labels (categories)



THE BASICS

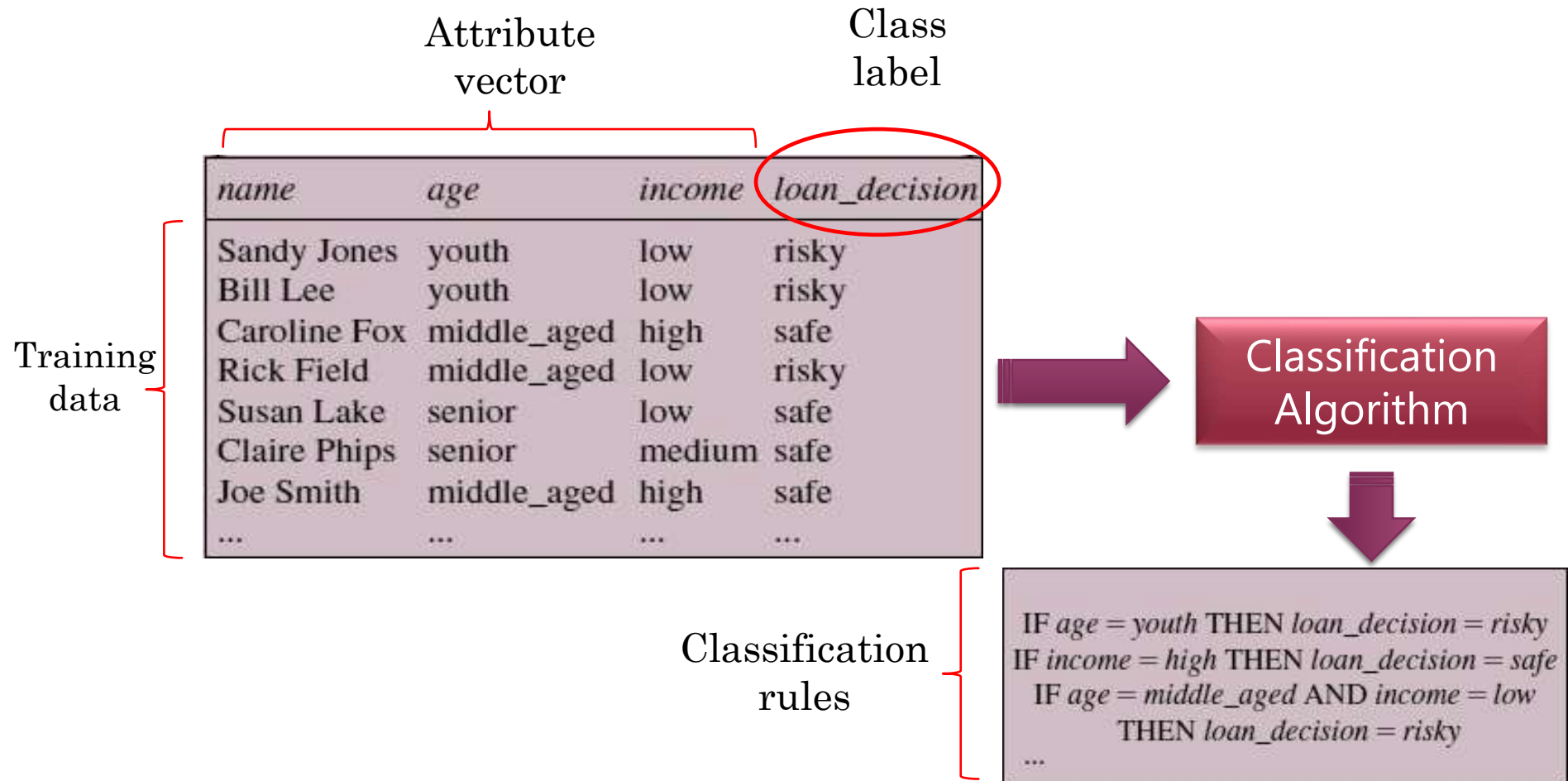
GENERAL APPROACH

- A two-step process:
- Learning (training) step → construct classification model
 - Build classifier for a predetermined set of classes
 - Learn from a training dataset (data tuples + their associated classes) → Supervised Learning
- Classification step → model is used to predict class labels for given data (test set)



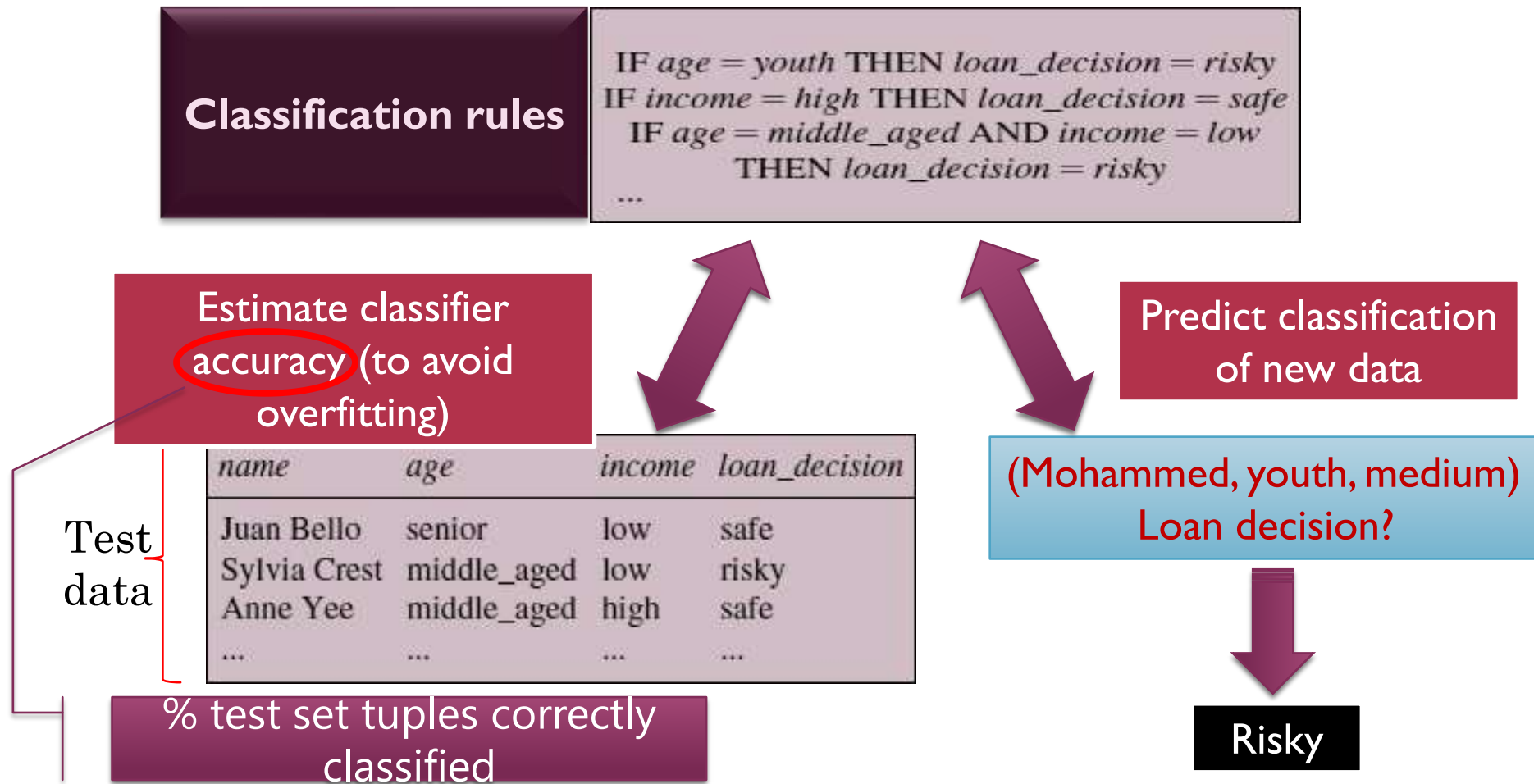
THE BASICS

GENERAL APPROACH



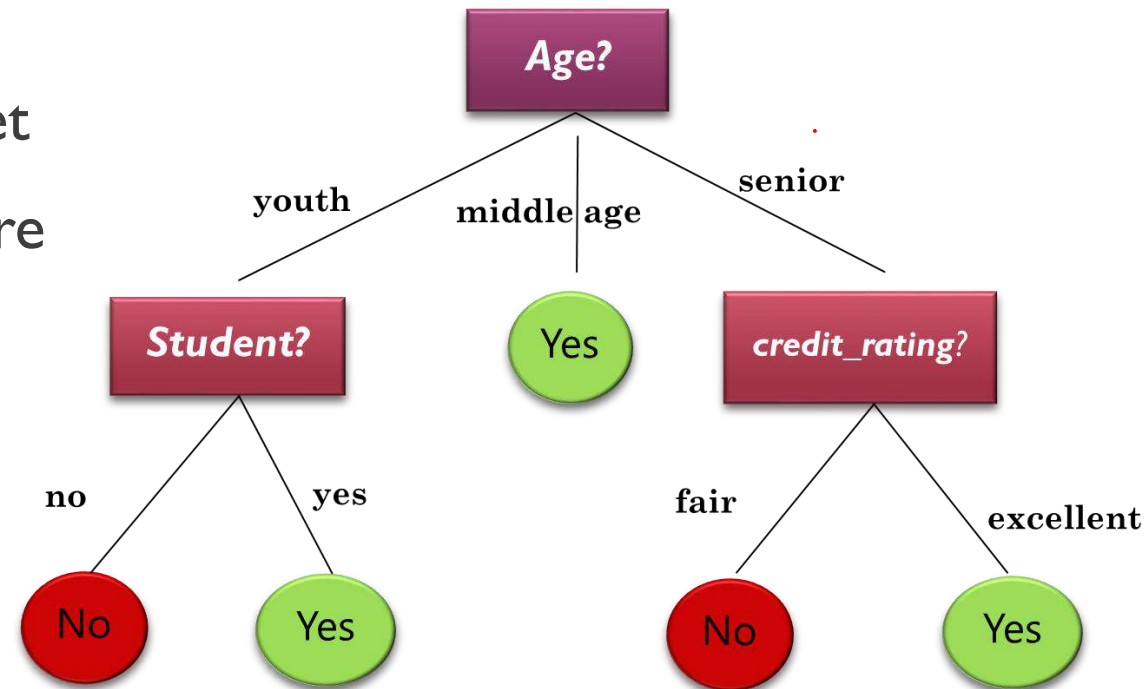
THE BASICS

GENERAL APPROACH



DECISION TREE INDUCTION

- Learning of decision trees from training dataset
- Decision tree → A flowchart-like tree structure
 - Internal node → a test on an attribute
 - Branch → a test outcome
 - Leaf node → a class label
- Constructed tree can be binary or otherwise



DECISION TREE INDUCTION

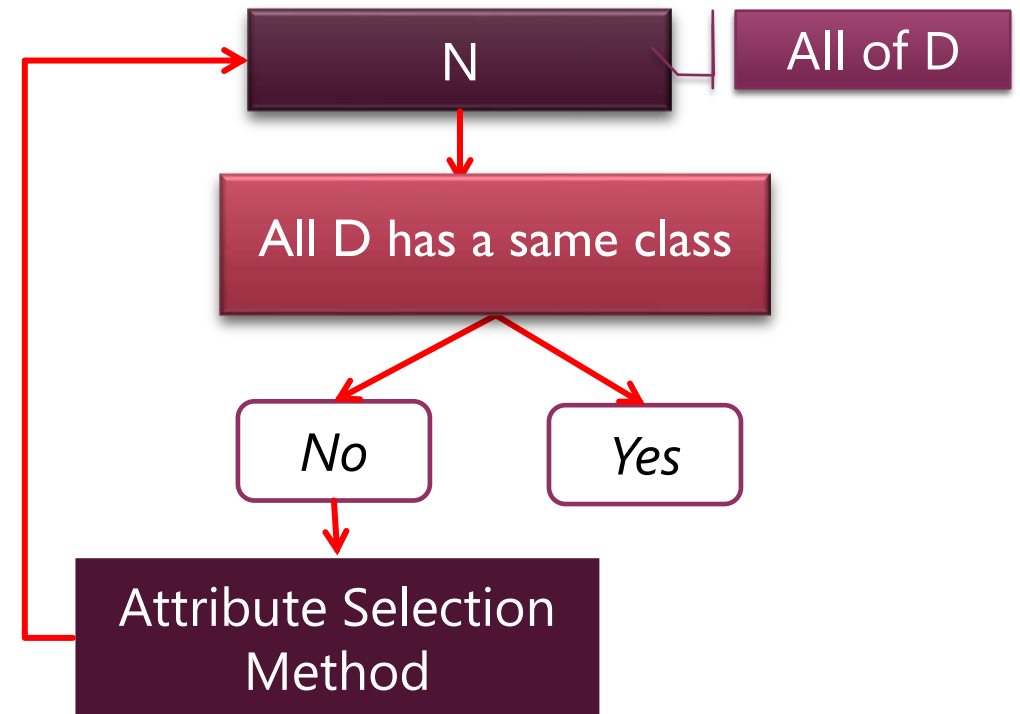
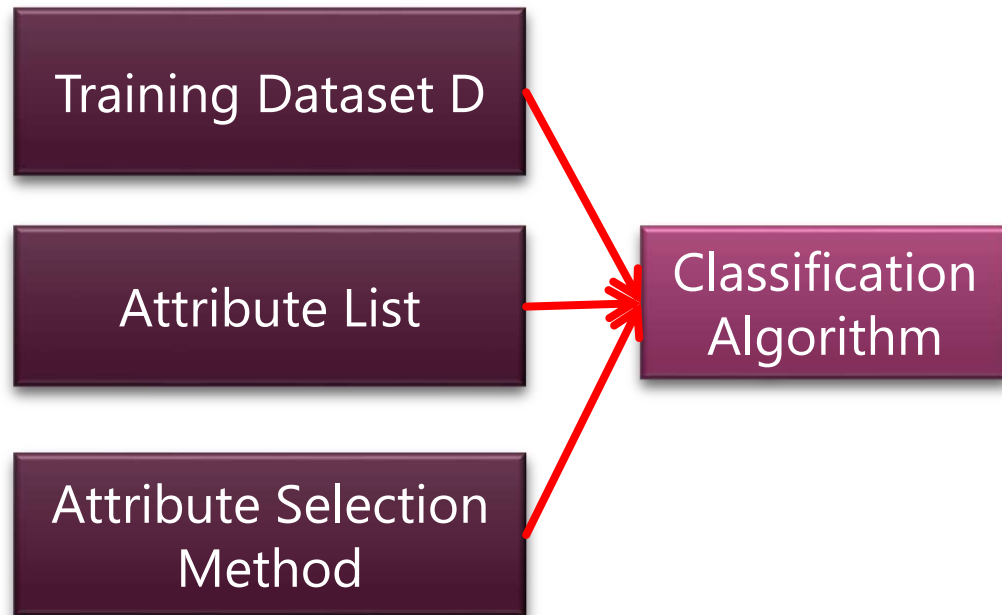
Benefits

- No domain knowledge required
- No parameter setting
- Can handle multidimensional data
- Easy-to-understand representation
- Simple and fast



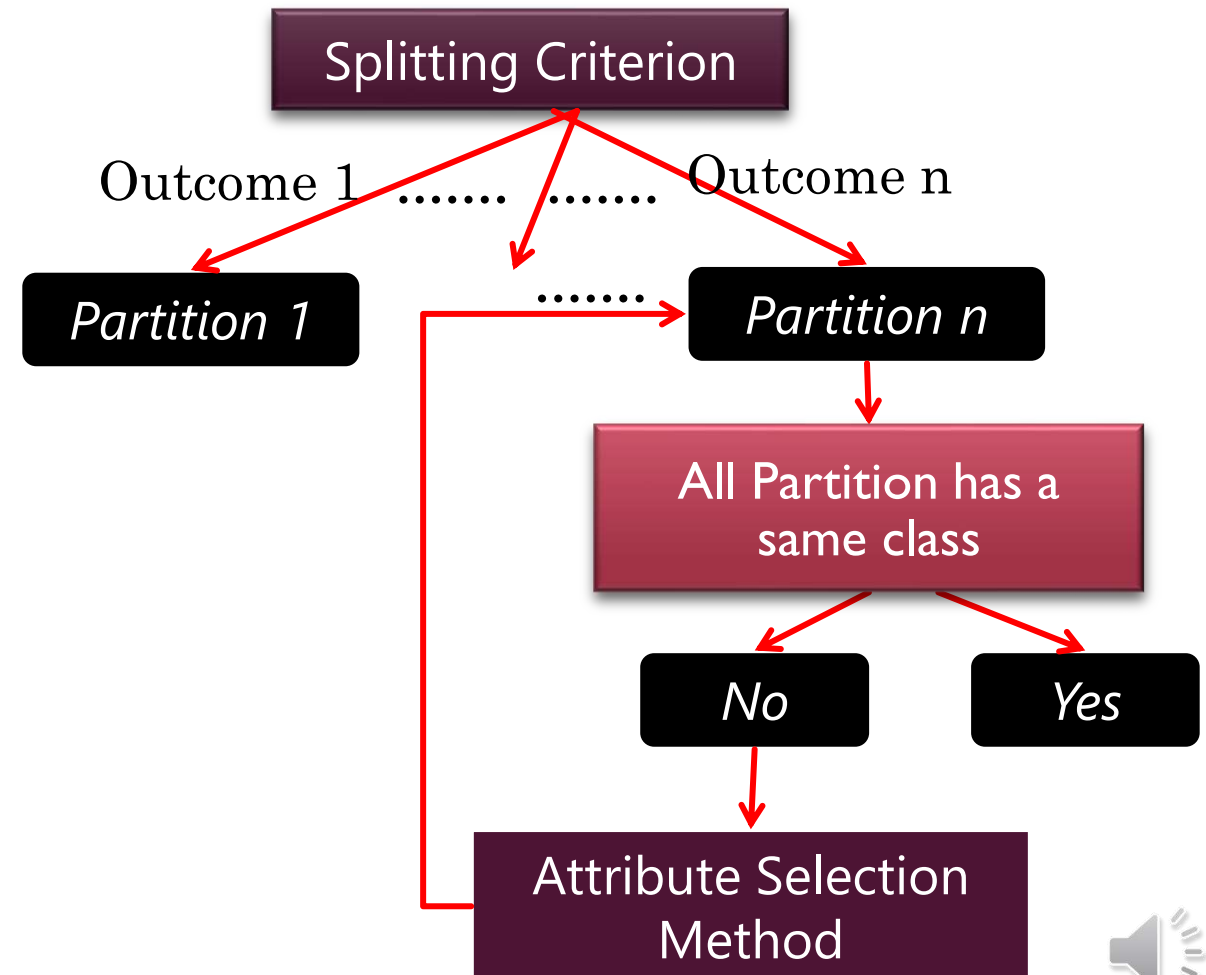
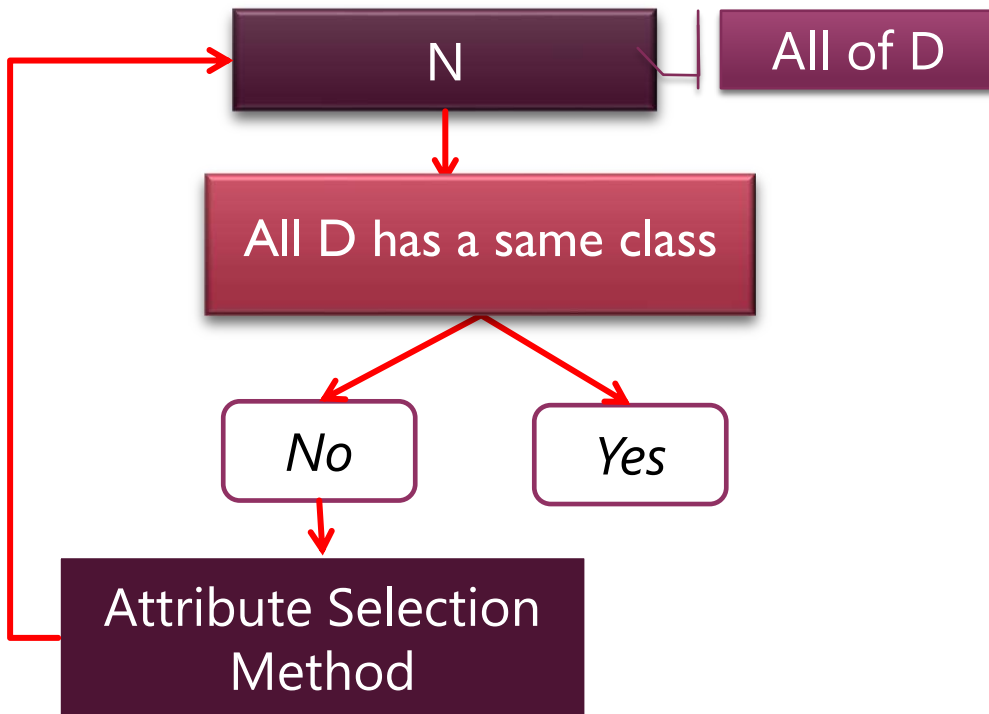
DECISION TREE INDUCTION

THE ALGORITHM



DECISION TREE INDUCTION

THE ALGORITHM



DECISION TREE INDUCTION

THE ALGORITHM

Splitting Attribute

Splitting Criterion

Partitioning scenarios

Examples

Discrete

Continuous

Discrete

Binary Tree

Outcome 1 Outcome n

Partition 1

Partition n

All Partition has a same class

No

Yes

Attribute Selection Method



DECISION TREE INDUCTION

THE ALGORITHM

- **Splitting Criterion is a test:**
 - Which attribute to test at node $N \rightarrow$ What is the “best” way to partition D into mutually exclusive classes
 - which (and how many) branches to grow from node N to represent the test outcomes
- Resulting partitions at each branch should be as “pure” as possible
 - A partition is “pure” if all its tuples belong to the same class
- When attribute is chosen to split training data set, it's removed from attribute list



DECISION TREE INDUCTION

THE ALGORITHM

- **Terminating conditions**

- All the tuples in D (represented at node N) belong to the same class
- There are no remaining attributes on which the tuples may be further partitioned
 - majority voting is employed $\rightarrow$ convert node into a leaf and label it with the most common class in data partition
- There are no tuples for a given branch
 - a leaf is created with the majority class in data partition



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- **Attribute selection measure** → a heuristic for selecting the splitting criterion that “best” splits a given data partition into smaller mutually exclusive classes
- Attributes are ranked according to a measure
 - attribute having the best score is chosen as the splitting attribute
 - split-point for continuous attributes
 - splitting subset for discrete attributes with binary trees
- Measures: Information Gain, Gain Ratio, Gini Index



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

Information Gain

- Based on *Shannon's information theory*
- Goal is to **minimize the expected number of tests needed to classify a tuple**
 - guarantee that a simple tree is found
- Attribute with the highest information gain is chosen as the splitting attribute
 - minimizes information needed to classify tuples in resulting partitions
 - reflects least “impurity” in resulting partitions



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- Given m class labels ($C_i, i = 1$ to m)
- Expected Information needed to classify a tuple in D
- $\text{Info}(D) = \text{entropy} = - \sum_{i=1}^m p_i \log_2(p_i)$
- $p_i \rightarrow$ probability that an arbitrary tuple in D belong to class C_i

$$p_i = \frac{|C_{i,D}|}{|D|}$$

- $C_{i,D} \rightarrow$ set of tuples having class label C_i in partition D



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- How much more Information would be needed after Partitioning to arrive at a “pure” classification”
 - Expected information required to classify a tuple from D based on the partitioning by attribute A:
 - $info_A(D) = \sum_{j=1}^v \frac{|D_j|}{D} \times info(D_j)$
 - The smaller the expected information still required, the greater the purity of the partitions



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- **Information gain** is the different between the original information required (based on proportion of classes) and the new requirement (after partitioning on A)
- $Gain(A) = info(D) - info_A(D)$
- $Gain(A)$ tells you how much would be gained by branching on A
 - Expected reduction in the information requirement caused by knowing the values of A
 - Attributes A with the highest $Gain(A)$ is chosen as the splitting attribute at node N



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
sales	Middle aged	low	junior	40
systems	youth	medium	junior	20
systems	Middle aged	high	senior	20
systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

$$CI(\text{Senior}) = 80, C2(\text{Junior}) = 120$$

$$\begin{aligned} \text{info}(D) = \text{entropy} &= -\sum_{i=1}^m p_i \log_2 p_i \\ &= -\frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97 \end{aligned}$$

$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
sales	Middle aged	low	junior	40
systems	youth	medium	junior	20
systems	Middle aged	high	senior	20
systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

$$CI(\text{Senior}) = 80, C2(\text{Junior}) = 120$$

$$\text{info}(D) = \text{entropy} = -\sum_{i=1}^m p_i \log_2 p_i$$

$$= -\frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97$$

$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right)$$

$$+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right)$$

$$= 0.92$$



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
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systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
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$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
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systems	youth	medium	junior	20
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marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
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$$CI(\text{Senior}) = 80, C2(\text{Junior}) = 120$$

$$\begin{aligned} \text{info}(D) = \text{entropy} &= -\sum_{i=1}^m p_i \log_2 p_i \\ &= -\frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97 \end{aligned}$$

$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$

$$\text{Gain}_{\text{department}} = 0.97 - 0.92 = 0.05$$



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
sales	Middle aged	low	junior	40
systems	youth	medium	junior	20
systems	Middle aged	high	senior	20
systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

$$CI(\text{Senior}) = 80, C2(\text{Junior}) = 120$$

$$\begin{aligned} \text{info}(D) = \text{entropy} &= -\sum_{i=1}^m p_i \log_2 p_i \\ &= -\frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97 \end{aligned}$$

$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$

$$\text{Gain}_{\text{department}} = 0.97 - 0.92 = 0.05 \text{ bits}$$

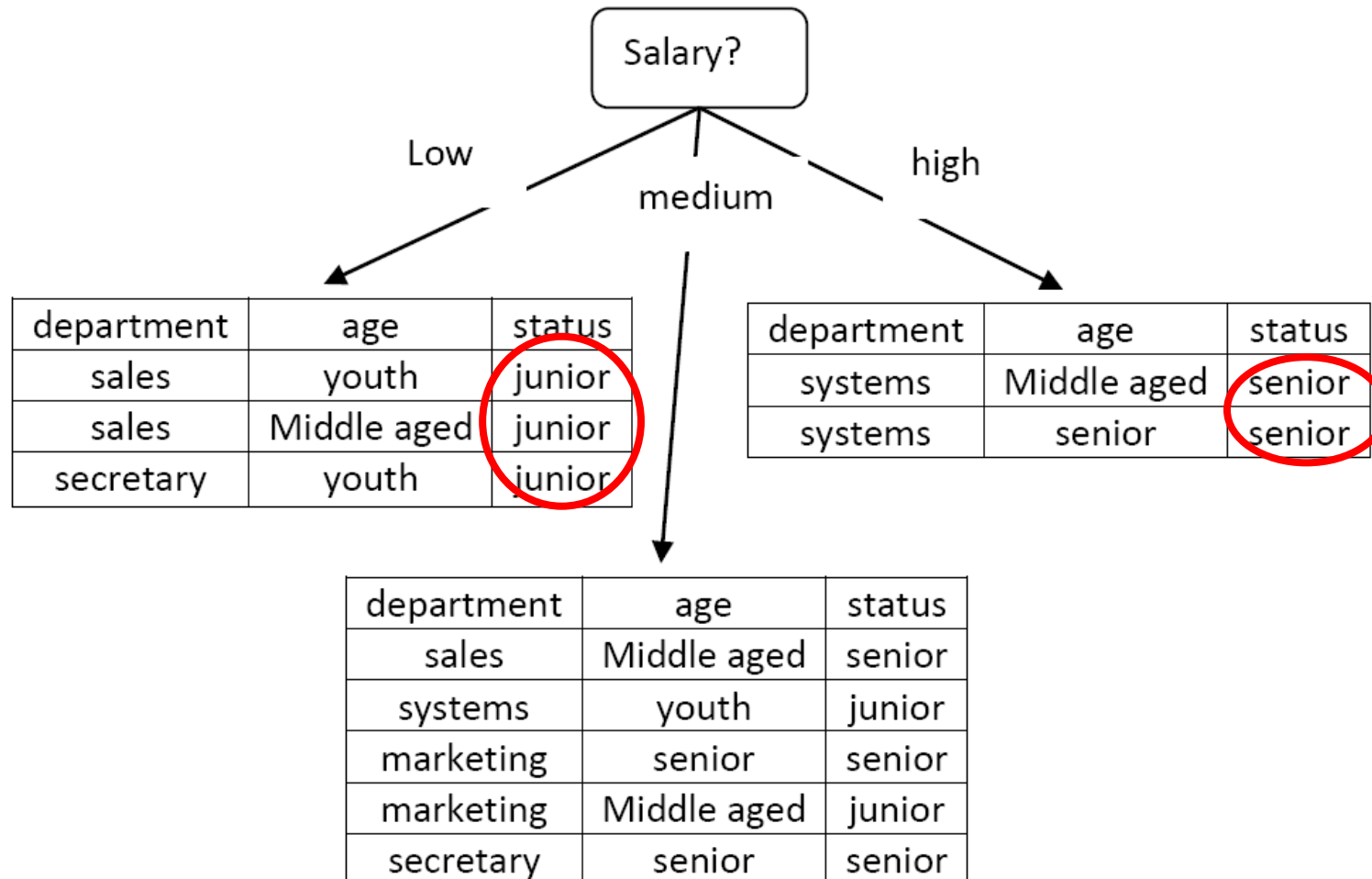
$$\text{Gain}_{\text{Age}} = 0.97 - 0.55 = 0.42 \text{ bits}$$

$$\text{Gain}_{\text{Salary}} = 0.97 - 0.45 = 0.52 \text{ bits}$$



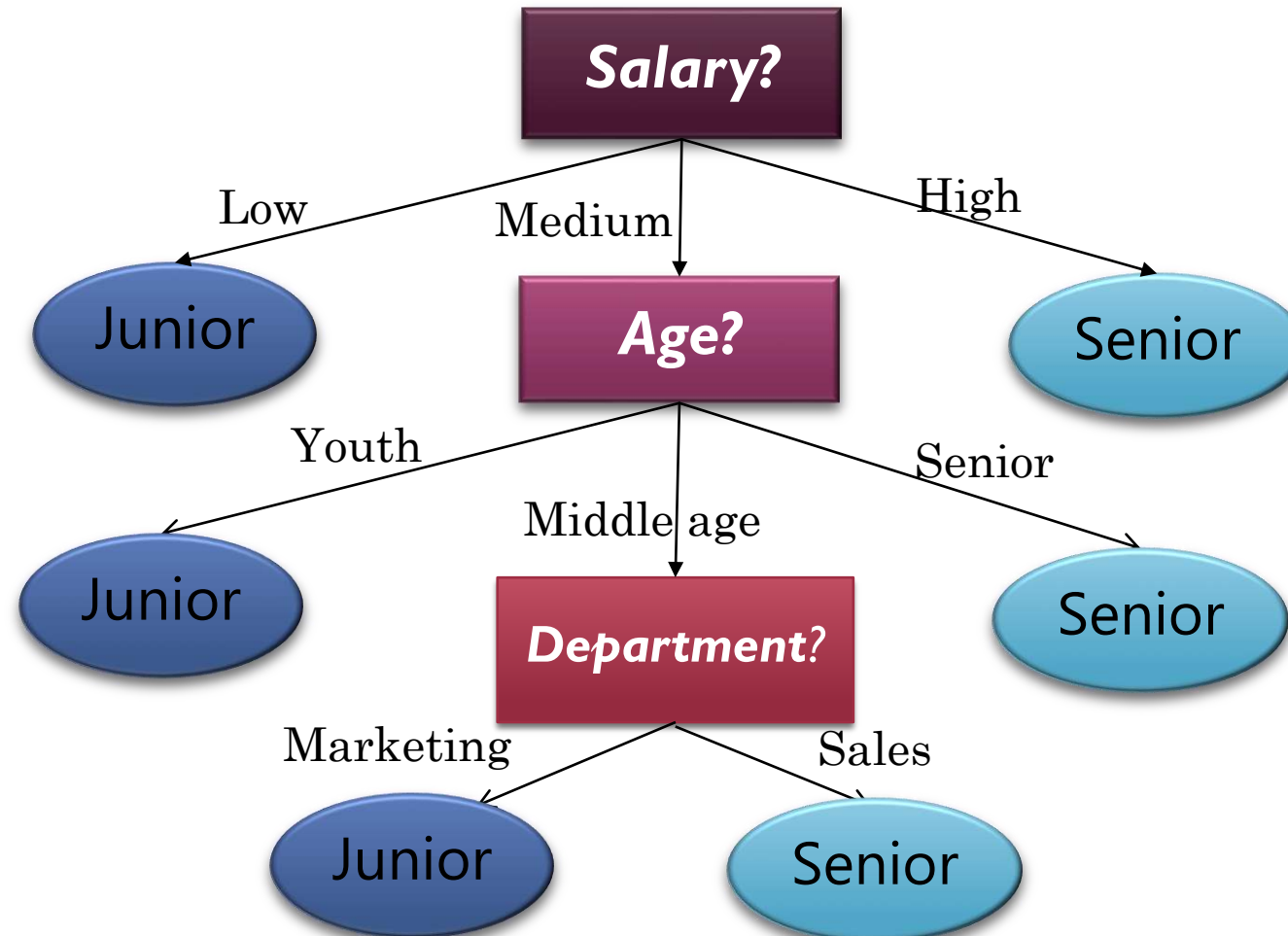
DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

*Information gain for **continuous attributes***

1. Sort values in increasing order
2. Each *midpoint* between two adjacent values can serve as *split-point* ■
3. Split-point between two values v_i and $v_{i+1} = \frac{v_i + v_{i+1}}{2}$
4. For each split-point, evaluate $info_A(D)$ with the number of partitions = 2 ($A \leq split-point$ & $A > split-point$)



DECISION TREE INDUCTION

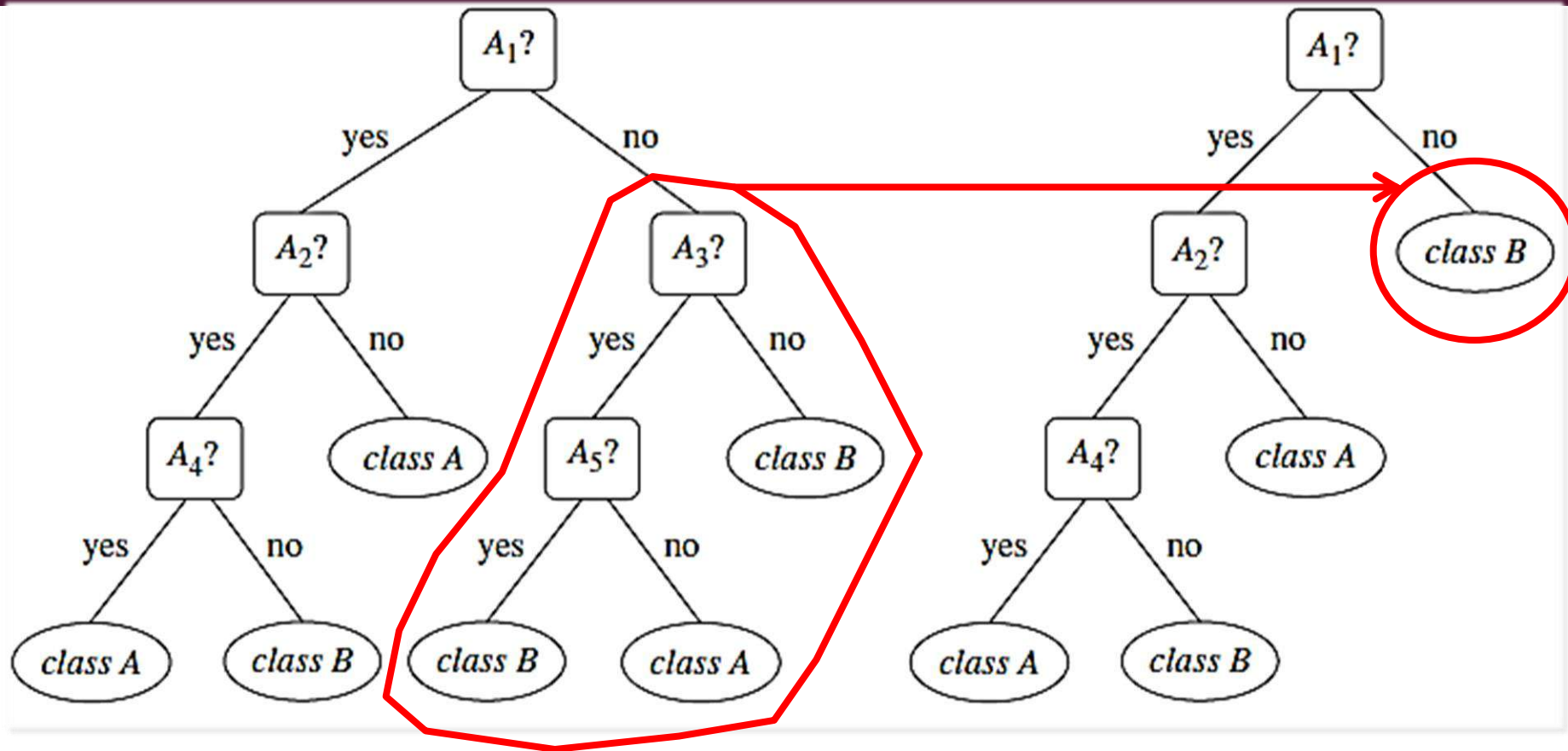
TREE PRUNING

- Data may be *overfitted* to dataset anomalies and outliers
- **Pruning** removes the least reliable branches
 - DT becomes less complex
- Prepruning → statistically assess the *goodness of a split* before it takes place
 - hard to choose *thresholds* for statistical significance
- Postpruning → remove sub-trees from already constructed trees
 1. remove sub-tree branches and replace with leaf node
 2. leaf is labeled with most frequent class in sub-tree



DECISION TREE INDUCTION

TREE PRUNING



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DECISION TREE – WHAT ARE RULES?

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
sales	Middle aged	low	junior	40
systems	youth	medium	junior	20
systems	Middle aged	high	senior	20
systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

RI: IF salary =medium AND age = youth
THEN Status = Junior

$$\text{coverage}(R) = \frac{n_{\text{covers}}}{|D|}$$

$$\text{accuracy}(R) = \frac{n_{\text{correct}}}{n_{\text{covers}}}$$

- coverage(RI) = 20/200=10% and
- accuracy(RI)= 20/20 = 100%.

X: (Department = system age = youth, salary = low)



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DT – RESOLVING RULES CONFLICTS

Rules conflicts are the result of a tuple firing more than one rule with different class predictions

Two resolution strategies

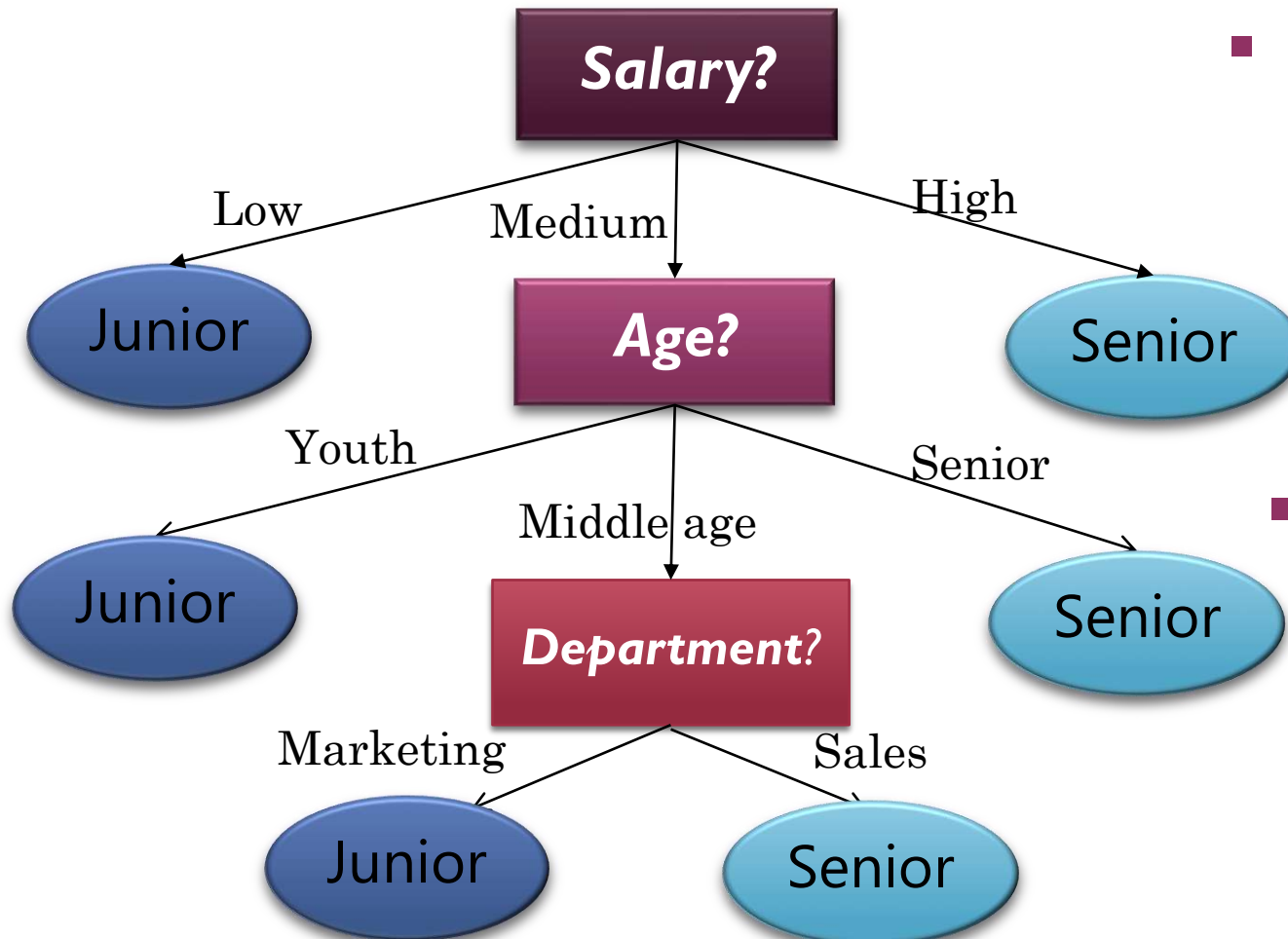
- **Size Ordering** → rule with **largest antecedent** (toughest) has highest priority fires and returns class prediction
- **Rule Ordering** → rules **prioritized apriori** according to
 - **Class-based ordering** → decreasing importance (most frequent are highest – order of prevalence)
 - **Rule-based ordering** → measures of rule quality (e.g. accuracy, size, domain expertise)

Fallback (default) rule when no rules are triggered



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DECISION TREE



- Create one rule for each path from root to leaf in the decision tree
 1. Each splitting criterion is ANDed to form rule antecedent (IF)
 2. Leaf node holds class prediction (THEN)
- RI: IF salary =medium AND age = youth THEN Status = Junior

Can the rules resulting from decision trees have conflicts?



QUESTION?

NEXT



IS422P - DATA MINING CLASSIFICATION (PART II)



AMIRA REZK
INFORMATION SYSTEM DEPARTMENT



AGENDA



The Basics

What is Classification?
General Approach



Decision Tree Induction

The Algorithm
Attribute Selection Measures
Tree Pruning
Extracting Rules from Decision Trees



Bayes Classification

Bayes' Theorem
Naïve Bayesian Classification



Lazy Learners

K-Nearest Neighbor Classifiers



Regression analysis

Linear regression



Model Evaluation

Metrics for Evaluating Classifiers Performance
Cross-Validation
Bootstrap



Improving Classification Accuracy

Bagging
Boost and AdaBoost

BAYES CLASSIFICATION METHODS

- **Naïve Bayesian classifier** → Statistical classifier that predicts the probability that a tuple belongs to a specific class
 - Based on Bayes Theorem → Bayes was an 18th century clergyman who worked on probability
 - *High accuracy*
 - *Speed*
 - *Class-conditional Independence* → Attributes' effect on class determination is independent



BAYES CLASSIFICATION METHODS

NAÏVE BAYESIAN CLASSIFICATION

- Given tuples with n attribute and m classes, Naïve Bayes predicts that X belong to class with highest posteriori probability
- $P(C_i | X) > P(C_j | X)$ for $1 \leq j \leq m, j \neq i$
- C_i is called the maximum posteriori hypothesis
- $$P(C_i | X) = \frac{P(X | C_i)P(C_i)}{P(X)}$$
- Since $P(X)$ is constant, maximize only numerator
- If $P(C_i)$ is unknown for all i , assume uniform probability
 - Then you only have to maximize $P(X | C_i)$
 - Otherwise,
$$P(C_i) = \frac{|C_i_D|}{|D|}$$



BAYES CLASSIFICATION METHODS

NAÏVE BAYESIAN CLASSIFICATION

- To reduce computation of $P(X|C_i)$, attributes are assumed to be independent → hence the “naïve” in the name
- $P(X|C_i) = \prod_{k=1}^n P(x_k|C_i) = P(x_1|C_i) \times P(x_2|C_i) \times \dots \times P(x_n|C_i)$
- If attribute is **categorical** → $P(x_k|C_i) = \frac{|C_{i,D,xk}|}{|C_i|}$
- If attribute is **numerical** → assume gaussian distribution →

$$P(x_k|C_i) = \frac{1}{\sqrt{2\pi\sigma_{ci}^2}} e^{-\frac{(x_k - \mu_{ci})^2}{2\sigma_{ci}^2}}$$

- Evaluate for each C_i , assign class label of class with max $P(X|C_i)$



BAYES CLASSIFICATION METHODS

NAÏVE BAYESIAN CLASSIFICATION - EXAMPLE

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marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

The individual has status Junior

$C1$ (Senior) = 80 , $C2$ (Junior) = 120
 $P(C1)=80/200$ $P(C2)=120/200$

X : {department = "marketing"; age= "youth", salary= low}

$$P(X|C1) = \frac{10}{80} \times \frac{0}{80} \times \frac{0}{80} = 0 \rightarrow P(X|C1)P(C1)=0$$

$$P(X|C2) = \frac{20}{120} \times \frac{60}{120} \times \frac{80}{120} = 0.055 \rightarrow$$

$$P(X|C2)P(C2)=0.033$$

Laplacian correction or
Laplace estimator

$$P(X|C1) = \frac{10}{80} \times \frac{1}{80} \times \frac{1}{80} = 0.00002 \rightarrow$$

$$P(X|C1)P(C1)=0.000008$$

$$P(X|C2) = \frac{20}{120} \times \frac{60}{120} \times \frac{80}{120} = 0.055 \rightarrow$$

$$P(X|C2)P(C2)=0.033$$

LAZY LEARNERS

K-NEAREST NEIGHBOR CLASSIFIERS

- Delay classification until new test data is available
 - Store training data meanwhile
- Use similarity measure to compute distance between test data tuple and each of the training data tuples (Euclidian, Manhattan, ...)
 - Remember to normalize if ranges vary between attributes
- k stands for the number of “closest” neighbors of a test data tuple according to *measured distance*
 - Majority voting of their class labels used to determine class of test tuple



LAZY LEARNERS

K-NEAREST NEIGHBOR - EXAMPLE

RID	age	Loan(\$)	Default
1	25	40000	No
2	35	60000	No
3	45	80000	No
4	20	20000	No
5	35	120000	No
6	52	18000	No
7	23	95000	Yes
8	40	62000	Yes
9	60	100000	Yes
10	48	220000	Yes
11	33	150000	Yes
	48	142000	?

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

K → No. of Neighbor **(Odd)**



LAZY LEARNERS

K-NEAREST NEIGHBOR - EXAMPLE

RID	age	Loan (\$)	Default	Distance
1	25	40000	No	102000
2	35	60000	No	82000
3	45	80000	No	62000
4	20	20000	No	122000
5	35	120000	No	22000
6	52	18000	No	124000
7	23	95000	Yes	47000
8	40	62000	Yes	80000
9	60	100000	Yes	42000
10	48	220000	Yes	78000
11	33	150000	Yes	8000
48	142000	?	fair	

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

K=1 → NN is RID 11

K=3 → NN is RID 5, 9, 11
Default = Yes



LAZY LEARNERS

K-NEAREST NEIGHBOR - EXAMPLE

RID	age	Loan (\$)	Default	Distance	Loan (\$)	Distance
1	25	40000	No	102000	24.4	30.6
2	35	60000	No	82000	28.3	20.8
3	45	80000	No	62000	32.3	12.7
4	20	20000	No	122000	20.4	37.0
5	35	120000	No	22000	40.2	13.7
6	52	18000	No	124000	20.0	24.9
7	23	95000	Yes	47000	35.2	26.7
8	40	62000	Yes	80000	28.7	17.8
9	60	100000	Yes	42000	36.2	14.6
10	48	220000	Yes	78000	60.0	15.4
11	33	150000	Yes	8000	46.1	15.1
48	142000	?	fair	44.6		

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

K=1 → NN is RID 3

K=3 → NN is RID 3, 5, 9
Default = NO



REGRESSION ANALYSIS

LINEAR REGRESSION

- Linear regression is a way to model a relationship between two sets of variables. The result is a **linear regression equation** that can be used to make **predictions** about data.
- First; make a scatter plot to see if data roughly fits a line. **Why?**
- The equation has the form $Y = a + bX$, where Y is the **dependent variable** (that's the variable that goes on the Y axis), X is the **independent variable** (i.e. it is plotted on the X axis), b is the **slope** of the line and a is the **y-intercept**.



REGRESSION ANALYSIS

LINEAR REGRESSION- EXAMPLE

SUBJECT	AGE X	GLUCOSE LEVEL Y	XY	X2	Y2
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

$$a = 65.1416$$

$$b = 0.385225$$

Insert the values into the equation.

$$y' = a + bx$$

$$y' = 65.14 + 0.385225x$$



QUESTION?

NEXT



IS422P - DATA MINING CLASSIFICATION (PART I)



AMIRA REZK
INFORMATION SYSTEM DEPARTMENT



AGENDA



The Basics

What is Classification?
General Approach



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THE BASICS

WHAT IS CLASSIFICATION

- Motivation: Prediction
 - Is a bank loan applicant “safe” or “risky”?
 - Which treatment is better for patient, “treatmentX” or “treatmentY”?
- **Classification** is a data analysis task where a model is constructed to **predict class labels (categories)**



THE BASICS

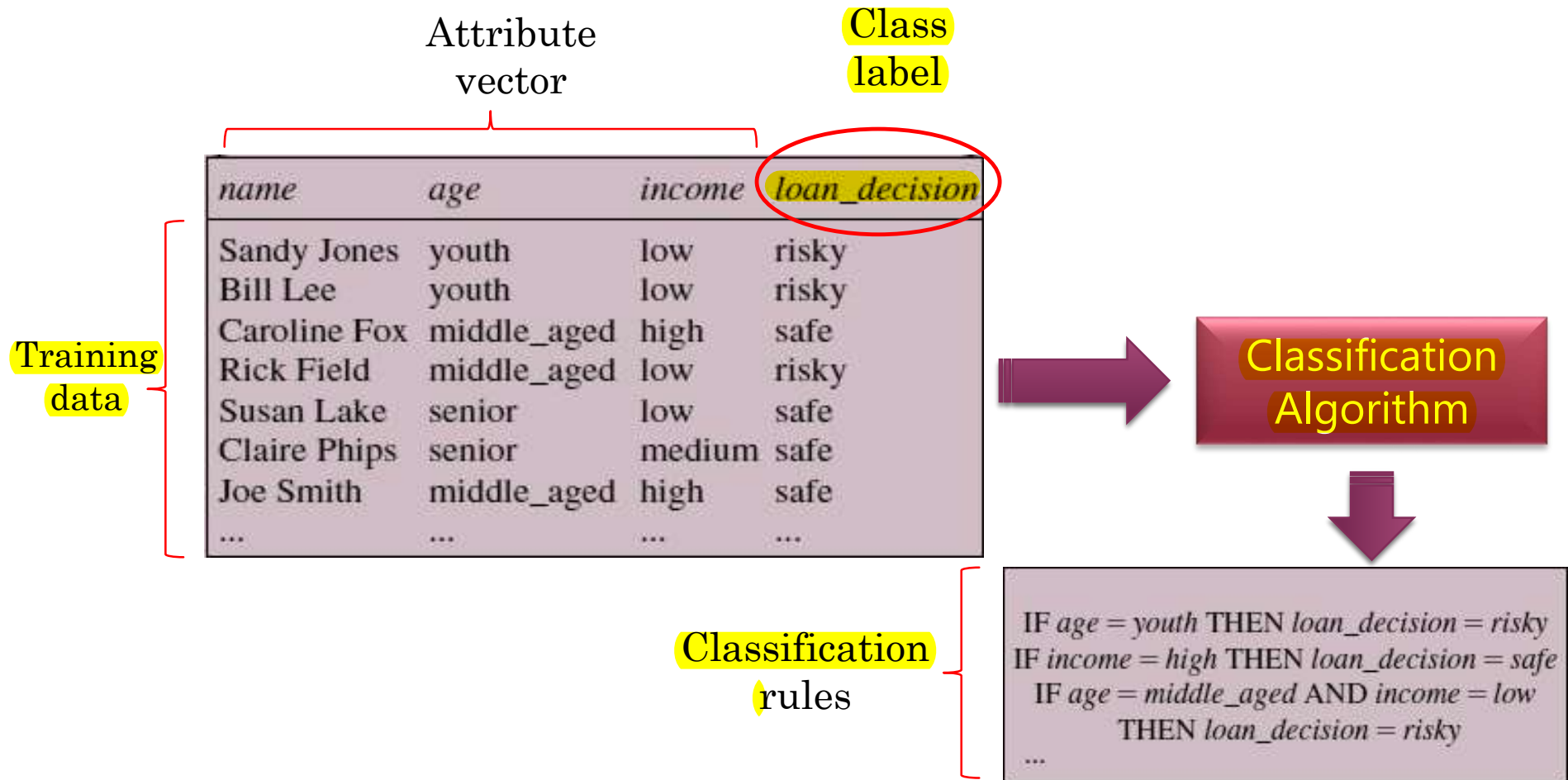
GENERAL APPROACH

- A two-step process:
- Learning (training) step → construct classification model
 - Build classifier for a predetermined set of classes
 - Learn from a training dataset (data tuples + their associated classes) → Supervised Learning
- Classification step → model is used to predict class labels for given data (test set)



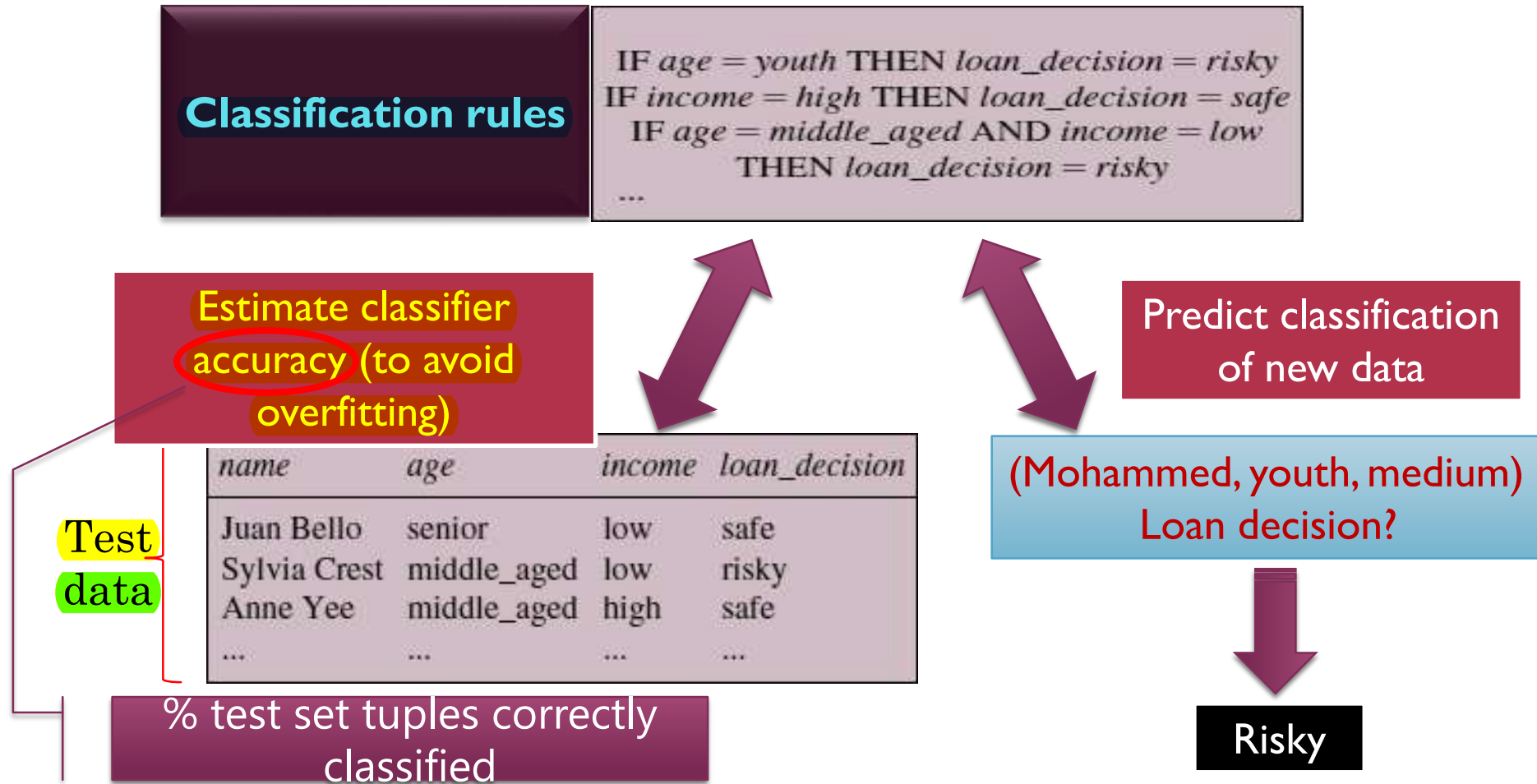
THE BASICS

GENERAL APPROACH



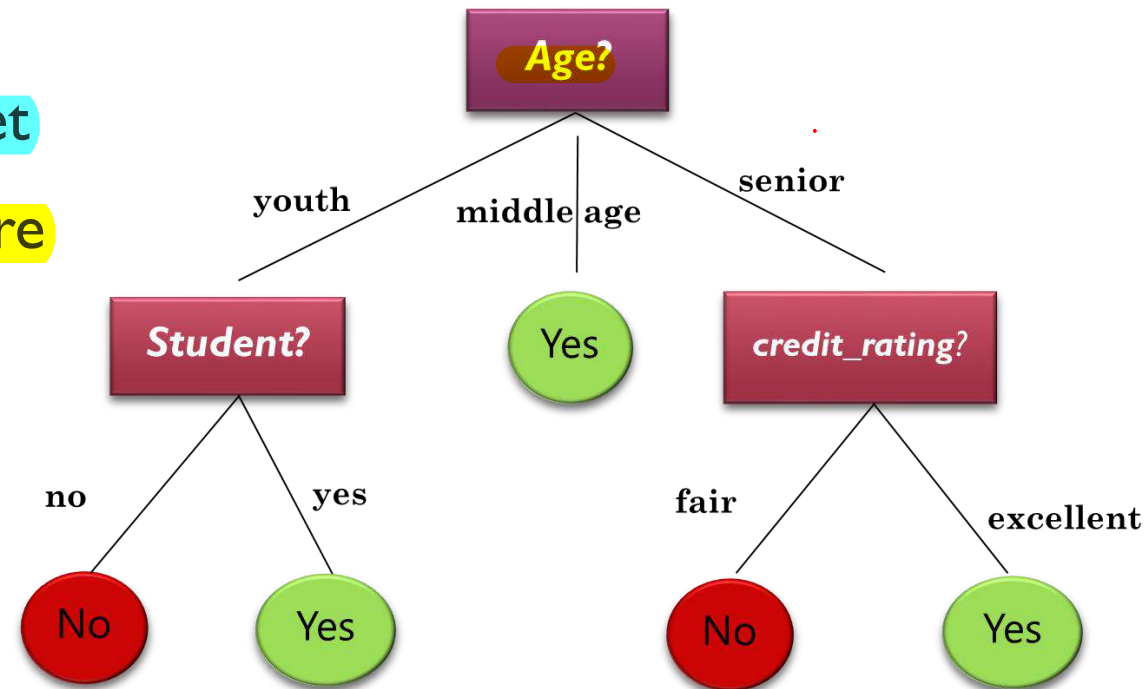
THE BASICS

GENERAL APPROACH



DECISION TREE INDUCTION

- Learning of decision trees from training dataset
- Decision tree → A flowchart-like tree structure
 - Internal node → a test on an attribute
 - Branch → a test outcome
 - Leaf node → a class label
- Constructed tree can be binary or otherwise



DECISION TREE INDUCTION

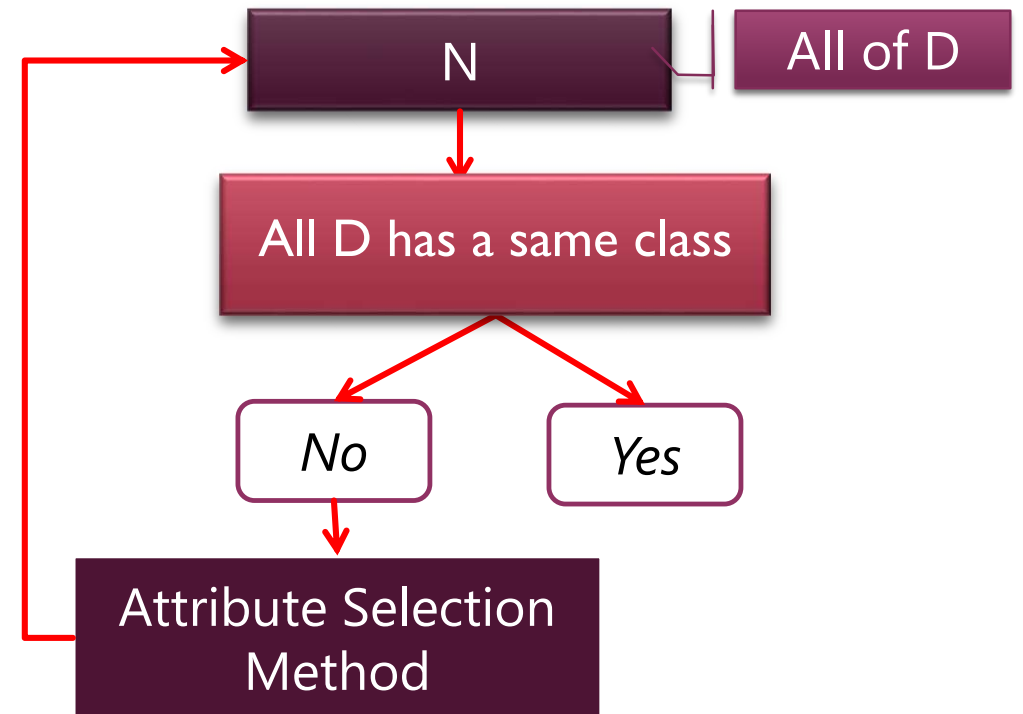
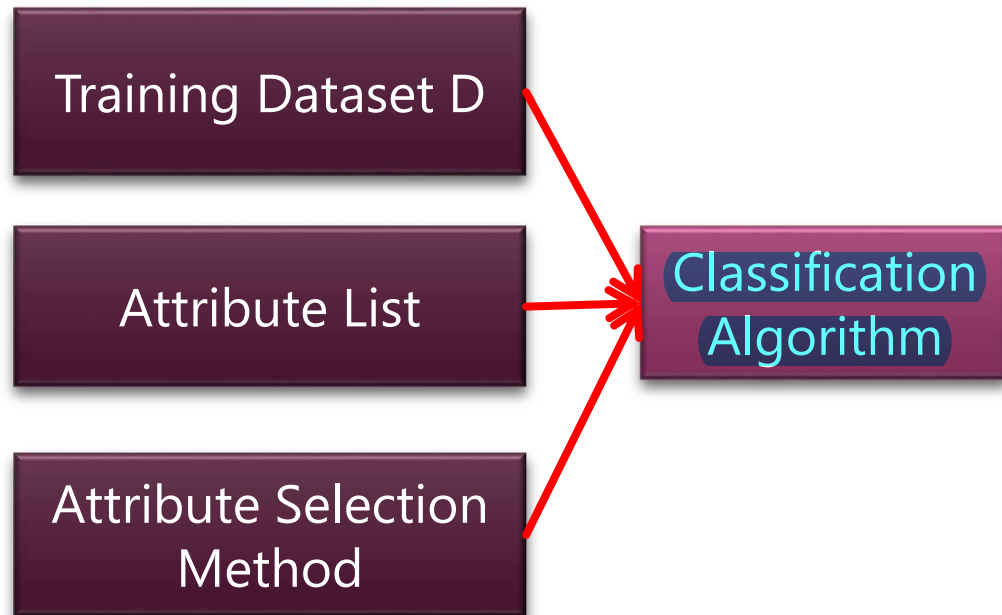
Benefits

- No domain knowledge required
- No parameter setting
- Can handle multidimensional data
- Easy-to-understand representation
- Simple and fast



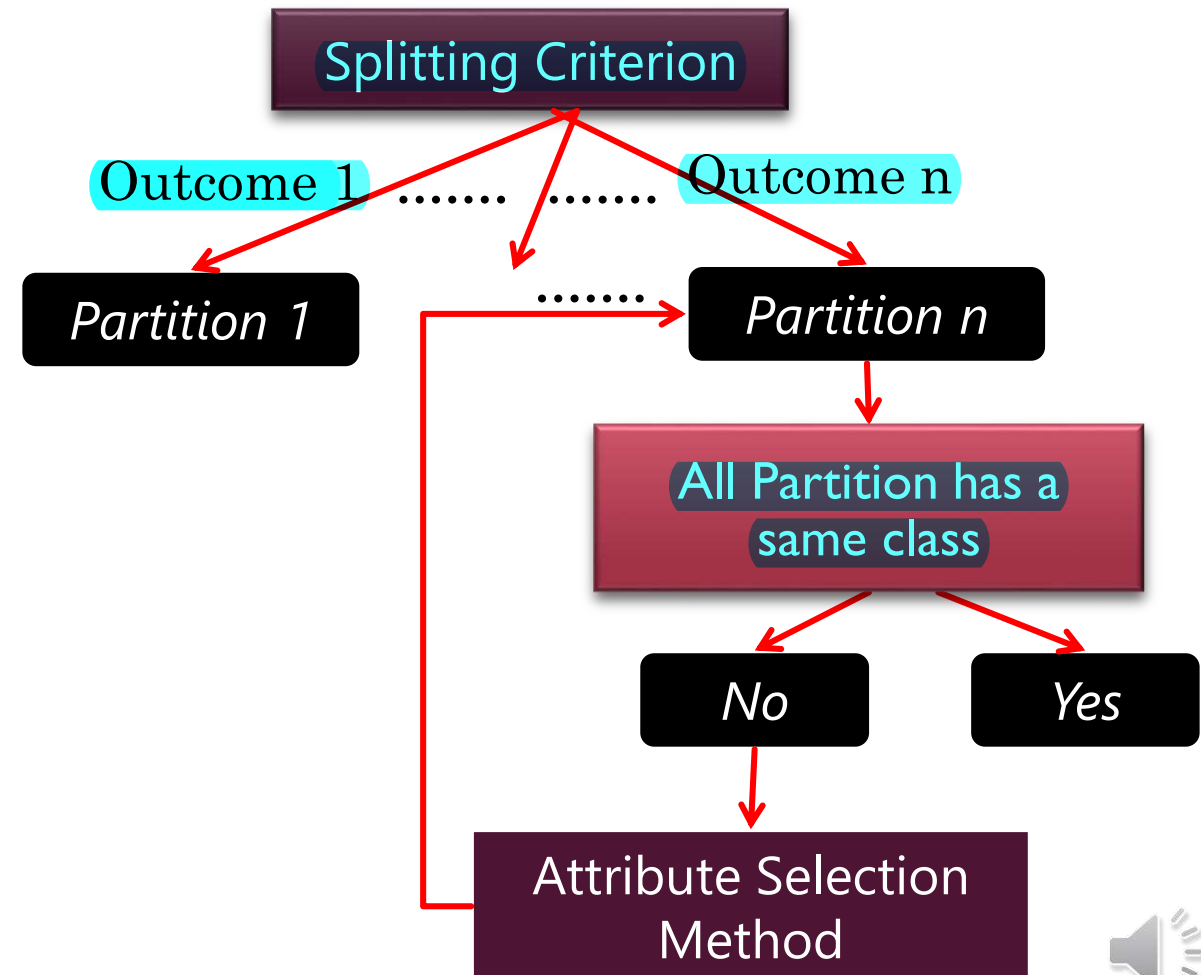
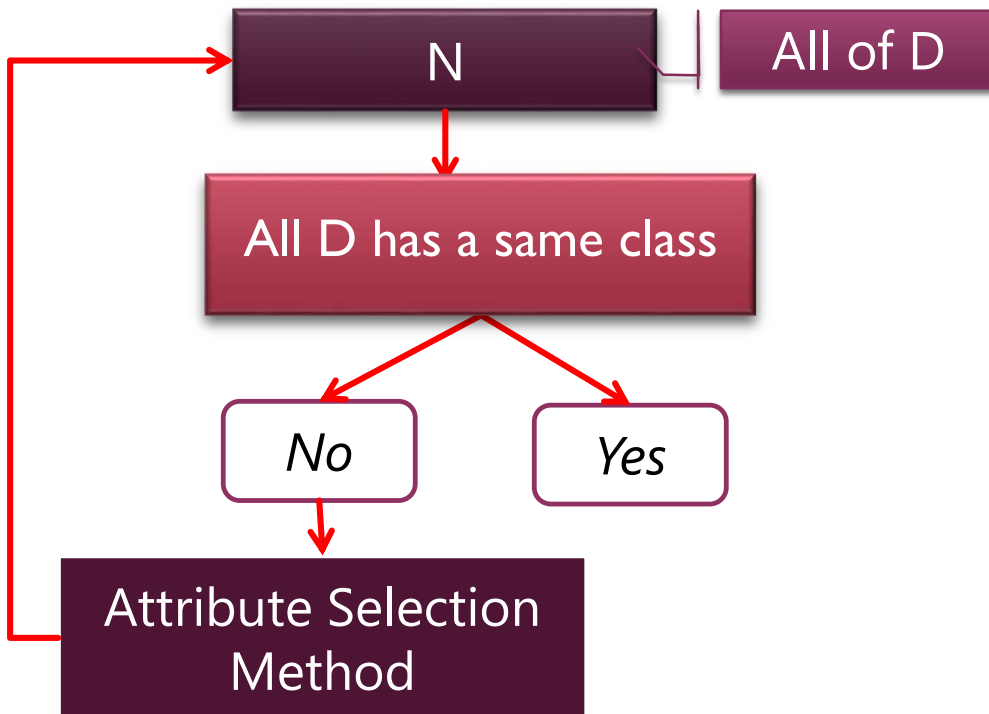
DECISION TREE INDUCTION

THE ALGORITHM



DECISION TREE INDUCTION

THE ALGORITHM



DECISION TREE INDUCTION

THE ALGORITHM

Splitting Attribute

Splitting Criterion

Partitioning scenarios

Examples

Discrete

Continuous

Discrete

Binary Tree

Outcome 1 Outcome n

Partition 1

Partition n

All Partition has a same class

No

Yes

Attribute Selection Method



DECISION TREE INDUCTION

THE ALGORITHM

- **Splitting Criterion is a test:**
 - Which attribute to test at node $N \rightarrow$ What is the “best” way to partition D into mutually exclusive classes
 - which (and how many) branches to grow from node N to represent the test outcomes
- Resulting partitions at each branch should be as “pure” as possible
 - A partition is “pure” if all its tuples belong to **the same class**
- When attribute is chosen to split training data set, it's removed from attribute list



DECISION TREE INDUCTION

THE ALGORITHM

- **Terminating conditions**

- All the tuples in D (represented at node N) belong to the same class
- There are no remaining attributes on which the tuples may be further partitioned
 - majority voting is employed $\rightarrow$ convert node into a leaf and label it with the most common class in data partition
- There are no tuples for a given branch
 - a leaf is created with the majority class in data partition



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- **Attribute selection measure** → a heuristic for selecting the splitting criterion that “best” splits a given data partition into smaller mutually exclusive classes
- Attributes are ranked according to a measure
 - attribute having the best score is chosen as the splitting attribute
 - split-point for continuous attributes
 - splitting subset for discrete attributes with binary trees
- Measures: Information Gain, Gain Ratio, Gini Index



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

Information Gain

- Based on *Shannon's information theory*
- Goal is to **minimize the expected number of tests needed to classify a tuple**
 - guarantee that a simple tree is found
- Attribute with the highest information gain is chosen as the splitting attribute
 - minimizes information needed to classify tuples in resulting partitions
 - reflects least “impurity” in resulting partitions



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- Given m class labels ($C_i, i = 1$ to m)
- Expected Information needed to classify a tuple in D
- $\text{Info}(D) = \text{entropy} = - \sum_{i=1}^m p_i \log_2(p_i)$
- $p_i \rightarrow$ probability that an arbitrary tuple in D belong to class C_i

$$p_i = \frac{|C_{i,D}|}{|D|}$$

- $C_{i,D} \rightarrow$ set of tuples having class label C_i in partition D



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- How much more Information would be needed after Partitioning to arrive at a “pure” classification”
 - Expected information required to classify a tuple from D based on the partitioning by attribute A:
 - $info_A(D) = \sum_{j=1}^v \frac{|D_j|}{D} \times info(D_j)$
 - The smaller the expected information still required, the greater the purity of the partitions



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- **Information gain** is the different between the original information required (based on proportion of classes) and the new requirement (after partitioning on A)
- $Gain(A) = info(D) - info_A(D)$
- $Gain(A)$ tells you how much would be gained by branching on A
 - Expected reduction in the information requirement caused by knowing the values of A
 - Attributes A with the highest $Gain(A)$ is chosen as the splitting attribute at node N



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
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secretary	youth	low	junior	10

$$CI(\text{Senior}) = 80, C2(\text{Junior}) = 120$$

$$\begin{aligned} \text{info}(D) = \text{entropy} &= -\sum_{i=1}^m p_i \log_2 p_i \\ &= -\frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97 \end{aligned}$$

$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$



DECISION TREE INDUCTION

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Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$

$$\text{Gain}_{\text{department}} = 0.97 - 0.92 = 0.05$$



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
sales	Middle aged	low	junior	40
systems	youth	medium	junior	20
systems	Middle aged	high	senior	20
systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

$$CI(\text{Senior}) = 80, C2(\text{Junior}) = 120$$

$$\begin{aligned} \text{info}(D) = \text{entropy} &= -\sum_{i=1}^m p_i \log_2 p_i \\ &= -\frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97 \end{aligned}$$

$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$

$$\text{Gain}_{\text{department}} = 0.97 - 0.92 = 0.05 \text{ bits}$$

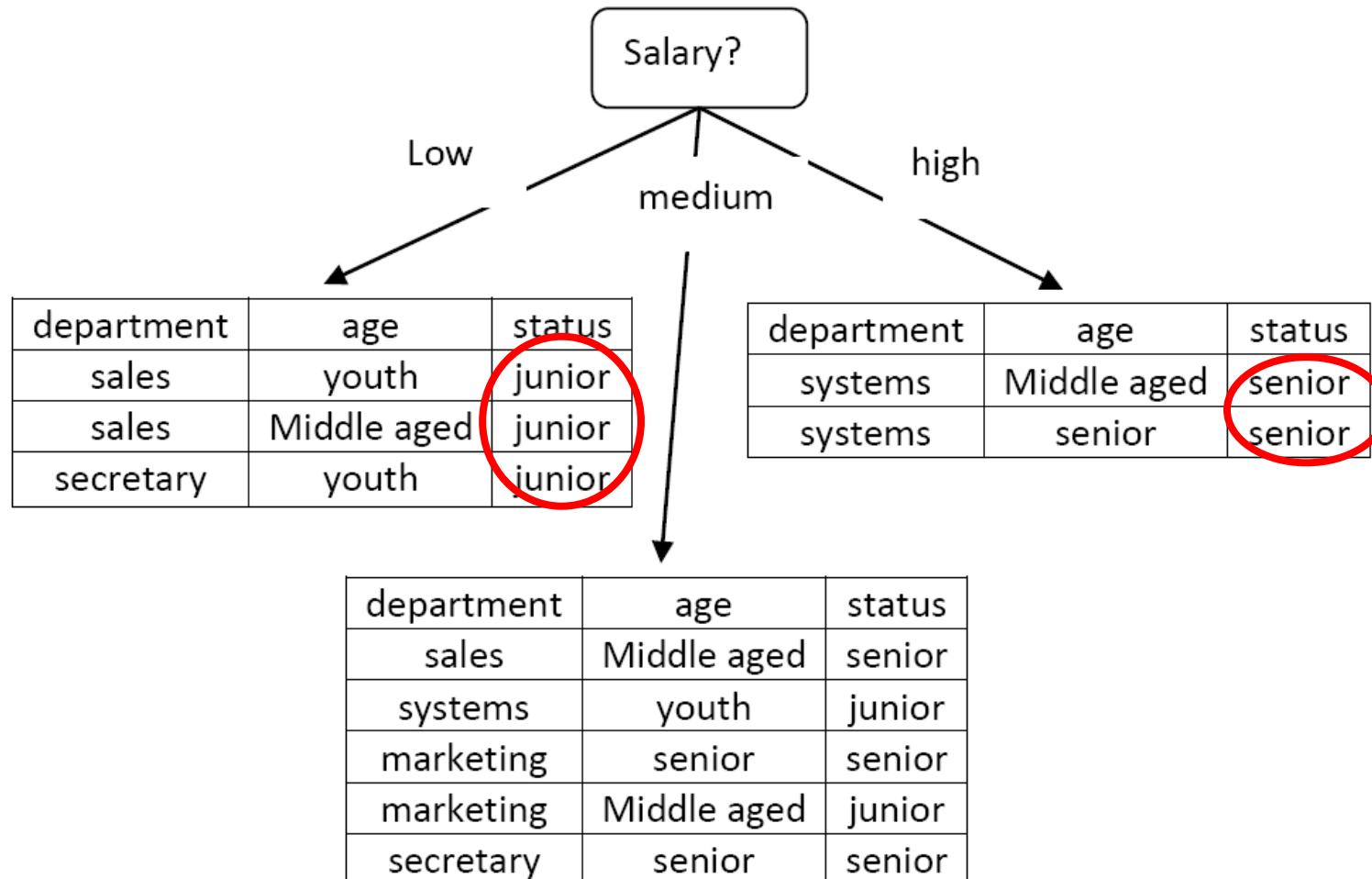
$$\text{Gain}_{\text{Age}} = 0.97 - 0.55 = 0.42 \text{ bits}$$

$$\text{Gain}_{\text{Salary}} = 0.97 - 0.45 = 0.52 \text{ bits}$$



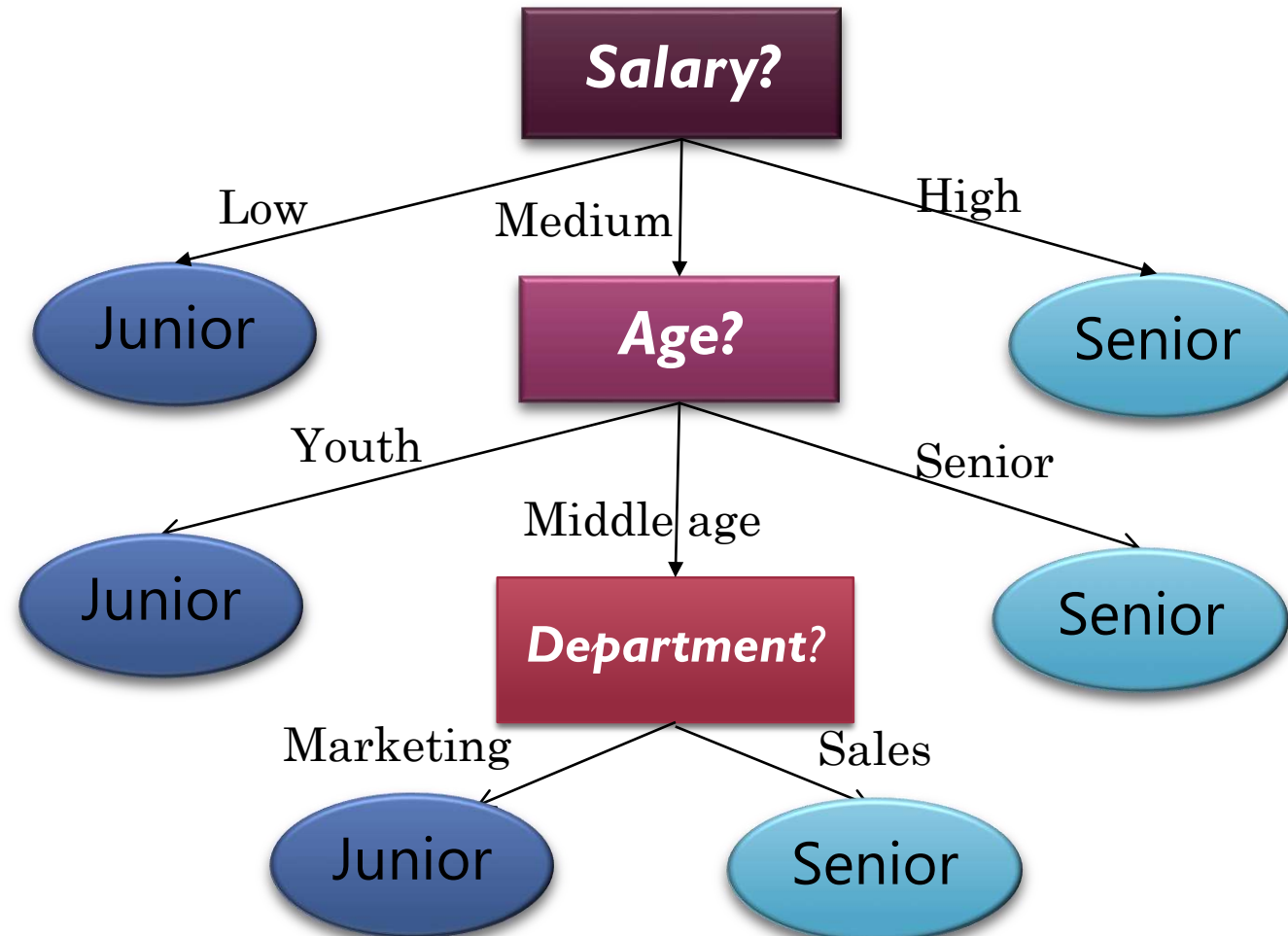
DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

*Information gain for **continuous attributes***

1. Sort values in increasing order
2. Each *midpoint* between two adjacent values can serve as *split-point*
3. Split-point between two values v_i and $v_{i+1} = \frac{v_i + v_{i+1}}{2}$
4. For each split-point, evaluate $info_A(D)$ with the number of partitions = 2 ($A \leq split-point$ & $A > split-point$)



DECISION TREE INDUCTION

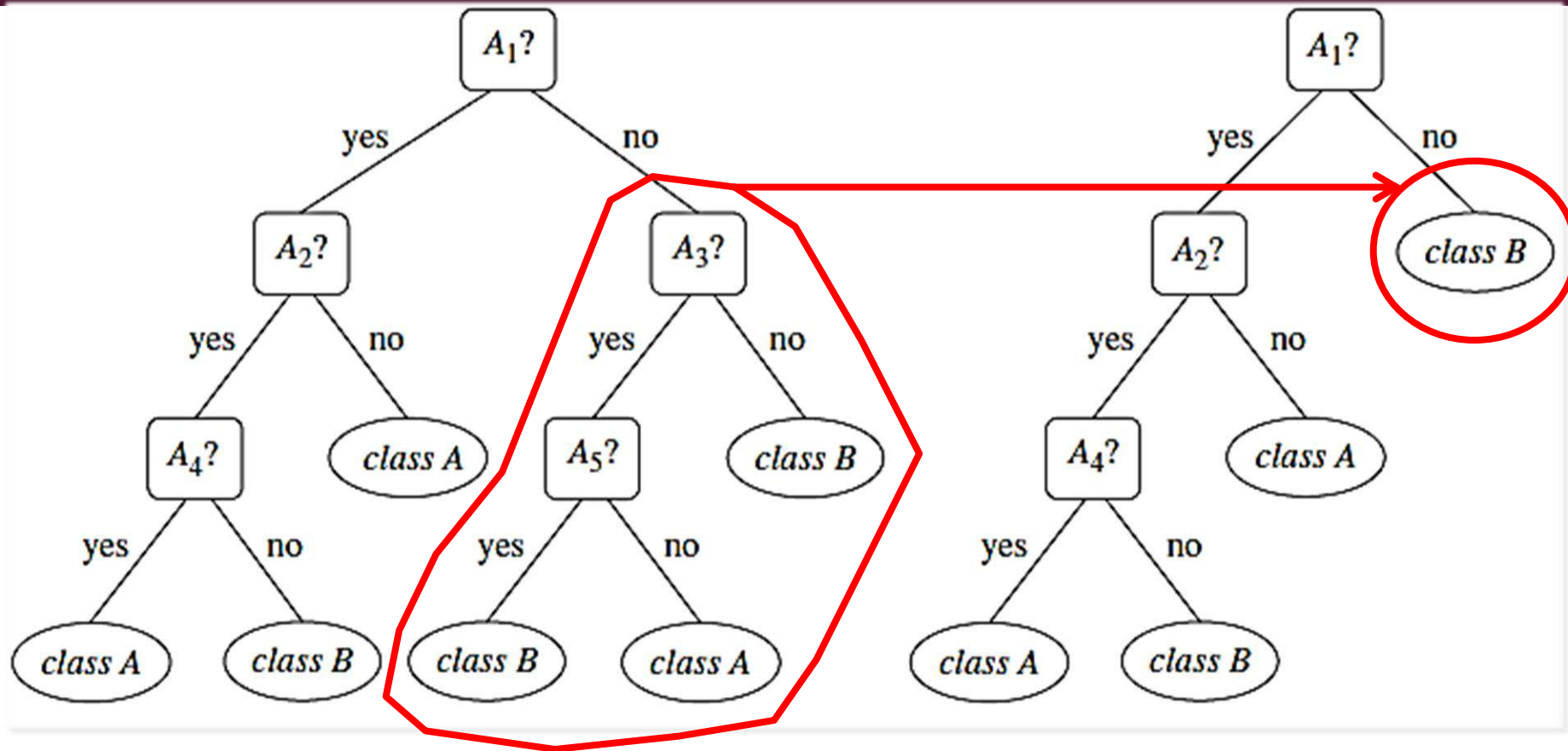
TREE PRUNING

- Data may be *overfitted* to dataset anomalies and outliers
- **Pruning** removes the least reliable branches
 - DT becomes less complex
- Prepruning → statistically assess the *goodness of a split* before it takes place
 - hard to choose *thresholds* for statistical significance
- Postpruning → remove sub-trees from already constructed trees
 1. remove sub-tree branches and replace with leaf node
 2. leaf is labeled with most frequent class in sub-tree



DECISION TREE INDUCTION

TREE PRUNING



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DECISION TREE – WHAT ARE RULES?

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
sales	Middle aged	low	junior	40
systems	youth	medium	junior	20
systems	Middle aged	high	senior	20
systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

RI: IF salary =medium AND age = youth
THEN Status = Junior

$$coverage(R) = \frac{n_{covers}}{|D|}$$

$$accuracy(R) = \frac{n_{correct}}{n_{covers}}$$

- coverage(RI) = 20/200=10% and
- accuracy(RI)= 20/20 = 100%.

X: (Department = system age = youth, salary = low)

?



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DT – RESOLVING RULES CONFLICTS

Rules conflicts are the result of a tuple firing more than one rule with different class predictions

Two resolution strategies

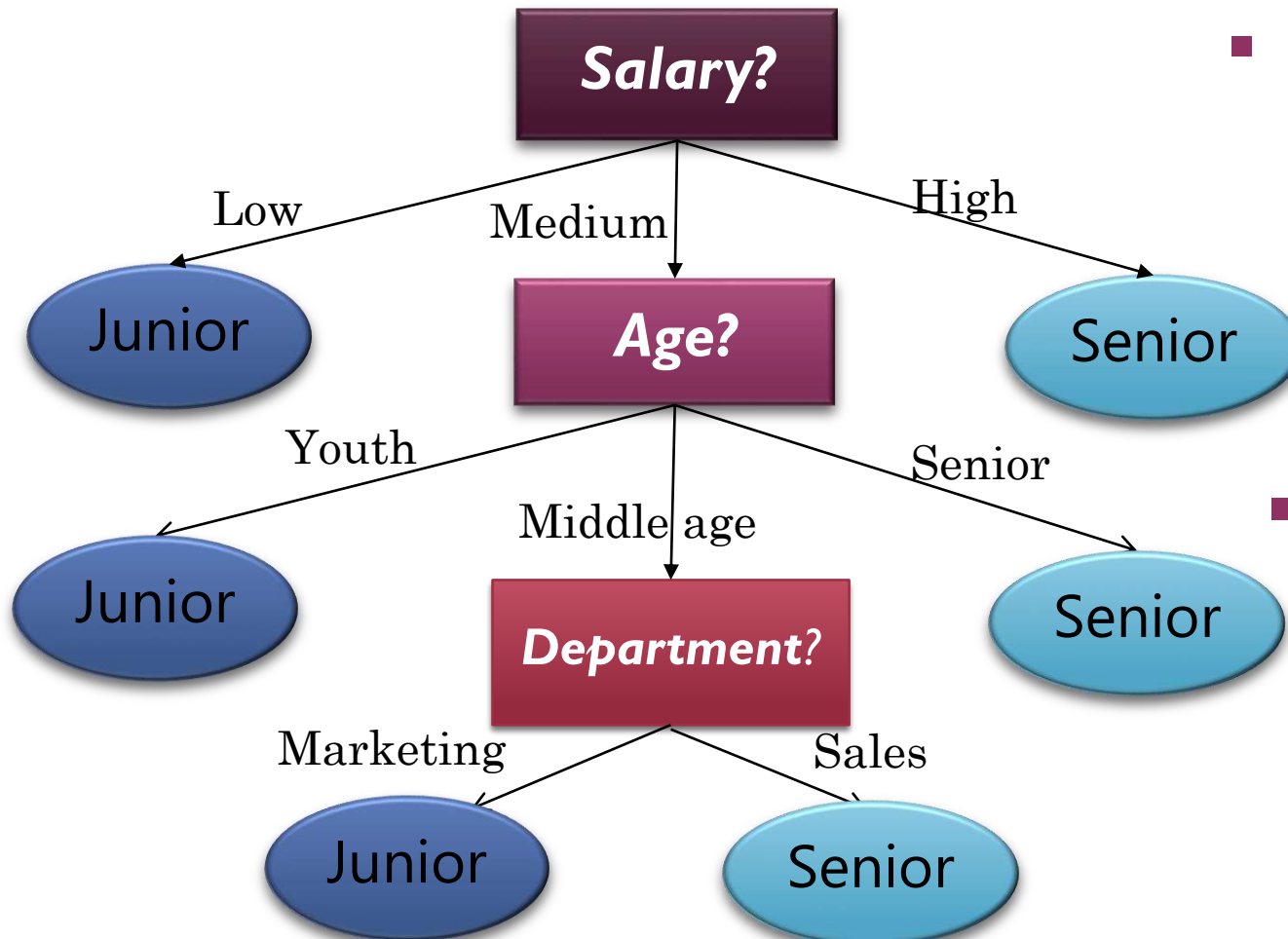
- **Size Ordering** → rule with **largest antecedent** (toughest) has highest priority fires and returns class prediction
- **Rule Ordering** → rules **prioritized apriori** according to
 - **Class-based ordering** → decreasing importance (most frequent are highest – order of prevalence)
 - **Rule-based ordering** → measures of rule quality (e.g. accuracy, size, domain expertise)

Fallback (default) rule when no rules are triggered



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DECISION TREE



- Create one rule for each path from root to leaf in the decision tree
 1. Each splitting criterion is ANDed to form rule antecedent (IF)
 2. Leaf node holds class prediction (THEN)
- RI: IF salary =medium AND age = youth THEN Status = Junior

Can the rules resulting from decision trees have conflicts?



QUESTION?

NEXT



IS422P - DATA MINING CLASSIFICATION (PART III)



AMIRA REZK
INFORMATION SYSTEM DEPARTMENT



AGENDA



The Basics

What is Classification?
General Approach



Decision Tree Induction

The Algorithm
Attribute Selection Measures
Tree Pruning
Extracting Rules from Decision Trees



Bayes Classification

Bayes' Theorem
Naïve Bayesian Classification



Lazy Learners

K-Nearest Neighbor Classifiers



Regression analysis

Linear regression



Model Evaluation

Metrics for Evaluating Classifiers Performance
Cross-Validation
Bootstrap



Improving Classification Accuracy

Bagging
Boost and AdaBoost



DECISION TREE INDUCTION

THE ALGORITHM

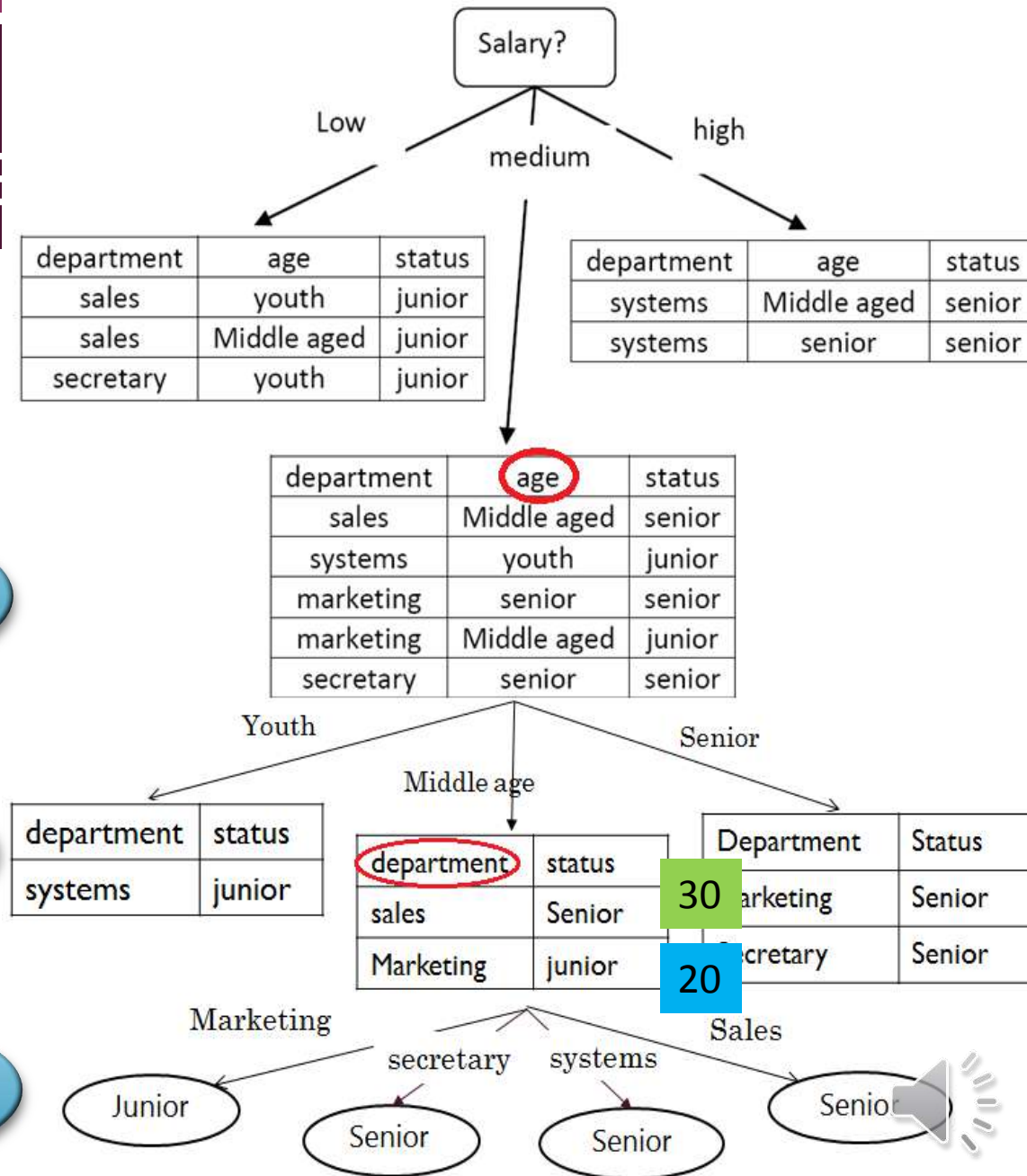
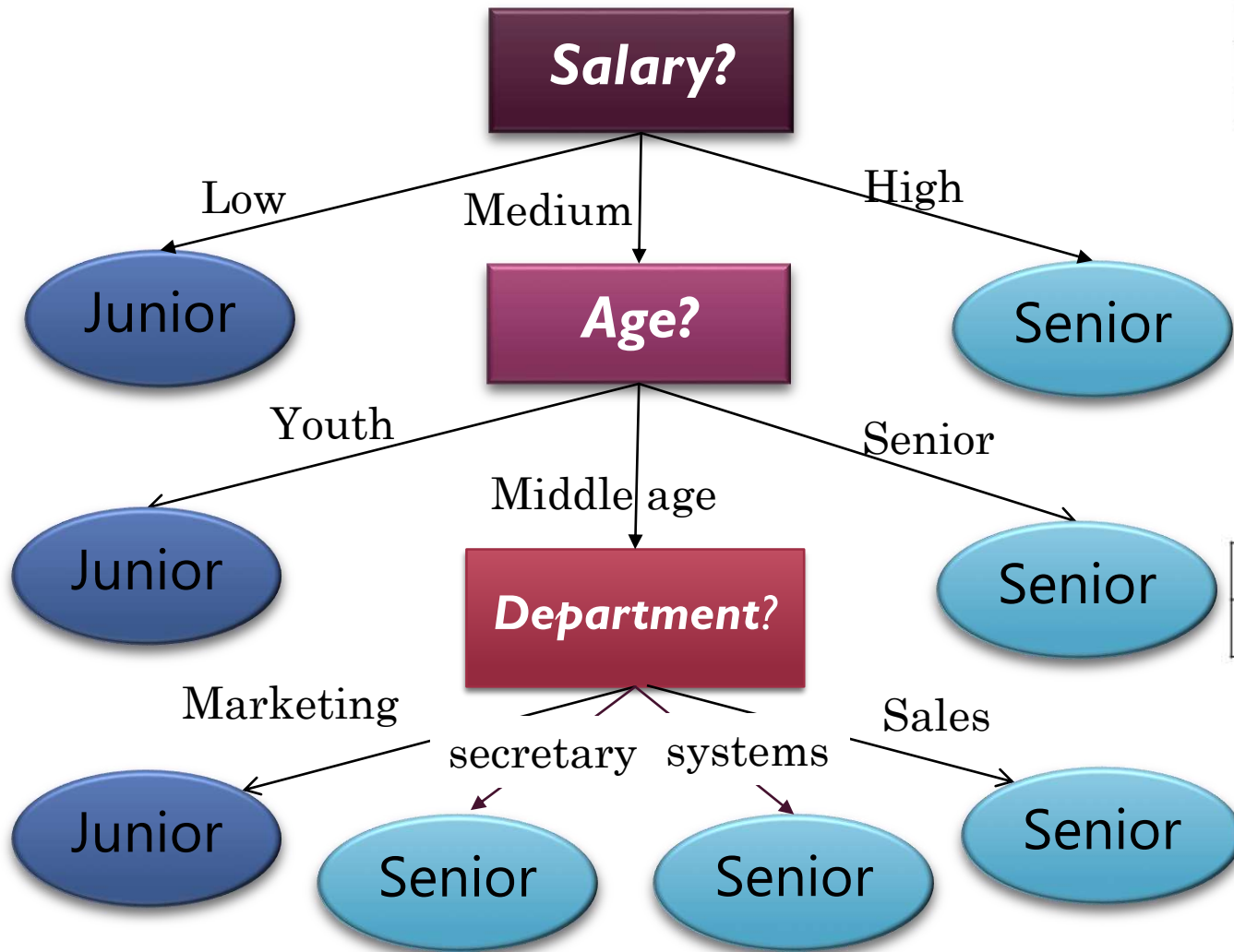
- **Terminating conditions**

- All the tuples in D (represented at node N) belong to the same class
- There are no remaining attributes on which the tuples may be further partitioned
 - majority voting is employed $\rightarrow$ convert node into a leaf and label it with the most common class in data partition
- There are no tuples for a given branch
 - a leaf is created with the majority class in data partition



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURE



MODEL EVALUATION

METRICS FOR EVALUATING CLASSIFIER PERFORMANCE

Measure	Formula
accuracy , recognition rate	$\frac{TP + TN}{P + N}$
error rate , misclassification rate	$\frac{FP + FN}{P + N}$
sensitivity , true positive rate, recall	$\frac{TP}{P}$
specificity , true negative rate	$\frac{TN}{N}$
precision	$\frac{TP}{TP + FP}$

Positives →

tuples representing class of interest

Negatives →

tuples representing other class(es)

True Positives →

positive tuples correctly labeled

False Positives →

negative tuples incorrectly labeled

True Negatives →

negative tuples correctly labeled

False Negatives →

positive tuples incorrectly labeled



MODEL EVALUATION

METRICS FOR EVALUATING CLASSIFIER PERFORMANCE

		Predicted		
		Yes	No	Total
Actual	Yes	<i>TP</i>	<i>FN</i>	<i>P</i>
	No	<i>FP</i>	<i>TN</i>	<i>N</i>
	Total	$\hat{P}$	$\hat{N}$	<i>P + N</i>

Confusion Matrix



MODEL EVALUATION

METRICS FOR EVALUATING CLASSIFIER PERFORMANCE

Balanced Classes

		Predicted			
Actual		Yes	No	Total	Recognition(%)
	Yes	6954	46	7000	$TP/P = 99.34$
	No	412	2588	3000	$TN/N = 86.27$
	Total	7366	2634	10000	$\frac{TP+TN}{P+N} = 95.42$

Example Buys_Computer Confusion Matrix

Use *sensitivity* (TPs or *recall*) and *specificity*



MODEL EVALUATION

METRICS FOR EVALUATING CLASSIFIER PERFORMANCE

Imbalanced Classes

Actual \ Predicted	Predicted			Accuracy (%)
	Yes	No	Total	
Yes	90	210	300	30
No	140	9560	9700	98.56
Total	230	9770	10000	96.4

Low

Example Cancer Confusion Matrix

Use *sensitivity* (TPs or *recall*) and *specificity*

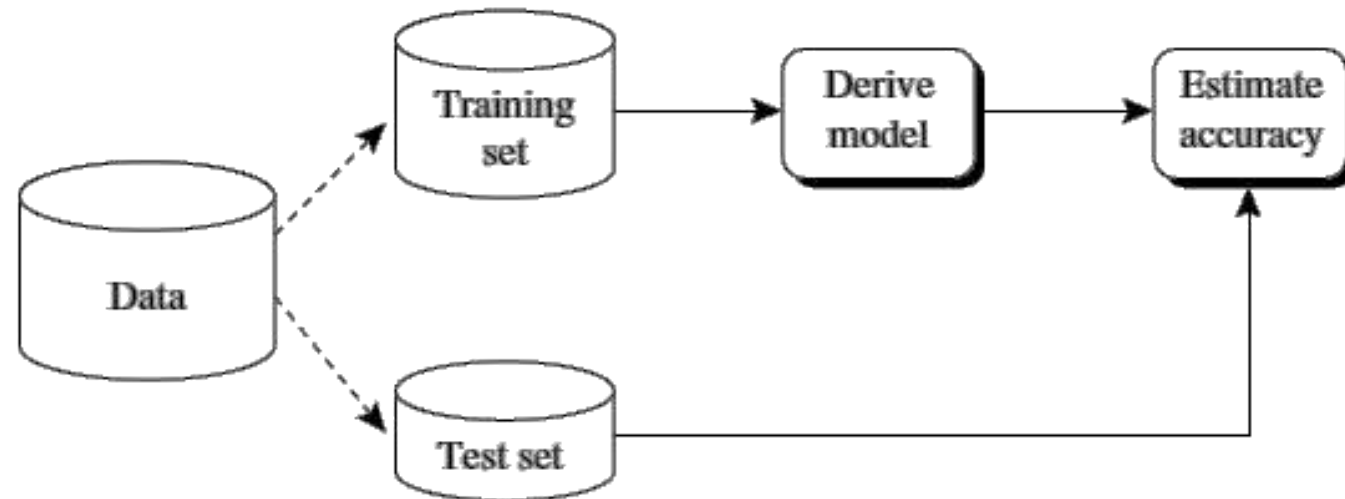
High



MODEL EVALUATION

HOLDOUT AND RANDOM SUBSAMPLING

- **Holdout** → RANDOMLY allocate 2/3 of data for training and remaining 1/3 for testing
- **Random Subsampling** → Repeat holdout k times and take average accuracy



MODEL EVALUATION

CROSS-VALIDATION

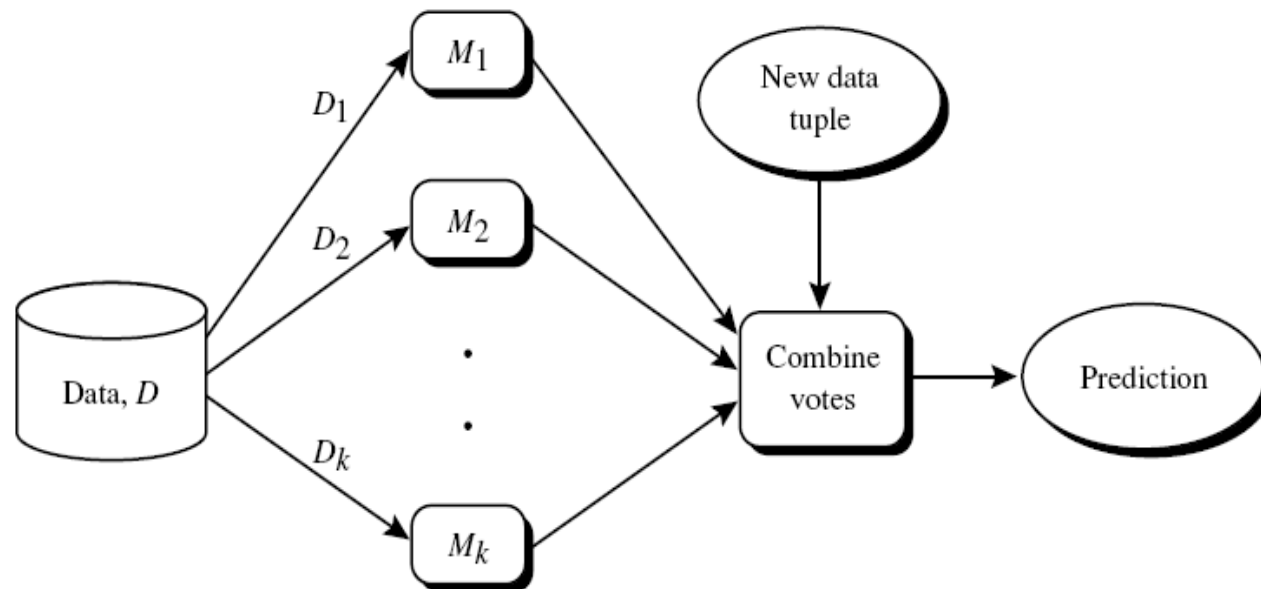
- ***k-fold cross-validation*** → randomly partition dataset into k mutually exclusive ***folds*** of approximately equal size
- In iteration i , $fold_i$ is test set and all other folds are training set
- Accuracy = $\frac{\sum \text{correct classifications for all } k \text{ iterations}}{\text{dataset size}}$
- **Stratified k-fold cross-validation** → class distribution in each fold is approximately the same as in initial dataset
 - **Stratified 10-fold cross-validation** is recommended



IMPROVING CLASSIFICATION ACCURACY

ENSEMBLE METHODS

- **Ensemble** → a set of classifiers, each with a vote for a class label
- Each base classifier is produced from a different partition of the dataset
 - Majority voting is used to compose an **aggregate classification**



IMPROVING CLASSIFICATION ACCURACY

ENSEMBLE METHODS – BAGGING / BOOTSTRAP

Algorithm: Bagging. The bagging algorithm—create an ensemble of classification models for a learning scheme where each model gives an equally weighted prediction.

Input:

- D , a set of d training tuples;
- k , the number of models in the ensemble;
- a classification learning scheme (decision tree algorithm, naïve Bayesian, etc.).

Output: The ensemble—a composite model, M^* .

Method:

- (1) for $i = 1$ to k do // create k models:
- (2) create bootstrap sample D_i , by sampling D with replacement;
- (3) use D_i and the learning scheme to derive a model, M_i ;
- (4) endfor

To use the ensemble to classify a tuple, X :

let each of the k models classify X and return the majority vote;

Bootstrap
→ same size
as dataset,
sampling
with
replacement

3
5
3
7
4
3
7
9
6
10



QUESTION?

NEXT

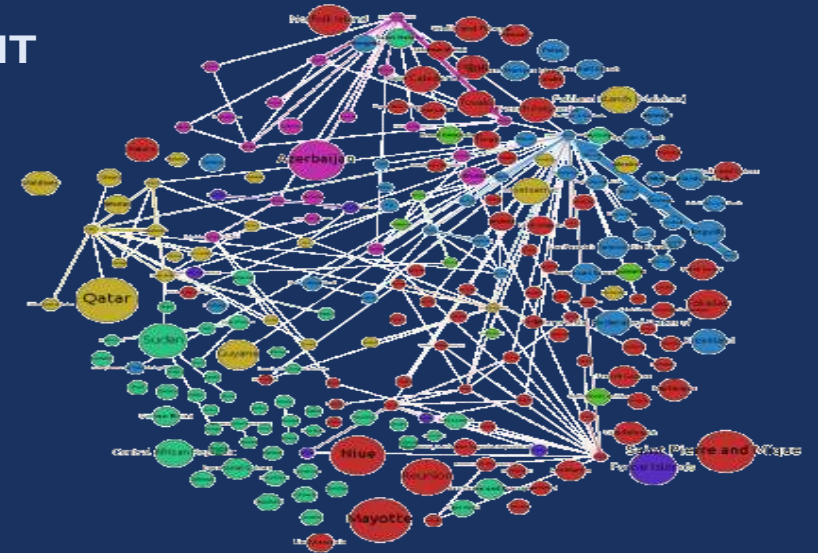


IS422P - DATA MINING

I0 - CLUSTERING



AMIRA REZK
INFORMATION SYSTEM DEPARTMENT



AGENDA



The Basics

What is Cluster Analysis?

Requirements for Cluster Analysis

Overview of methods



Partitioning Methods

K-Means



Hierarchical Methods

Agglomerative vs. Divisive

Distance Measures



Density-Based Methods

DBSCAN



Grid-Based Methods



Evaluation of Clustering

Assessing Clustering Tendency

Measuring Clustering Quality

AGENDA



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- DBSCAN



Grid-Based Methods



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WHAT IS CLUSTER ANALYSIS?

- Partitioning a set of data objects into subsets or clusters
 - objects in a cluster are similar, yet dissimilar to objects in other clusters
- **Goal**: discovery of previously unknown groups within the data
- Clusters are implicit classes
- **Applications** → business intelligence, image pattern recognition, web search, biology, security
- Clustering can be used for pre-processing and outlier detection

REQUIREMENTS FOR CLUSTER ANALYSIS

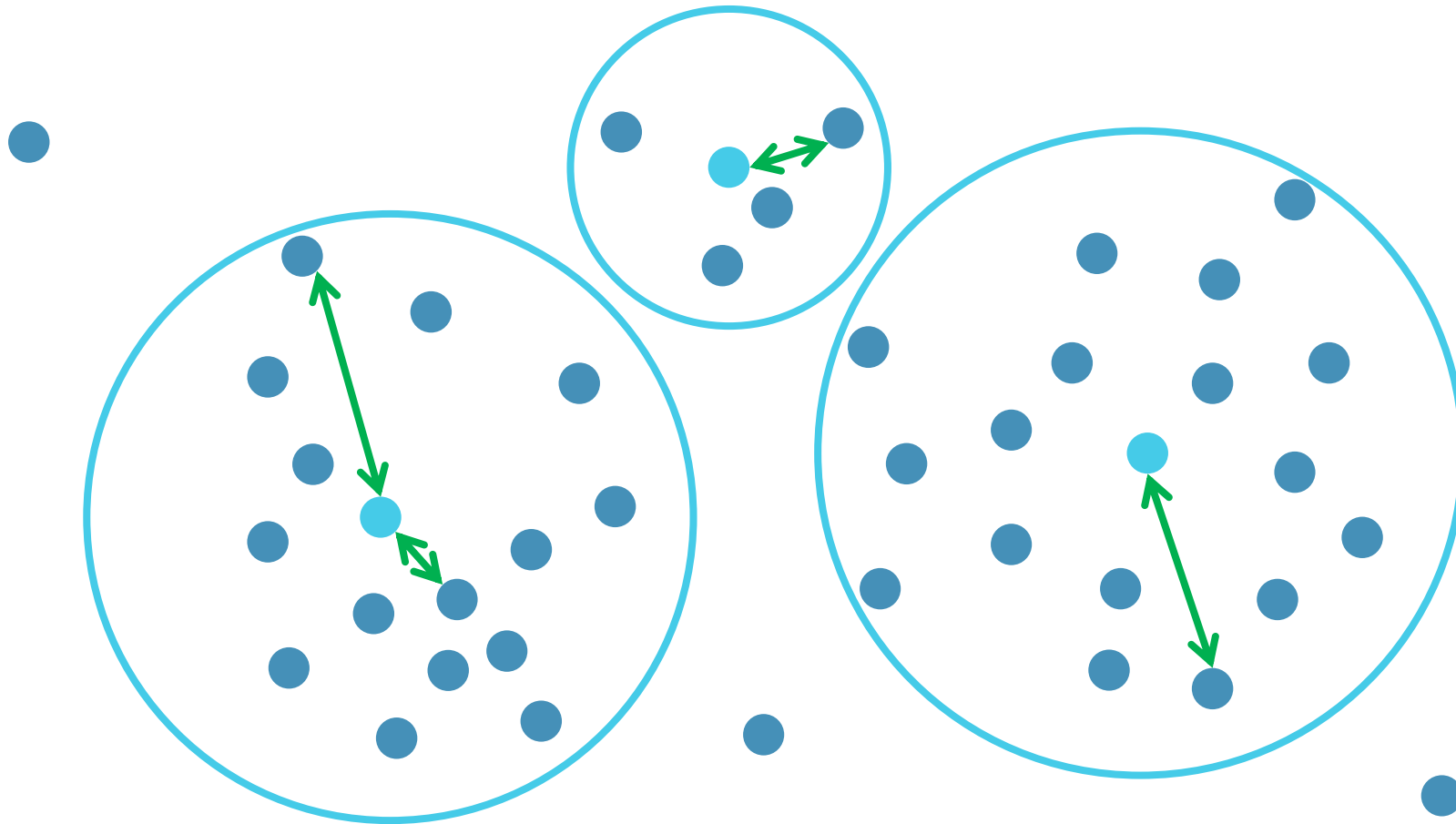
- Scalability → currently handles small datasets, uses sampling
- Handling different attribute types → mostly numerical
- Discovering clusters with arbitrary shape → currently mostly spherical
- Domain knowledge & input parameters → # clusters & clustering results
- Handling noisy data → currently sensitive to noise
- Incremental clustering & insensitivity to input order → new data requires re-computing clusters from scratch – sensitive to order
- Handling high-dimensionality data → mostly low Dimensionality
- Constraint-based clustering → little support for domain constraints
- Interpretability & usability → are results comprehensible & usable?

COMPARING CLUSTER ANALYSIS METHODS

- The partitioning criteria – **flat** or **hierarchical**?
- Separation of clusters – **mutually exclusive** or **overlapping**?
- Similarity measure – **distance** or **connectivity/density**?

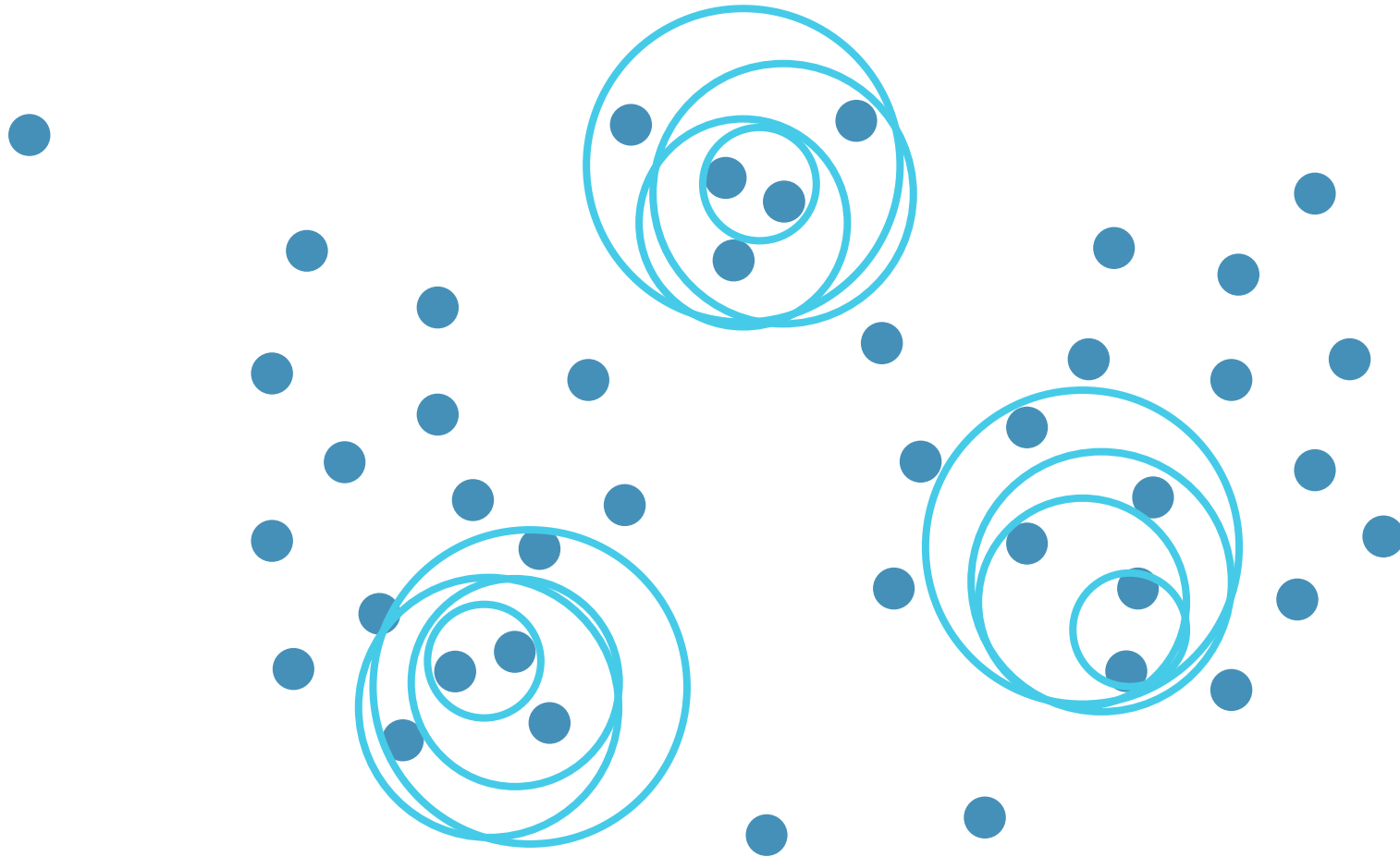
OVERVIEW OF CLUSTER ANALYSIS METHODS

PARTITIONING



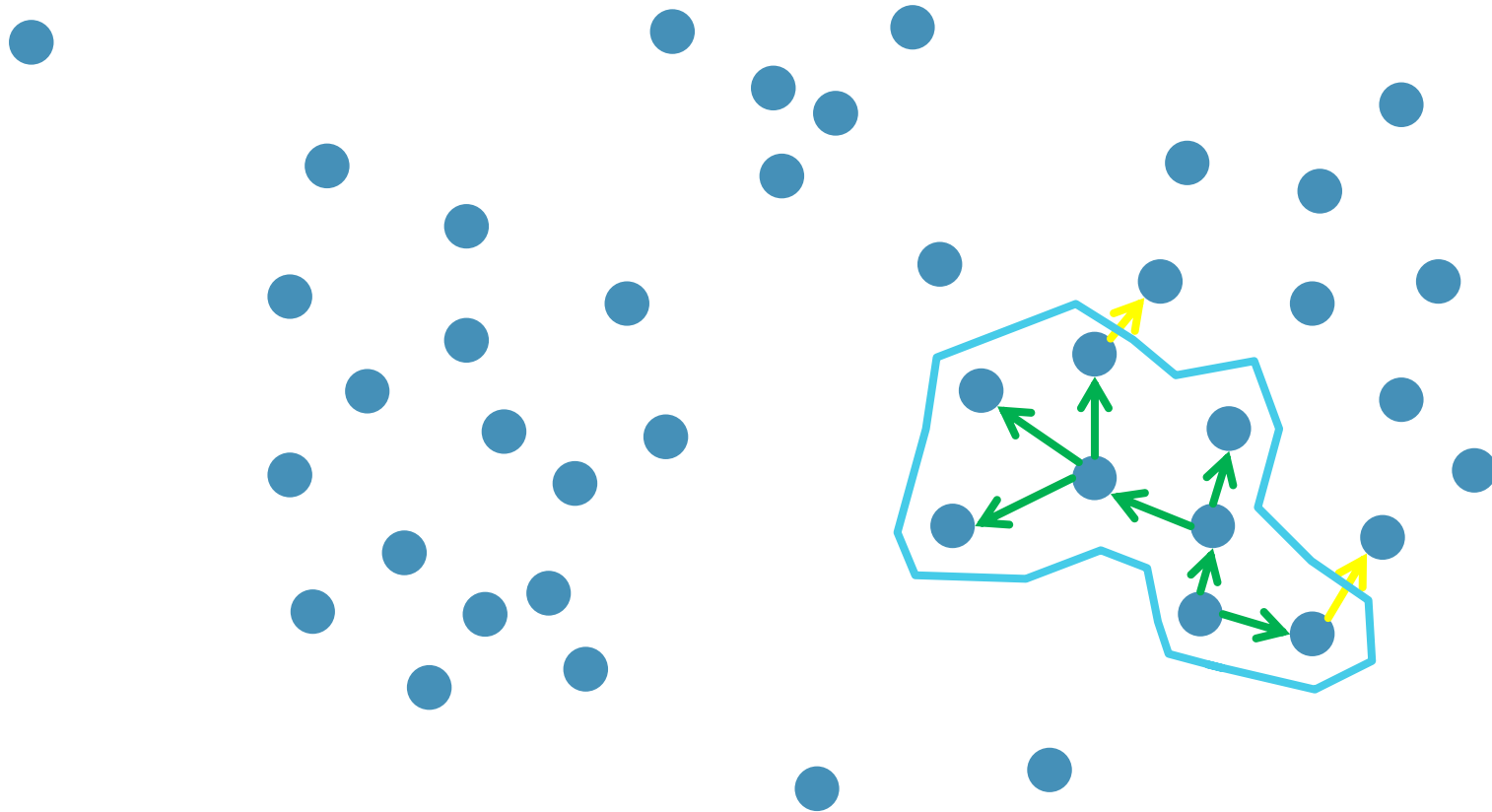
OVERVIEW OF CLUSTER ANALYSIS METHODS

HIERARCHICAL



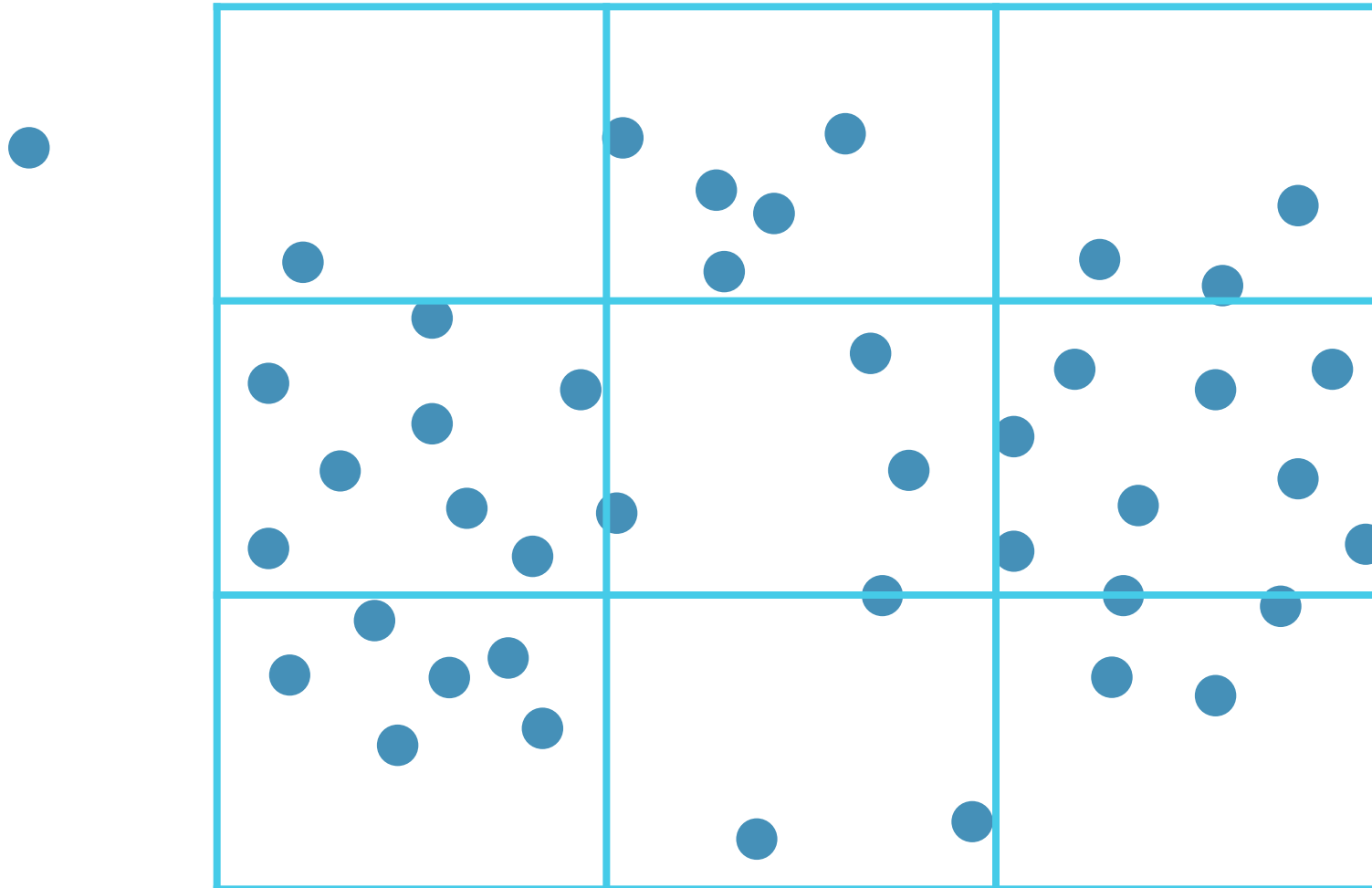
OVERVIEW OF CLUSTER ANALYSIS METHODS

DENSITY-BASED



OVERVIEW OF CLUSTER ANALYSIS METHODS

GRID-BASED



OVERVIEW OF CLUSTER ANALYSIS METHODS

Method	Characteristics
Partitioning methods	— Find <u>mutually exclusive</u> clusters of <u>spherical shape</u> — <u>Distance-based</u> — May <u>use mean or medoid</u> to represent cluster center — Effective for <u>small- to medium-size data sets</u>
Hierarchical methods	— Clustering is <u>hierarchy</u> involving multiple levels — Cannot correct <u>erroneous merges/splits</u> — May consider object "<u>linkages</u>"
Density-based methods	— Can find <u>arbitrarily shaped clusters</u> — Clusters are <u>dense regions</u> separated by <u>low-density regions</u> — Each point must have a <u>minimum number of points within its "neighborhood"</u> — May <u>filter out outliers</u>
Grid-based methods	— Use a multi-resolution <u>grid data structure</u> — <u>Fast processing time</u>



AGENDA



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Density-Based Methods

DBSCAN



Grid-Based Methods



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Measuring Clustering Quality



PARTITIONING METHODS

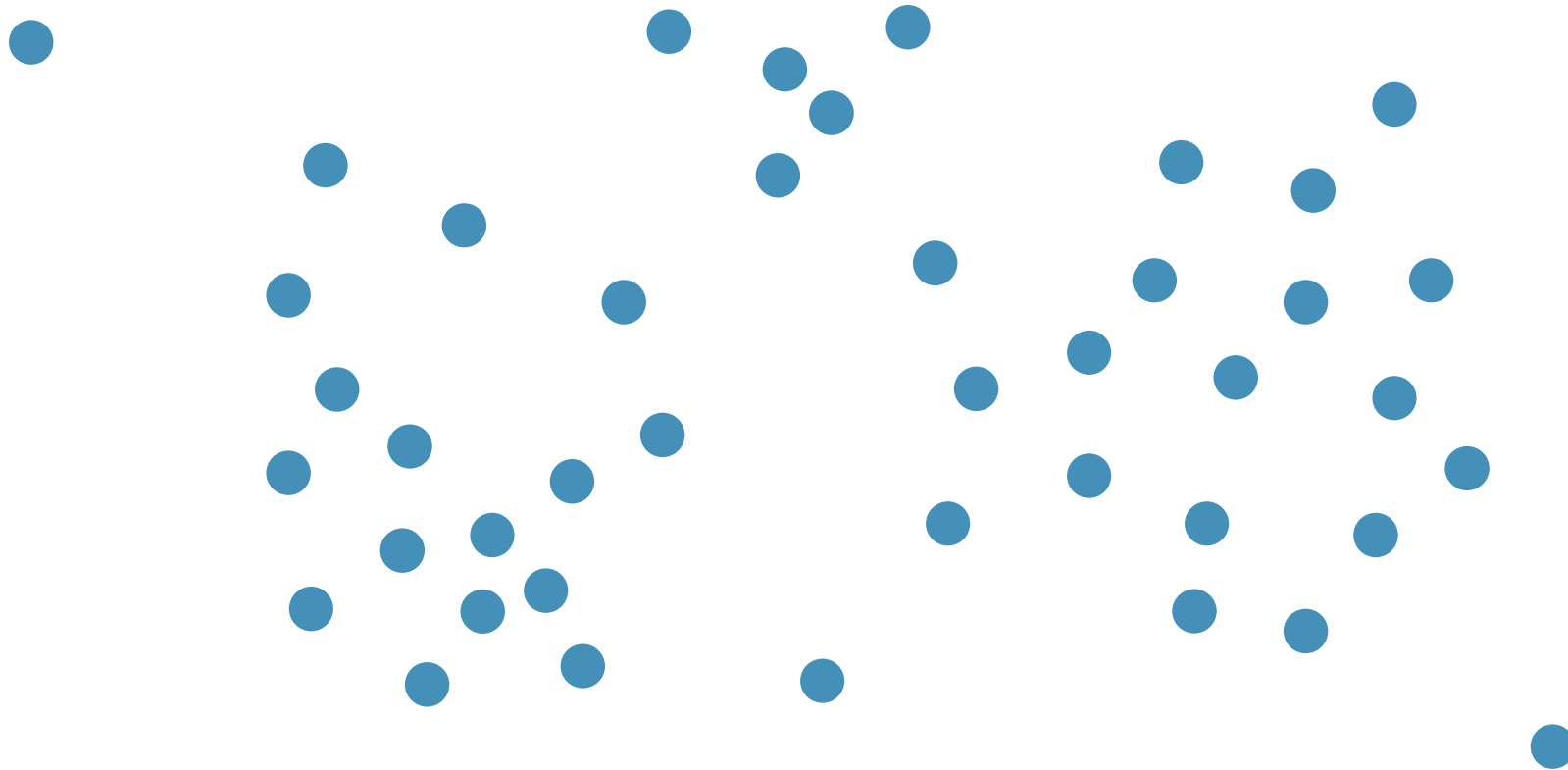
K-MEANS – A CENTROID-BASED TECHNIQUE

- Divide dataset into ***k* mutually exclusive** clusters
- Clusters are represented by their **centroids**
 - A centroid is a **cluster's center point**
- In *k*-means → centroid is **mean** of points within cluster
 - Each object ***x*** in cluster has a distance from centroid ***c_i*** → $dist(x, c_i)$
- - ***x*** is assigned to most similar cluster → ***C_i*** with **min** $dist(x, c_i)$
 - Cluster means are updated, then assignment is repeated
- To measure cluster quality → minimize sum of squared errors

$$E = \sum_{i=1}^k \sum_{x \in C_i} dist(x, c_i)^2$$

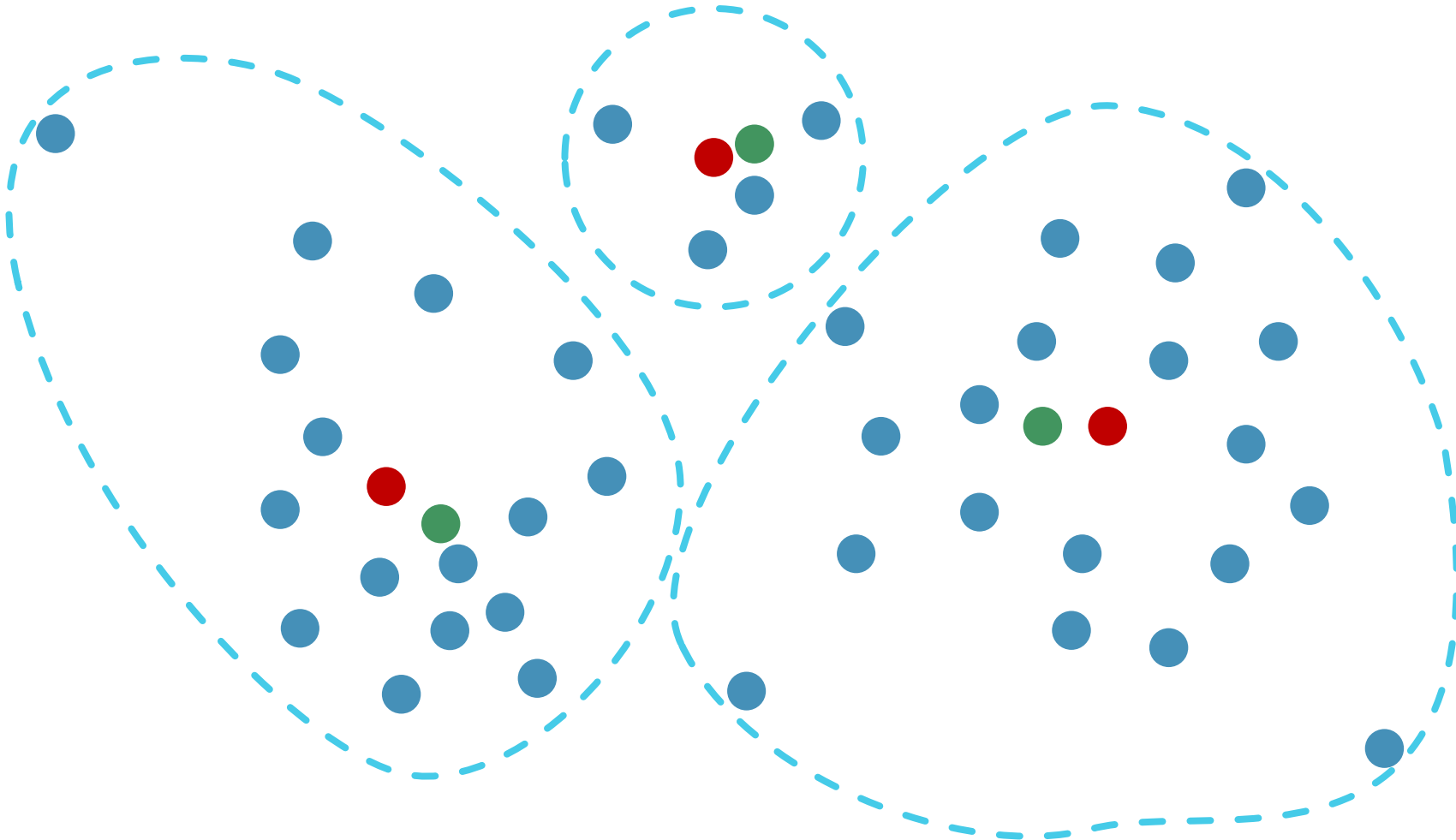
PARTITIONING METHODS

K-MEANS



PARTITIONING METHODS

K-MEANS



PARTITIONING METHODS

K-MEANS

Algorithm: k -means. The k -means algorithm for partitioning, where each cluster's center is represented by the mean value of the objects in the cluster.

Input:

- k : the number of clusters,
- D : a data set containing n objects.

Output: A set of k clusters.

Method:

- (1) arbitrarily choose k objects from D as the initial cluster centers;
- (2) repeat
- (3) (re)assign each object to the cluster to which the object is the most similar,
 based on the mean value of the objects in the cluster;
- (4) update the cluster means, that is, calculate the mean value of the objects for
 each cluster;
- (5) until no change;

PARTITIONING METHODS

K-MEANS

Cluster the eight points in table using k-means. Assume that $k = 3$ and that initially the points are assigned to clusters as follows: $C1 = \{x1, x2, x3\}$, $C2 = \{x4, x5, x6\}$, $C3 = \{x7, x8\}$.

- Apply the k-means algorithm until convergence (i.e., until the clusters do not change), using the Manhattan distance.

(Hint: The Manhattan distance is: $d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{in} - x_{jn}|$.) Make sure you clearly identify the final clustering and show your steps.

	A1	A2
x1	2	10
x2	2	5
x3	8	4
x4	5	8
x5	7	5
x6	6	4
x7	1	2
x8	4	9



PARTITIONING METHODS

K-MEANS

- $C1 = \{x1, x2, x3\} = \{(2, 10), (2, 5), (8, 4)\}$
 - Mean of $C1 = (\frac{2+2+8}{3}, \frac{10+5+4}{3}) = (4, 6\frac{1}{3})$
- $C2 = \{x4, x5, x6\} = \{(5, 8), (7, 5), (6, 4)\}$
 - Mean of $C2 = (6, 5\frac{2}{3})$
- $C3 = \{x7, x8\} = \{(1, 2), (4, 9)\}$
 - Mean of $C3 = (2\frac{1}{2}, 5\frac{1}{2})$

	A1	A2
x1	2	10
x2	2	5
x3	8	4
x4	5	8
x5	7	5
x6	6	4
x7	1	2
x8	4	9



PARTITIONING METHODS

K-MEANS

	X1 (2,10)	X2 (2,5)	X3 (8,4)	X4 (5,8)	X5 (7,5)	X6 (6,4)	X7 (1,2)	X8 (4,9)	NEW MEAN
C1 (4, 6 $\frac{1}{3}$)	5 $\frac{2}{3}$	3 $\frac{1}{3}$	6 $\frac{1}{3}$	2 $\frac{2}{3}$	4 $\frac{1}{3}$	4 $\frac{1}{3}$	7 $\frac{1}{3}$	2 $\frac{2}{3}$	4 $\frac{1}{2}$, 8 $\frac{1}{2}$
C2 (6, 5 $\frac{2}{3}$)	8 $\frac{1}{3}$	4 $\frac{2}{3}$	3 $\frac{2}{3}$	3 $\frac{1}{3}$	1 $\frac{2}{3}$	1 $\frac{2}{3}$	8 $\frac{2}{3}$	5 $\frac{1}{3}$	7, 4 $\frac{1}{3}$
C3 (2 $\frac{1}{2}$, 5 $\frac{1}{2}$)	5	1	7	5	5	5	5	5	1 $\frac{2}{3}$, 5 $\frac{2}{3}$

C1 = {x1, x4, x8} = {(2,10), (5,8), (4,9)} Mean of C1 = (2 $\frac{2}{3}$, 9)

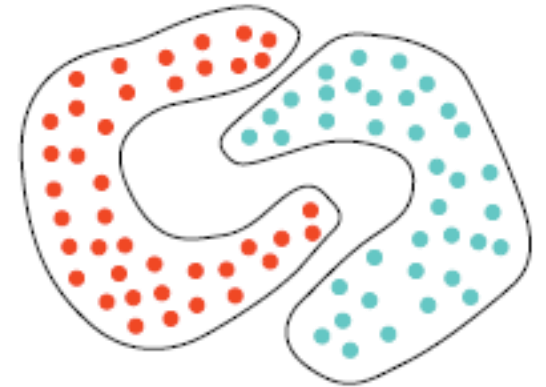
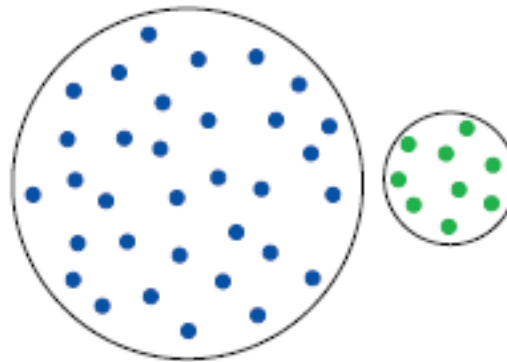
C2 = {x3, x5, x6} = {(8,4), (7,5), (6,4)} Mean of C2 = (7, 4 $\frac{1}{3}$)

C3 = {x2, x7} = {(2,5), (1,2)} Mean of C3 = (1 $\frac{1}{2}$, 3 $\frac{1}{2}$)

PARTITIONING METHODS

K-MEANS

- Factors to consider:
- Selection of k
- Selection of initial centroids
- Calculation of dissimilarity
- Calculation of cluster means
- When it fails!
- Clusters with very different sizes & with concave shapes



AGENDA



What is Cluster Analysis?

Requirements for Cluster Analysis

Overview of methods

The Basics



K-Means

Partitioning Methods



Agglomerative vs. Divisive

Distance Measures

Hierarchical Methods



DBSCAN

Density-Based Methods



Grid-Based Methods



Assessing Clustering Tendency

Measuring Clustering Quality

Evaluation of Clustering

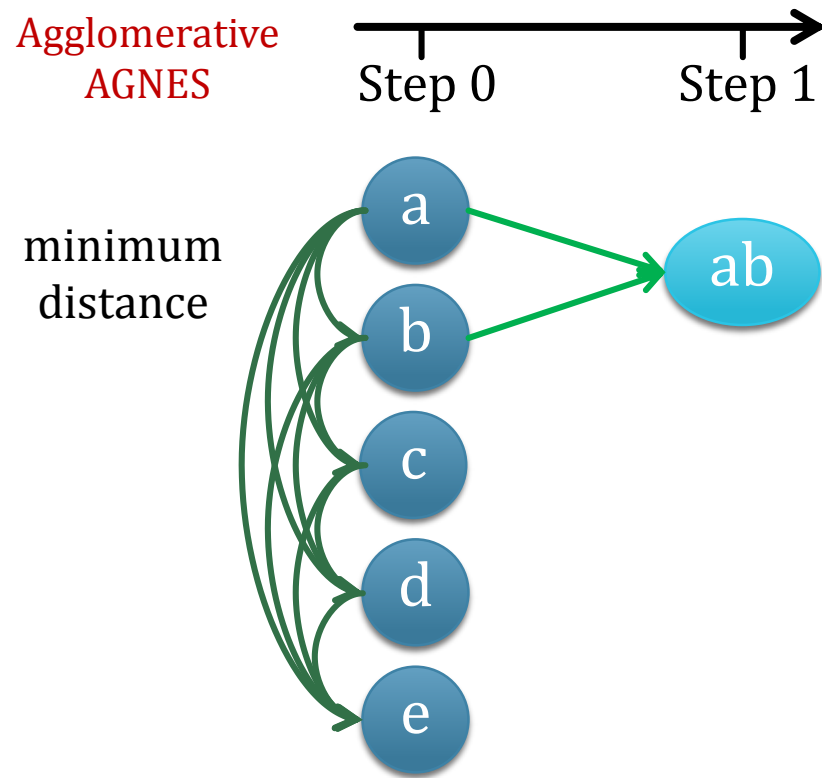
HIERARCHICAL METHODS

AGGLOMERATIVE VERSUS DIVISIVE CLUSTERING

- Hierarchical clustering → group data objects into a hierarchy or “tree” of clusters
- Agglomerative → bottom-up (merge) composition
 - Each object has its own cluster
 - Two clusters that are closest merged into a bigger cluster
 - Iteratively merge till termination condition or single cluster is formed
- Divisive → top-down (split) composition
 - All objects in one big cluster
 - Divide into subclusters
 - Recursively divide subclusters into even smaller subclusters
 - Terminate when each object has his own cluster or objects in clusters are similar “enough”

HIERARCHICAL METHODS

AGGLOMERATIVE CLUSTERING

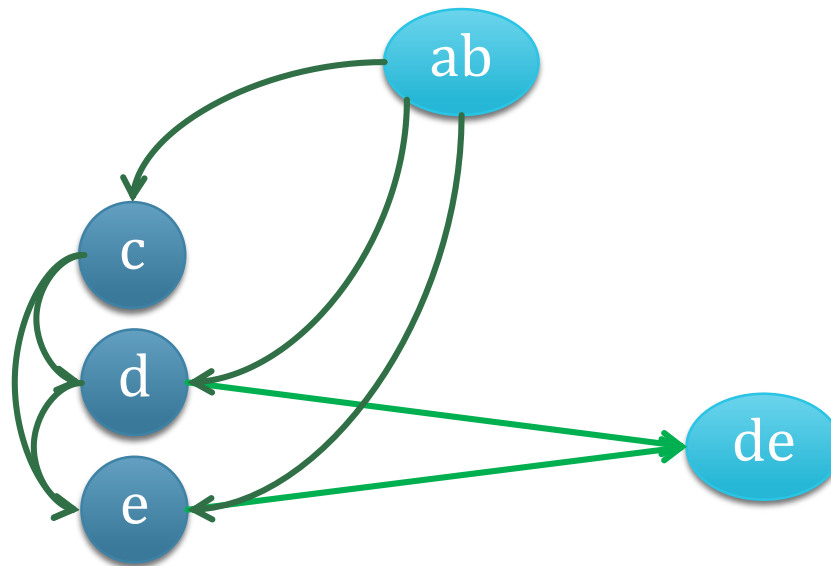


HIERARCHICAL METHODS

AGGLOMERATIVE CLUSTERING

Agglomerative
AGNES

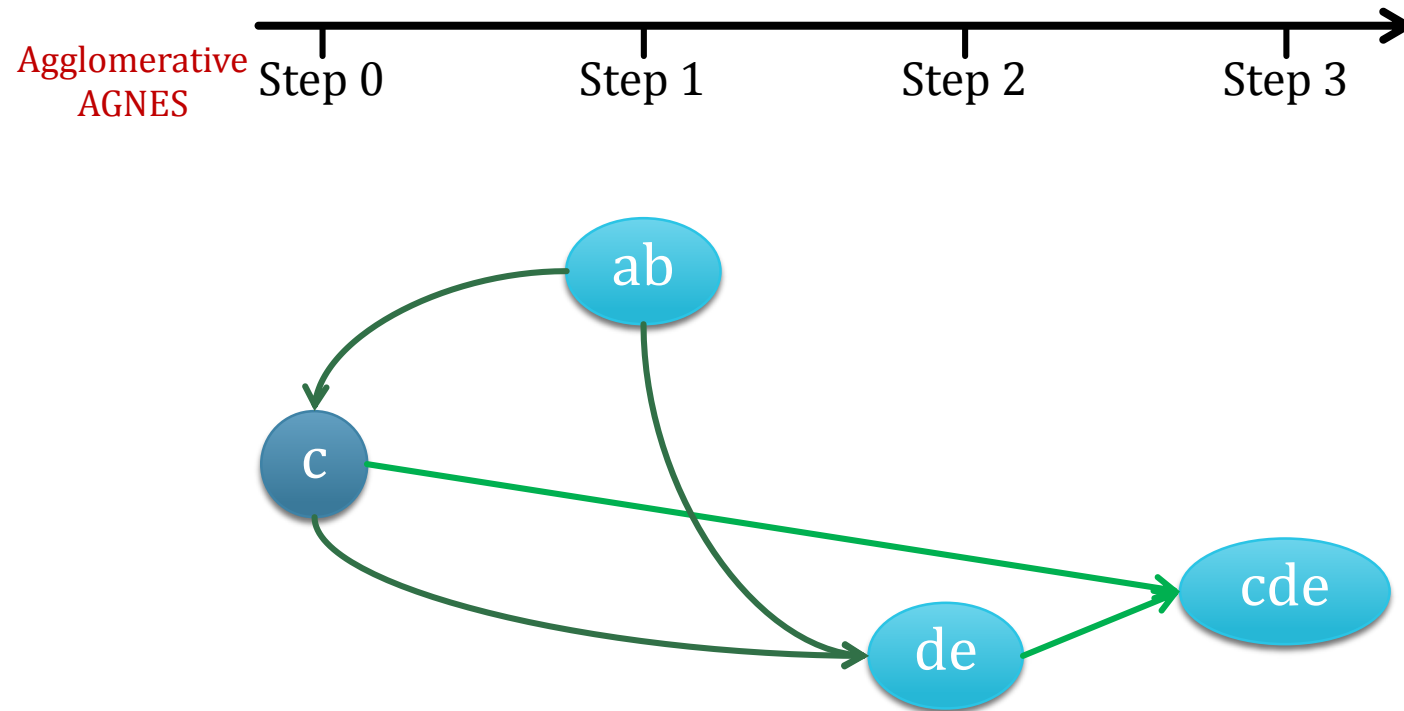
Step 0 Step 1 Step 2



Measure distance between c, d, e and individual elements in cluster {a,b}, choose any with minimum distance (single linkage)

HIERARCHICAL METHODS

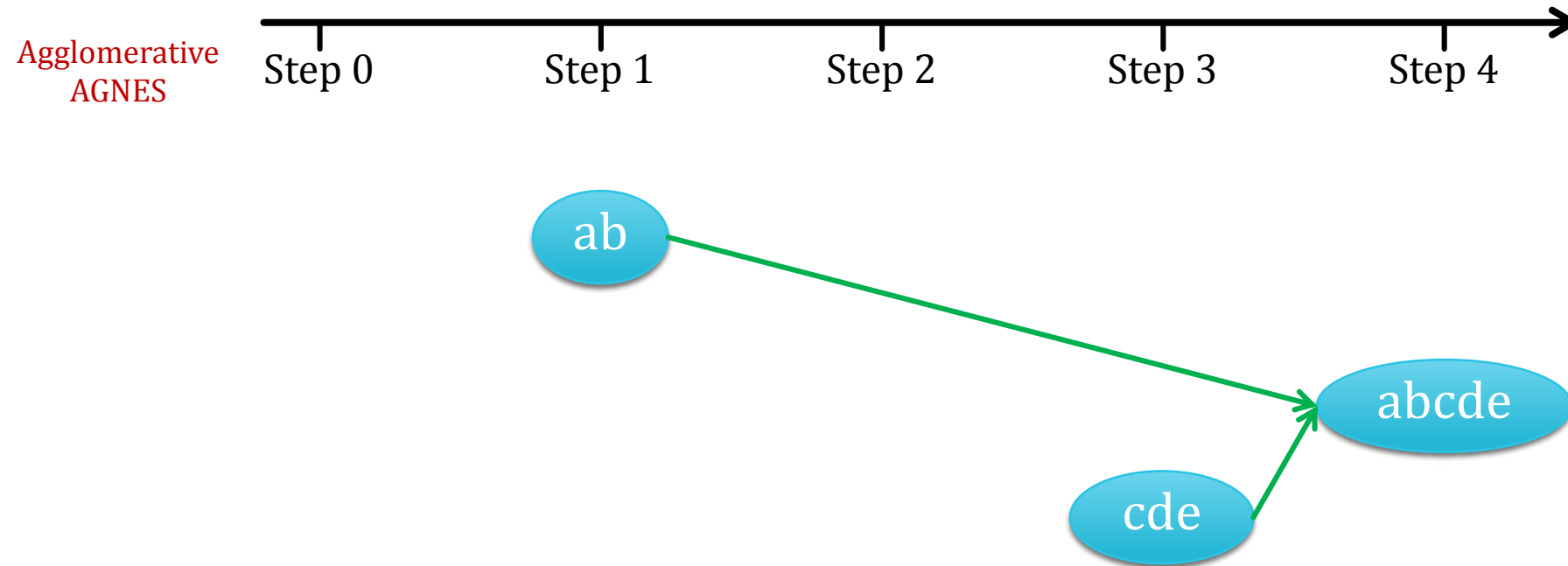
AGGLOMERATIVE CLUSTERING



Measure distance between c and individual elements in cluster $\{a,b\}$ and $\{d,e\}$, as well as distance between pairs in $\{a,b\}$ and $\{d,e\}$, choose any with minimum distance (single linkage)

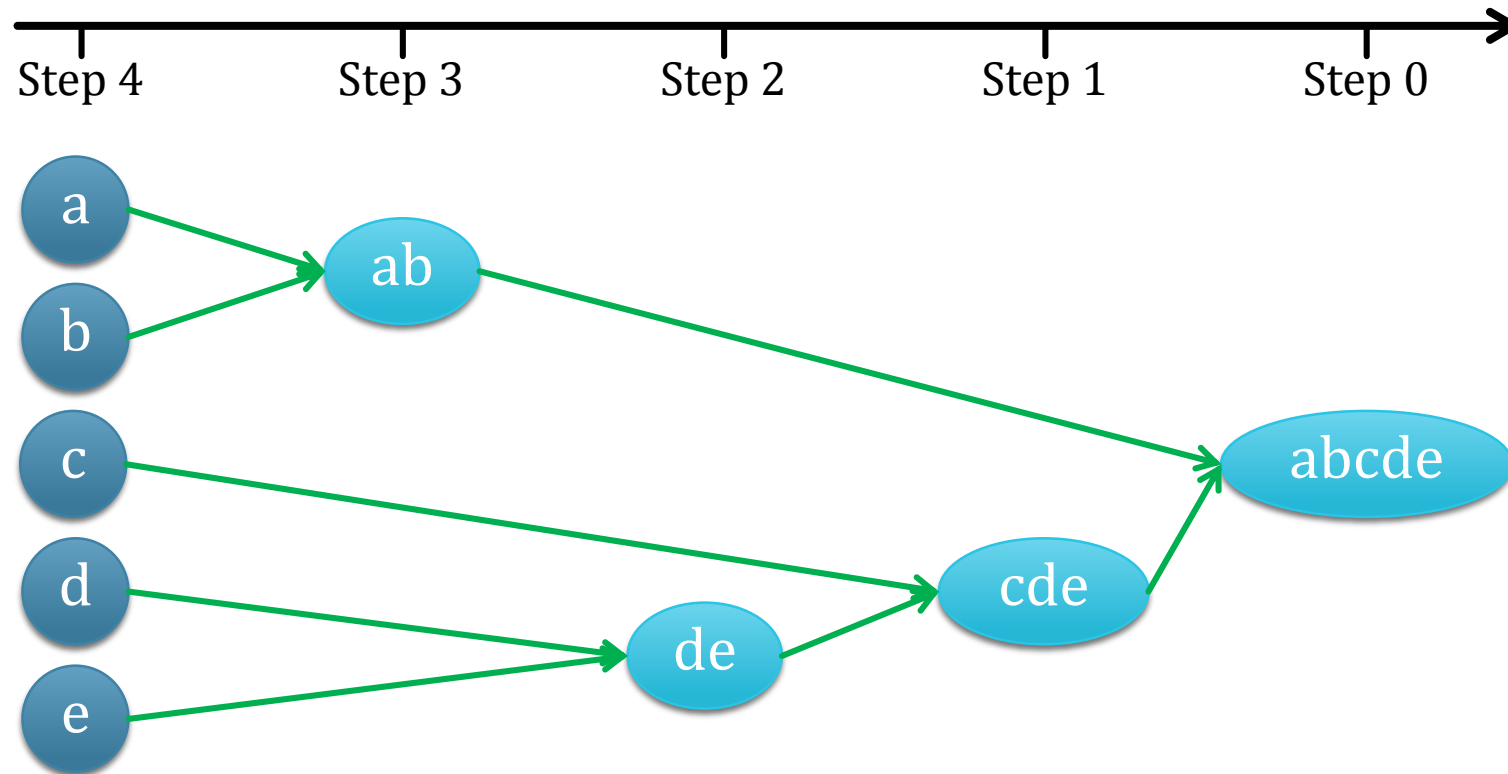
HIERARCHICAL METHODS

AGGLOMERATIVE CLUSTERING



HIERARCHICAL METHODS

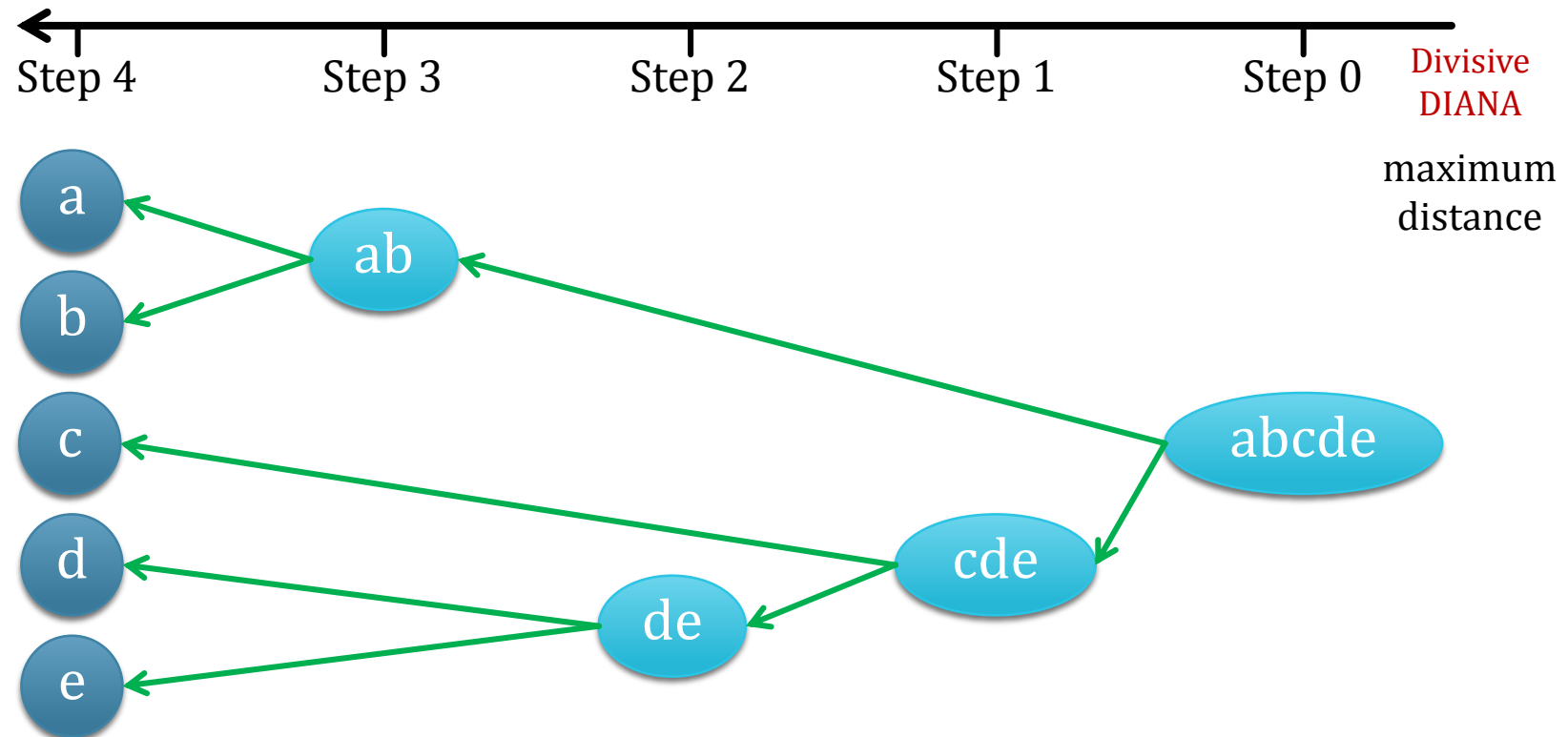
AGGLOMERATIVE CLUSTERING



Minimal spanning tree!

HIERARCHICAL METHODS

DIVISIVE CLUSTERING



How to divide a cluster is a challenge! Heuristic approaches may be used

QUESTION?

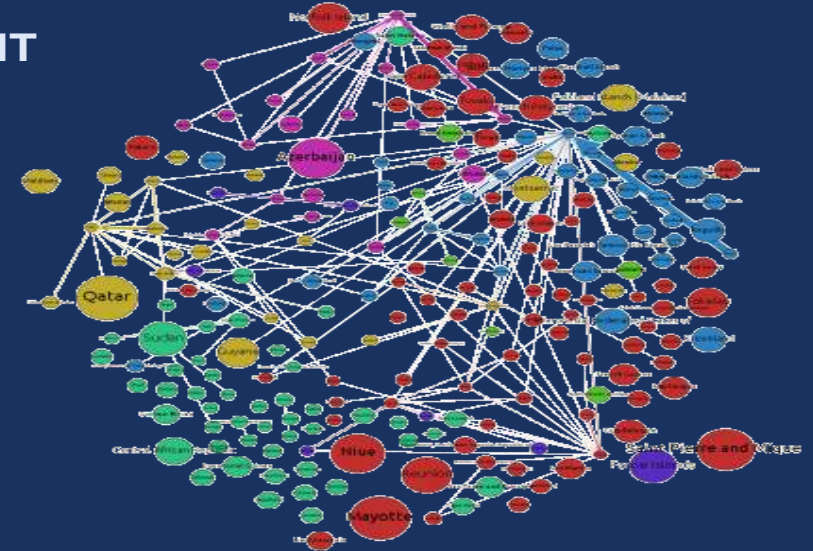
NEXT



IS422P - DATA MINING CLUSTERING-PART2



AMIRA REZK
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AGENDA



The Basics

What is Cluster Analysis?
Requirements for Cluster Analysis
Overview of methods



Partitioning Methods

K-Means



Hierarchical Methods

Agglomerative vs. Divisive
Distance Measures



Density-Based Methods

DBSCAN



Evaluation of Clustering

Assessing Clustering Tendency
Measuring Clustering Quality

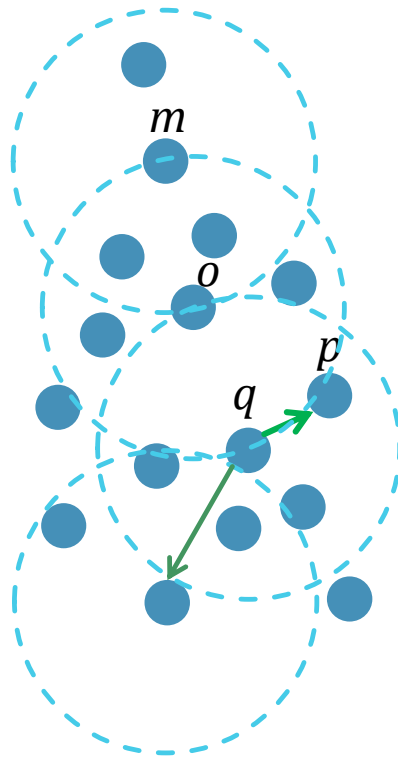
DENSITY-BASED METHODS

DBSCAN: DENSITY-BASED SPATIAL CLUSTERING OF APPLICATIONS WITH NOISE

- Find *core objects* (with dense neighborhoods)
- Connect core objects to form dense clusters
- User provides:
 - **ϵ -neighborhood** of object **o** $\rightarrow$ space within a radius **ϵ** centered at **o**
 - **Neighborhood density** $\rightarrow$ # objects in that neighborhood
 - **$MinPts$** $\rightarrow$ **density threshold** for a neighborhood
- **Core object** $\rightarrow$ object whose ϵ -neighborhood contains at least *MinPts* objects

DENSITY-BASED METHODS

DBSCAN



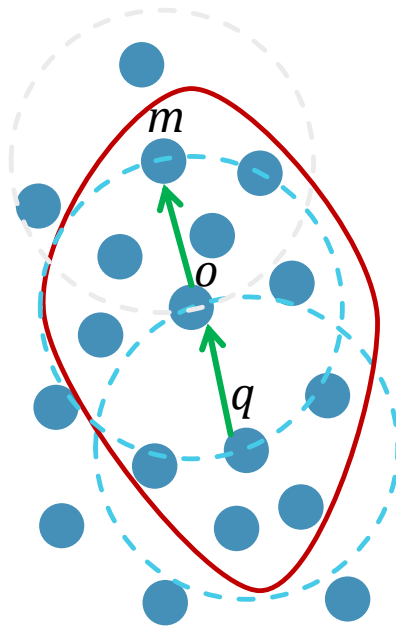
Given $\epsilon = 4$ and
 $MinPts = 5$

an object p is directly **density-reachable** from another object q if and only if q is a core object and p is in the ϵ -neighborhood of q

DENSITY-BASED METHODS

DBSCAN

Given $\epsilon = 4$ and
 $MinPts = 5$



an object p is directly **density-reachable** from another object q if and only if q is a core object and p is in the ϵ -neighborhood of q

objects q & m are **density-connected** if there is an object o such that q & m are both **density-reachable** from o

Algorithm: DBSCAN: a density-based clustering algorithm.

Input:

- D : a data set containing n objects,
- ϵ : the radius parameter, and
- $MinPts$: the neighborhood density threshold.

Output: A set of density-based clusters.

Method:

- (1) mark all objects as unvisited;
- (2) **do**
- (3) randomly select an unvisited object p ;
- (4) mark p as visited;
- (5) if the ϵ -neighborhood of p has at least $MinPts$ objects
- (6) create a new cluster C , and add p to C ;
- (7) let N be the set of objects in the ϵ -neighborhood of p ;
- (8) **for** each point p' in N
- (9) **if** p' is unvisited
- (10) mark p' as visited;
- (11) if the ϵ -neighborhood of p' has at least $MinPts$ points,
 add those points to N ;
- (12) **if** p' is not yet a member of any cluster, add p' to C ;
- (13) **end for**
- (14) output C ;
- (15) **else** mark p as noise;
- (16) **until** no object is unvisited;

DENSITY- BASED METHODS DBSCAN



AGENDA



The Basics

What is Cluster Analysis?
Requirements for Cluster Analysis
Overview of methods



Partitioning Methods

K-Means



Hierarchical Methods

Agglomerative vs. Divisive
Distance Measures



Density-Based Methods

DBSCAN



Evaluation of Clustering

Assessing Clustering Tendency
Measuring Clustering Quality

EVALUATION OF CLUSTERING

ASSESSING CLUSTERING TENDENCY

- Determines whether a given data set has a non-random structure
- **Hopkins Statistic** → Statistical tests for spatial randomness

- Sample n points, $p_1, \dots, p_n$ uniformly from D
- For each point, p_i find its nearest neighbor in D

$$\underline{\text{distance } x_i} = \min_{v \in D} \{\text{dist}(p_i, v)\}$$

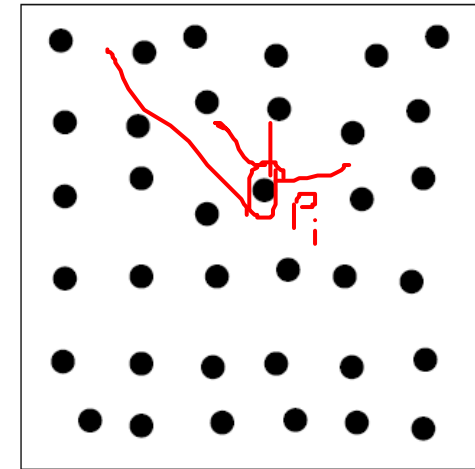
- Sample n points, $q_1, \dots, q_n$ uniformly from D
- For each q_i find its nearest neighbor of q_i in $D - \{q_i\}$

$$\underline{\text{distance } y_i} = \min_{v \in D, v \neq q_i} \{\text{dist}(q_i, v)\}$$

- Calculate the Hopkins Statistic H :

$$\underline{H} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i + \sum_{i=1}^n y_i}$$

- If D is uniformly distributed, $\sum_{i=1}^n x_i$ and $\sum_{i=1}^n y_i$ are roughly equal, and $H \approx 0.5$



EVALUATION OF CLUSTERING

MEASURING CLUSTERING QUALITY – EXTRINSIC METHODS

Extrinsic methods → compare clustering against **ground truth** (*supervision*)

- Assign a score $Q(C, C_g)$ to capture:
 - **Cluster homogeneity** → the purer the better – clusters represent separate class labels
 - **Cluster completeness** → an object with a class label belongs to the cluster representing that class label
 - **Rag bag** → objects that can't be merged into clusters belong to a *rag bag* – penalize a *misc. object* when put in a *pure cluster* more than in a rag bag
 - **Small cluster preservation** → splitting a small category is *more harmful* than splitting a large category
- Ex. **BCubed** *precision* and *recall* of every object in dataset:
 - Precision → how many objects in the same cluster ∈ the same category as the object
 - Recall → how many objects of the same category are assigned to the same cluster

EVALUATION OF CLUSTERING

MEASURING CLUSTERING QUALITY – INTRINSIC METHODS

- **Intrinsic methods** → measure how well the clusters are separated
- Ex. **The silhouette coefficient** → difference between:
 - average distance between object o and all other objects in the cluster to which o belongs (captures cluster correctness) – smaller is better (more compact)
 - minimum average distance from o to all clusters to which o does not belong (captures degree of separation from other clusters) – larger is better
- Compute average silhouette coefficient for all objects in a cluster or over all of the dataset
 - +ve → clustering is good
 - -ve → clustering is bad

EVALUATION OF CLUSTERING

MEASURING CLUSTERING QUALITY – INTRINSIC METHODS

- Compute **the silhouette coefficient** for object **x1**.

What is the meaning of the computed value?

$C1 = \{x1, x4, x8\} = \{(2,10), (5,8), (4,9)\}$ Mean of $C1 = (2\frac{2}{3}, 9)$
 $C2 = \{x3, x5, x6\} = \{(8,4), (7,5), (6,4)\}$ Mean of $C2 = (7, 4\frac{1}{3})$
 $C3 = \{x2, x7\} = \{(2,5), (1,2)\}$ Mean of $C3 = (1\frac{1}{2}, 3\frac{1}{2})$

- $a(o) = \frac{\sum_{o' \in c_i} dis(o, o')}{|c_i| - 1} = \frac{5+3}{2} = 4$
- $b(o) = \min \left\{ \frac{\sum_{o' \in c_j} dis(o, o')}{|c_j|} \right\} = \min \left\{ \frac{12+10+10}{3}, \frac{5+9}{2} \right\} = 7$
- $S(o) = \frac{b(o) - a(o)}{\max\{a(o), b(o)\}} = \frac{7-4}{7} \rightarrow +ve$
- This mean the cluster containing o is compact and o is far from other cluster



QUESTION?

NEXT

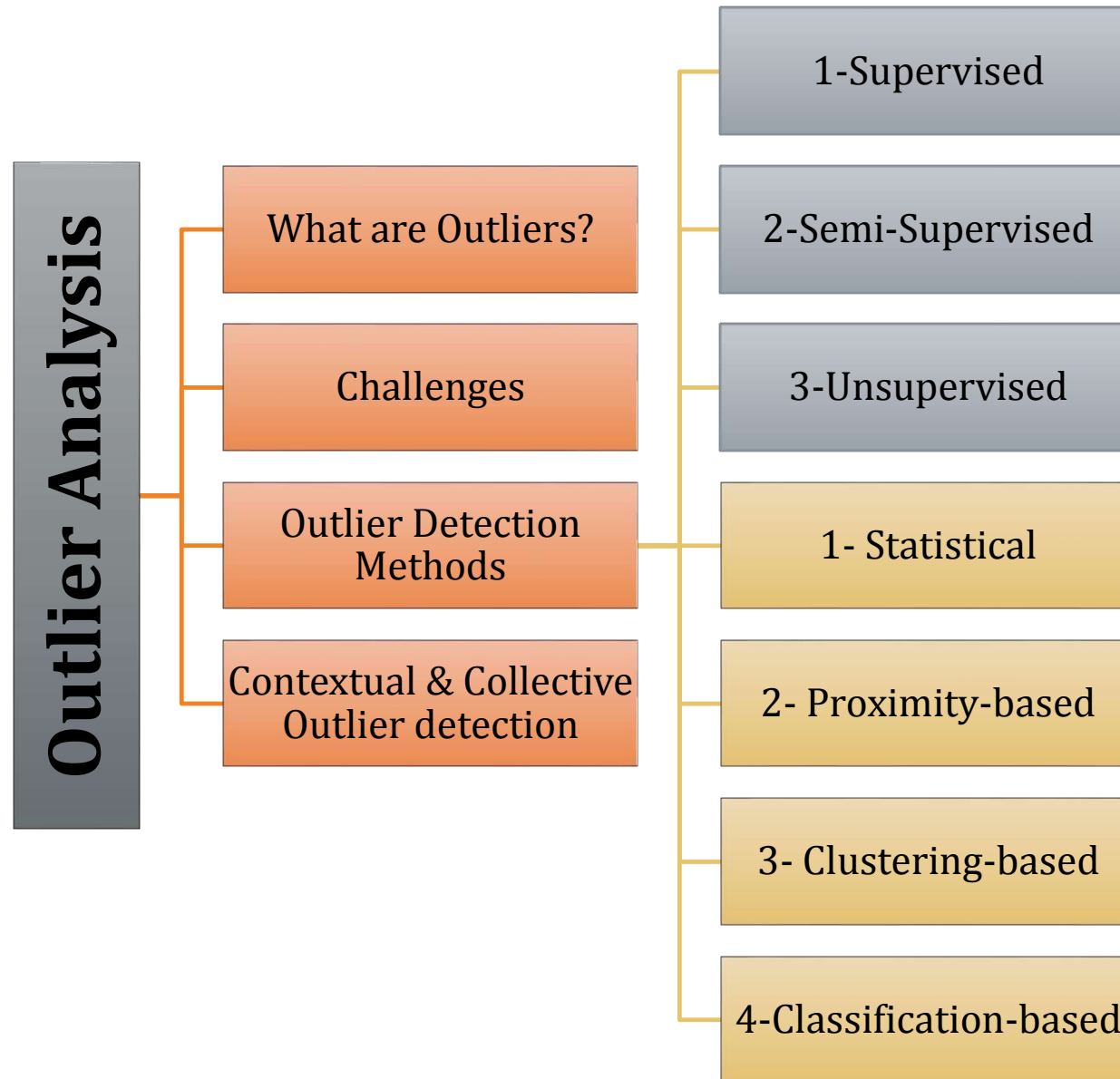


IS422P - DATA MINING OUTLIER ANALYSIS



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WHAT ARE OUTLIERS?

- Outlier → a data object that deviates significantly from the normal objects
 - Ex: a student with exceptionally high grades
 - Normal versus anomalous data objects → how to define normalcy?
- Outliers are different from the noise data
 - Noise is random error or variance in a measured variable
 - Noise should be removed before outlier detection
- Outlier detection vs. novelty detection:
 - early stage, outlier; but later merged into the model
- Applications:
 - Credit card fraud detection, Telecom fraud detection, Customer segmentation



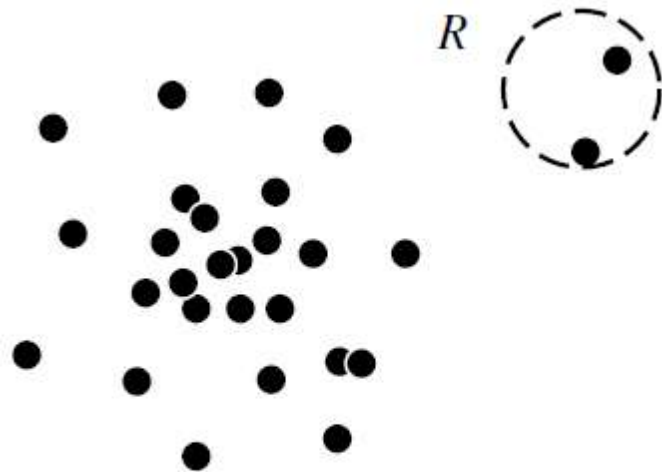
TYPES OF OUTLIERS

- Global
 - Deviate significantly from the rest of the dataset, Ex: Intrusion Detection
 - How to measure deviation?
- Contextual
 - Deviate with respect to context (time, location), Ex: Temperature values
 - Contextual attributes used to evaluate context
 - Behavioral attributes used to evaluate outlier behavior
- Collective
 - A subset of objects collectively deviates significantly from the dataset
 - Ex: Multiple order delays, DoS attacks

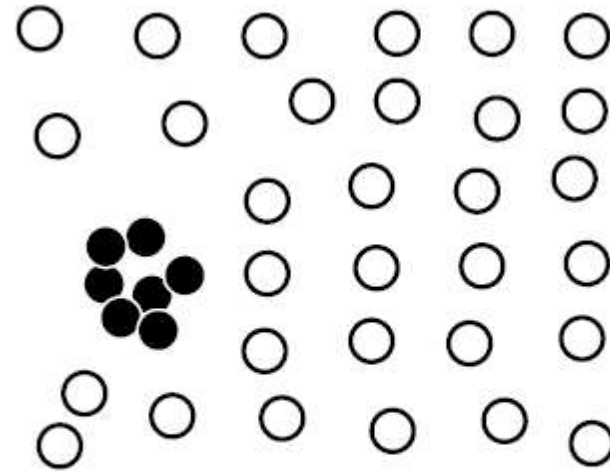


TYPES OF OUTLIERS

EXAMPLES



The objects in region R are outliers.



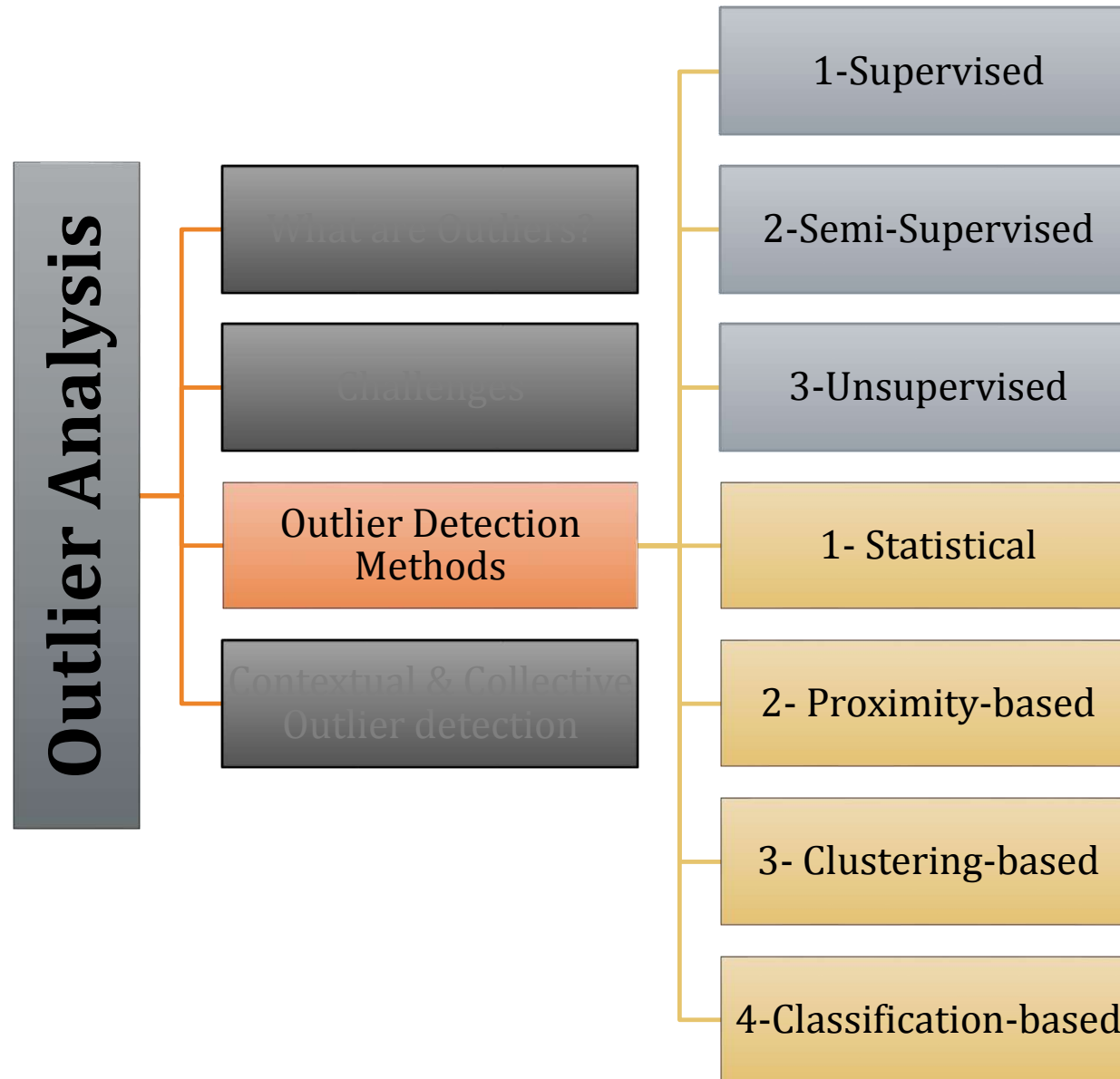
The black objects form a collective outlier.



CHALLENGES FOR OUTLIER DETECTION

- Modeling normal objects & outliers
 - Normal models are challenging to build
 - Distinction between normalcy and anomaly is ambiguous
- Application-specific outlier detection
 - How much deviation is considered an anomaly/outlier?
- Handling noise in outlier detection
 - How to detect outliers in the presence of noise?
 - Noise sometimes “hides” outliers!
- Understandability
 - Understand why these are outliers
 - Specify the degree of an outlier





OUTLIERS DETECTION METHODS

- Two ways to categorize outlier detection methods:
 - Based on whether user-labeled examples of outliers can be obtained:
 - Supervised, semi-supervised vs. unsupervised methods
 - Based on assumptions about normal data and outliers:
 - Statistical, proximity-based, and clustering-based methods



OUTLIERS DETECTION METHODS

SUPERVISED METHODS

- Modeling outlier detection as a classification problem
 - Samples examined by domain experts used for training & testing
- Methods for Learning a classifier effectively:
 - Model normal objects & report those not matching the model as outliers, or
 - Model outliers and treat those not matching the model as normal
- Challenges
 - Imbalanced classes, i.e., outliers are rare: Boost the outlier class and make up some artificial outliers
 - Catch as many outliers as possible, i.e., recall is more important than accuracy (i.e., not mislabeling normal objects as outliers)



OUTLIERS DETECTION METHODS

UNSUPERVISED METHODS

- Normal objects are somewhat ``clustered'' into multiple groups, each having some distinct features
 - An outlier is expected to be far away from any groups of normal objects
- Weakness: Cannot detect collective outlier effectively
 - Normal objects may not share any strong patterns, but the collective outliers may share high similarity in a small area
- Ex. In some intrusion or virus detection, normal activities are diverse
 - Unsupervised methods may have a high false positive rate but still miss many real outliers.
 - Supervised methods can be more effective, e.g., identify attacking some key resources



OUTLIERS DETECTION METHODS

SEMI-SUPERVISED METHODS

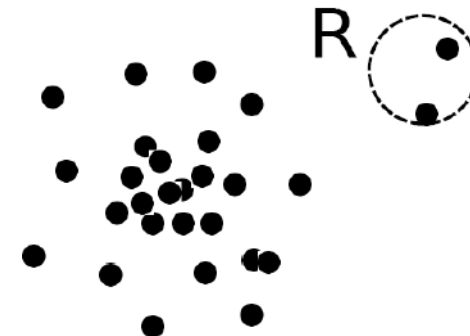
- In many applications, the number of labeled data is often small:
 - Labels could be on outliers, normal objects, or both
- If some labeled normal objects are available
 - Use the labeled examples and the proximate unlabeled objects to train a model for normal objects
 - Those not fitting the model of normal objects are detected as outliers
- If only some labeled outliers are available
 - a small number of labeled outliers may not cover the possible outliers well
 - To improve the quality of detection → models for normal objects learned from unsupervised methods



OUTLIERS DETECTION METHODS

STATISTICAL METHODS

- known as model-based methods: the normal data follow some statistical model (a stochastic model)
 - The data not following the model are outliers.
- Effectiveness: highly depends on whether the assumption of statistical model holds in the real data
- There are rich alternatives to use various statistical models
 - E.g., parametric vs. non-parametric



OUTLIERS DETECTION METHODS

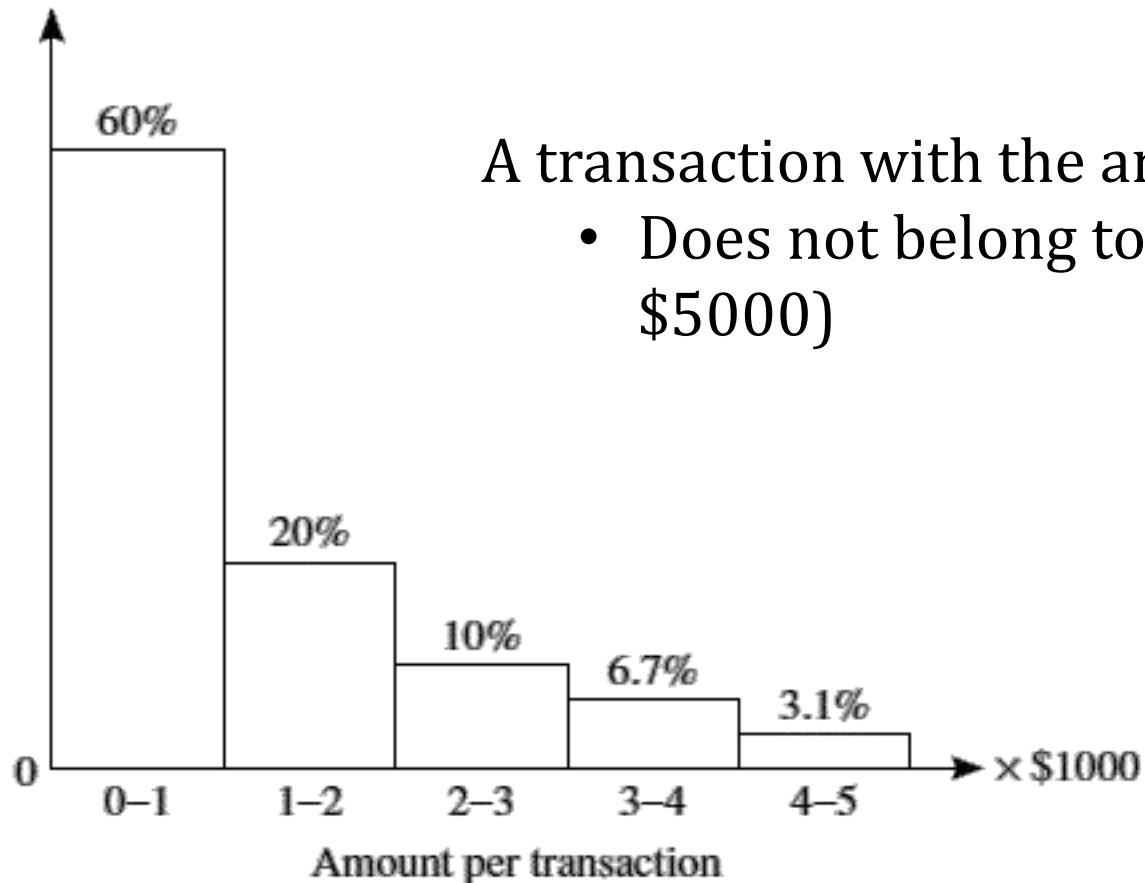
STATISTICAL METHODS

- **Parametric** → assume normal data is generated by a distribution with parameter θ
 - PDF of distribution $f(x, \theta)$ yields probability that x is generated by distribution → *smaller means outlier*
- - For **univariate outliers** → *boxplots* → parameters are mean and IQR
 - For multivariate outliers → *χ^2 -statistic* → parameter is mean
- **Nonparametric** → learn normal model from input data
 - *histograms*



OUTLIERS DETECTION METHODS

STATISTICAL METHODS



A transaction with the amount of \$7500 is considered an outlier

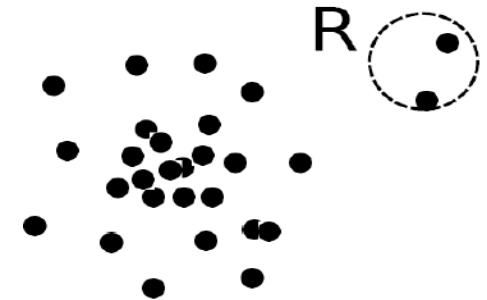
- Does not belong to any of the bins (0.2% of transactions > \$5000)



OUTLIERS DETECTION METHODS

PROXIMITY-BASED METHODS

- An object is an outlier if the nearest neighbors of the object are far away, i.e., the proximity of the object is significantly deviates from the proximity of most of the other objects in the same data set
- Effectiveness: highly relies on the proximity measure.
- Challenges:
 - proximity or distance measures cannot be obtained easily.
 - finding a group of outliers which stay close to each other
- Two major types of proximity-based outlier detection
 - Distance-based vs. density-based



OUTLIERS DETECTION METHODS

PROXIMITY-BASED METHODS

- Distance-based $\rightarrow$ for an object o , examine the number of other objects in its r -neighborhood
 - r is a distance threshold
 - π is a fraction threshold $\rightarrow$ min # objects needed in neighborhood
- Density-based $\rightarrow$ for an object o , examine its density relative to the density of its local neighbors
 - A local outlier factor (LOF) is computed in terms of the K-NN of an object in comparison to its neighbors



Algorithm: Distance-based outlier detection.

Input:

- a set of objects $D = \{o_1, \dots, o_n\}$, threshold r ($r > 0$) and π ($0 < \pi \leq 1$);

Output: $DB(r, \pi)$ outliers in D .

Method:

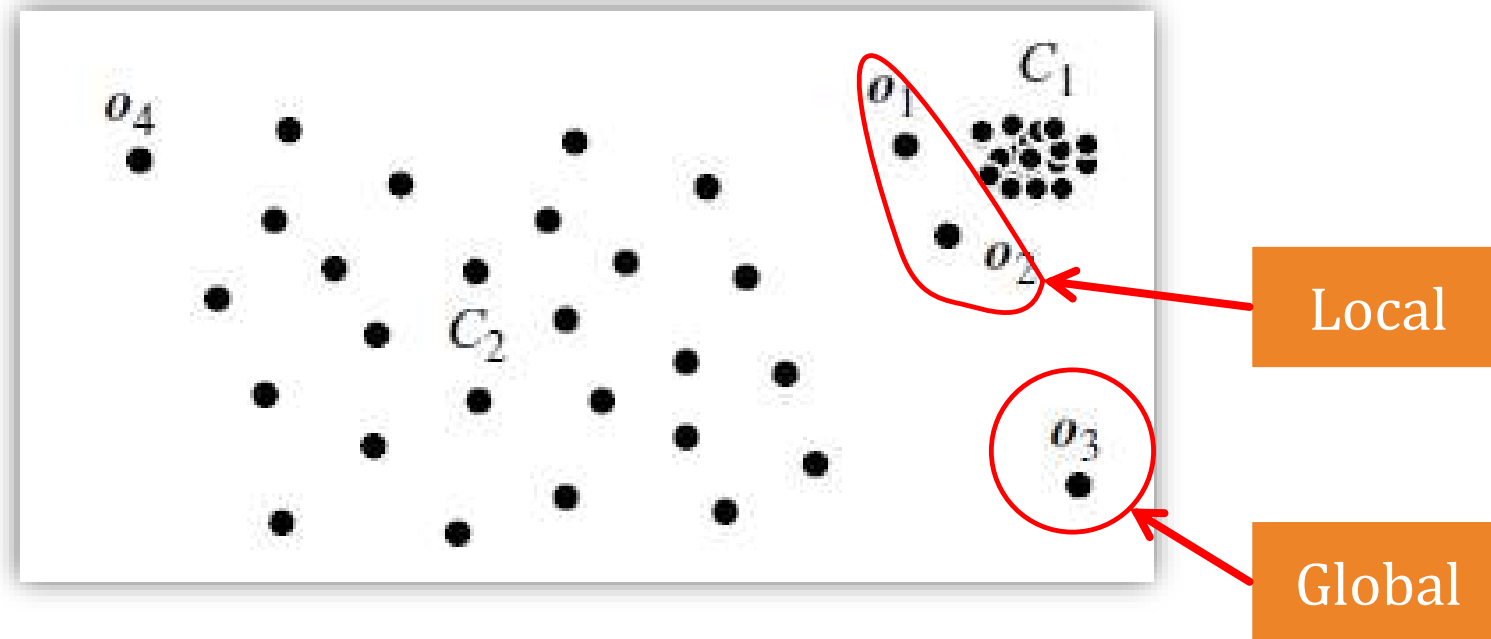
```
for  $i = 1$  to  $n$  do
   $count \leftarrow 0$ 
  for  $j = 1$  to  $n$  do
    if  $i \neq j$  and  $dist(o_i, o_j) \leq r$  then
       $count \leftarrow count + 1$ 
      if  $count \geq \pi \cdot n$  then
        exit { $o_i$  cannot be a  $DB(r, \pi)$  outlier}
      endif
    endif
  endfor
  print  $o_i$  { $o_i$  is a  $DB(r, \pi)$  outlier according to (Eq. 12.10)}
endfor;
```

OUTLIERS DETECTION METHODS PROXIMITY-BASED METHODS



OUTLIERS DETECTION METHODS

PROXIMITY-BASED METHODS



OUTLIERS DETECTION METHODS

CLUSTER-BASED METHODS

- Normal data belong to large and dense clusters, whereas outliers belong to small or sparse clusters, or do not belong to any clusters
- Since there are many clustering methods, there are many clustering-based outlier detection methods as well
- Clustering is expensive:
 - straightforward adaption of a clustering method for outlier detection can be costly and does not scale up well for large data sets



OUTLIERS DETECTION METHODS

CLUSTERING-BASED METHODS

- Either an object:

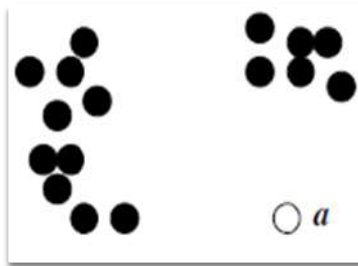
Ratio of dist (object, centroid) to average (distances between centroid and its assigned objects)

Rank clusters according to size

Assign a LOF to objects:

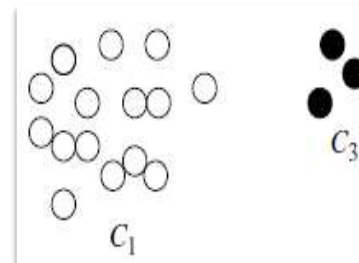
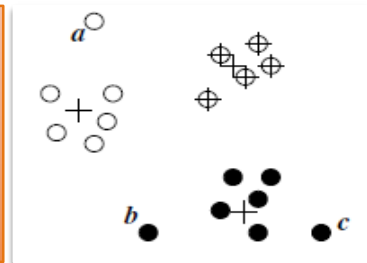
For large clusters: $size(C) \times similarity(o, C)$

For small clusters: $size(C) \times similarity(o, \text{closest large } C)$



does not belong to any cluster

Has a large distance to the closest cluster



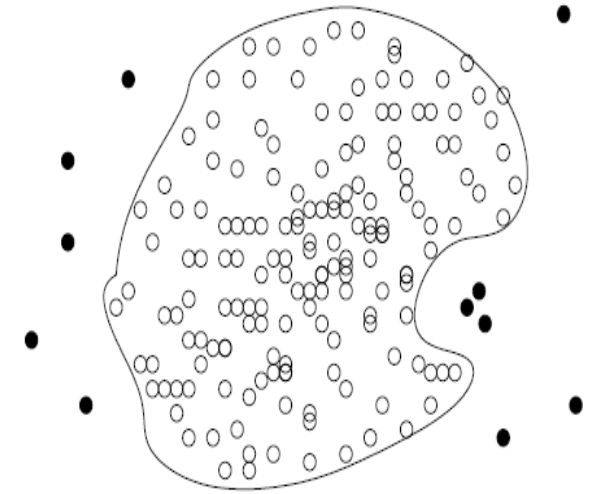
belongs to a small and remote cluster



OUTLIERS DETECTION METHODS

CLASSIFICATION-BASED METHODS

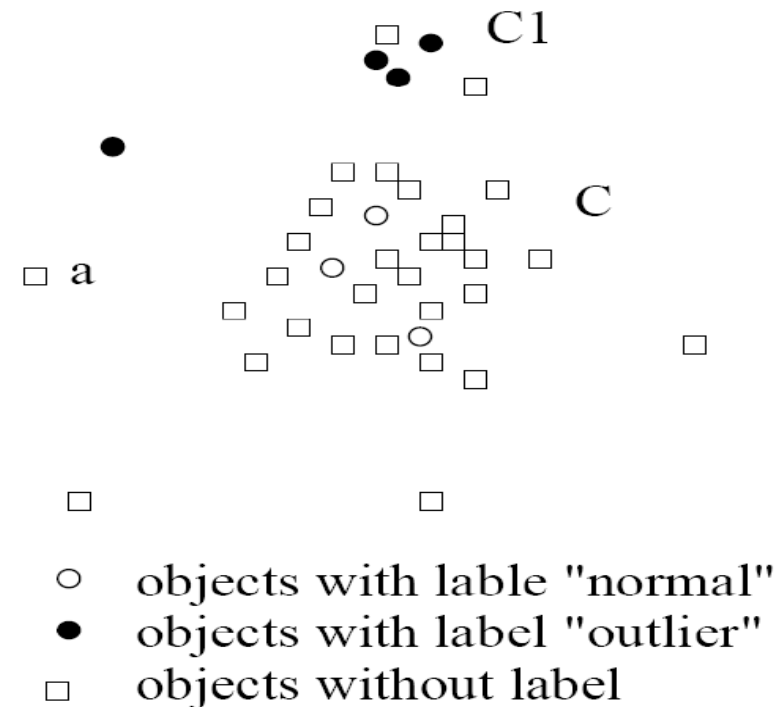
- One-class model: describe only the normal class.
 - Learn the decision boundary of the normal class using classification methods such as SVM
 - Any samples that do not belong to the normal class (not within the decision boundary) are declared as outliers
 - Advantage : detect new outliers
 - Extension: Normal objects may belong to multiple classes
- A brute-force approach: a training set contains samples labeled as “normal” and others labeled as “outlier”
 - But, the training set is typically heavily biased: # of “normal” samples likely far exceeds # of outlier samples
 - Cannot detect unseen anomaly



OUTLIERS DETECTION METHODS

CLASSIFICATION-BASED METHODS

- Semi-supervised learning: Combining classification-based and clustering-based methods
- **Strength:** Outlier detection is fast
- **Bottleneck:**
 - Quality heavily depends on the availability and quality of the training set, (difficult to obtain high-quality training data)



MINING CONTEXTUAL AND COLLECTIVE OUTLIERS

- Contextual outliers:
 - Determine contextual attributes
 - Group those attributes' values
 - Determine context of object (its group)
 - Do conventional outlier detection within that group
 - Ex → customers with similar age who live within same area may have similar behavior → a customer in that age group and living in that area with a different behavior is an outlier
- Collective outliers → challenging and advanced area



EXAMPLE

- For the following nine points (2,2), (2,7), (3,1), (3,5), (4,3), (4,8), (5,2), (6,2), (6,5)
- Assume that $k = 2$ and initially assign (2,2) and (2,7) as the center of each cluster.
- Apply the k-means algorithm using the Manhattan distance and show the new cluster centers after the first round execution (Hint: The Manhattan distance is: $d(i, j) = |x_i1 - x_j1| + |x_i2 - x_j2| + \dots + |x_in - x_jn|$.)
- Sol:
- Cluster 1: (2,2), (3,1), (4,3), (5,2), (6,2) new mean of C1 = (4,2)
- Cluster2: (2,7), (3,5), (4,8), (6,5) new mean of C2 = (3.75, 6.25)

	(3,1)	(3,5)	(4,3)	(4,8)	(5,2)	(6,2)	(6,5)
(2,2)	<u>2</u>	4	<u>3</u>	8	<u>3</u>	<u>4</u>	7
(2,7)	7	<u>3</u>	6	<u>3</u>	8	9	<u>6</u>



EXAMPLE

- Suppose that your result in (a) is the final cluster, how can you use it to detect the outlier? What is the object which likely be an outlier?
- **Sol:** calculate Ratio of $\text{dist}(\text{object}, \text{centroid})$ to average (distances between centroid and its assigned objects), if object has a large distance to the closest cluster then it is local outlier.

Average of C1= 1.6

Average of C2= 2.6

(6, 5) is likely an outlier

C1	(2,2)	(3,1)	(4,3)	(5,2)	(6,2)
(4,2)	2	2	1	1	2

C2	(2,7)	(3,5)	(4,8)	(6,5)
(3.75, 6.25)	2.5	2	2	4



QUESTION?

NEXT



IS422P - DATA MINING

DATA MINING TRENDS



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AGENDA

Data
mining via
different
approaches

BUSINESS INTELLIGENCE

BIG DATA

MACHINE LEARNING

WEB MINING

DATA WAREHOUSING

DATA ANALYSIS

NATURAL LANGUAGE PROCESSING

DATA VISUALIZATION



DATA MINING VS. BUSINESS INTELLIGENCE

- **Business intelligence** (BI) refers to the procedural and technical infrastructure that collects, stores, and analyzes the data produced by a company's activities.
- BI is a broad term that encompasses data mining, process analysis, performance benchmarking, and descriptive analytics.
- BI parses all the data generated by a business and presents easy-to-digest reports, performance measures, and trends that inform management decisions.



DATA MINING VS. BIG DATA

- **Big Data** refers to a huge volume of data that can be structured, semi-structured and unstructured. It comprises of 5 Vs i.e.
- • **Volume**: It refers to an amount of data or size of data that can be in quintillion when comes to big data.
- • **Variety**: It refers to different types of data like social media, web server logs etc.
- • **Velocity**: It refers to how fast data is growing, data is exponentially growing and at a very fast rate.
- • **Veracity**: It refers to an uncertainty of data like social media means if the data can be trusted or not.
- • **Value**: It refers to the data which we are storing and processing is worth and how we are getting benefit from this huge amount of data.



DATA MINING VS. MACHINE LEARNING

- Both data mining and machine learning are rooted in data science and generally fall under that umbrella.
- They often intersect or are confused with each other, but there is a key distinction between the two.
 - **Machine learning** can look at patterns and learn from them to **adapt** behavior for future incidents,
 - while **data mining** doesn't learn and apply knowledge on its own without human interaction. Data mining also can't automatically see the relationship between existing pieces of data with the same depth that machine learning can.



DATA MINING VS. WEB MINING

- Web mining comes under data mining but this is limited to web related data and identifying the patterns.
- It is the process of performing data mining on the web, extracting the web documents and discovering the patterns from it.
- **Web content mining** → discover different patterns that give a significant insight
- **Web Structure mining** → Data from hyperlinks that lead to different pages are gathered and prepared in order to discover a pattern.
- **Web usage mining** → user's web activity through the application logs are monitored and data mining is applied to it.



DATA MINING VS. DATA WAREHOUSING

- A Data Warehouse is an environment where essential data from multiple sources is stored under a single schema. It is then used for reporting and analysis.
 - Historical Data
 - Generalization
 - ETL
 - OLAP



DATA MINING VS. DATA ANALYSIS

- **Data Mining** studies are mostly on structured data. **Data Analysis** can be done on both structured, semi-structured or unstructured data.
- **Data mining** generally doesn't involve visualization tool, **Data Analysis** is always accompanied by visualization of results.



TEXT MINING VS. NATURAL LANGUAGE PROCESSING (NLP)

- Both **Text Mining** and **Natural Language** Processing trying to extract information from unstructured data.
- **Text mining** is concentrated on text documents and mostly depends on a statistical and probabilistic model to derive a representation of documents. Popular applications of Text Mining : Contextual Advertising, Content enrichment, Social media data analysis, Spam filtering, and Fraud detection through claims investigation.
- **Natural language** is what we use for communication. Techniques for processing such data to understand underlying meaning is collectively called as **Natural Language Processing (NLP)**. The data could be speech, text or even an image and approach involve applying Machine Learning (ML) techniques on data to build applications involving classification, extracting structure, summarizing and translating data.



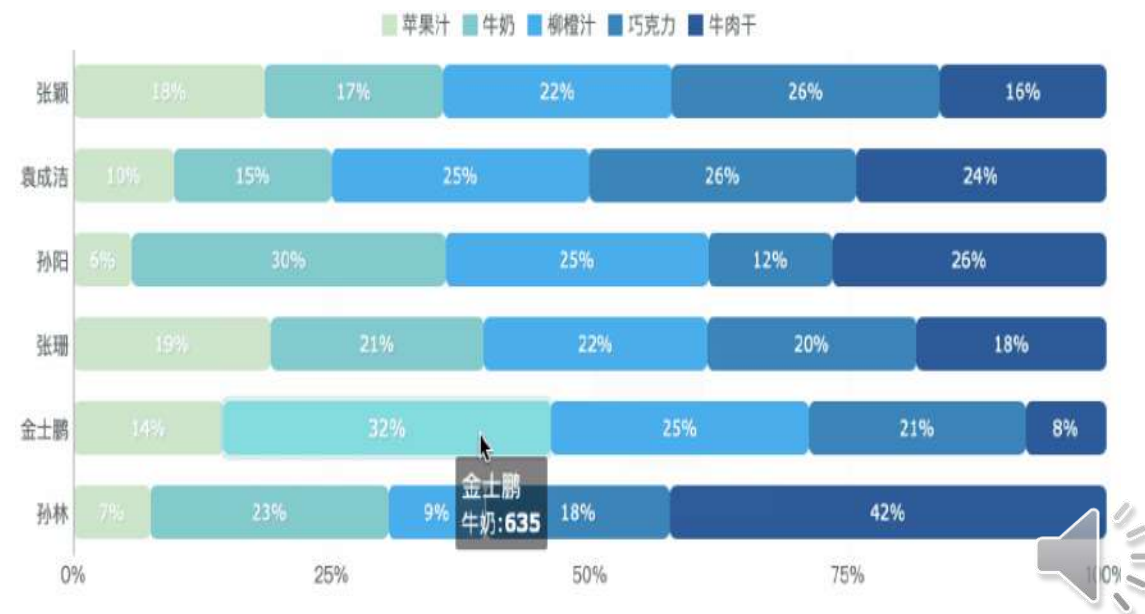
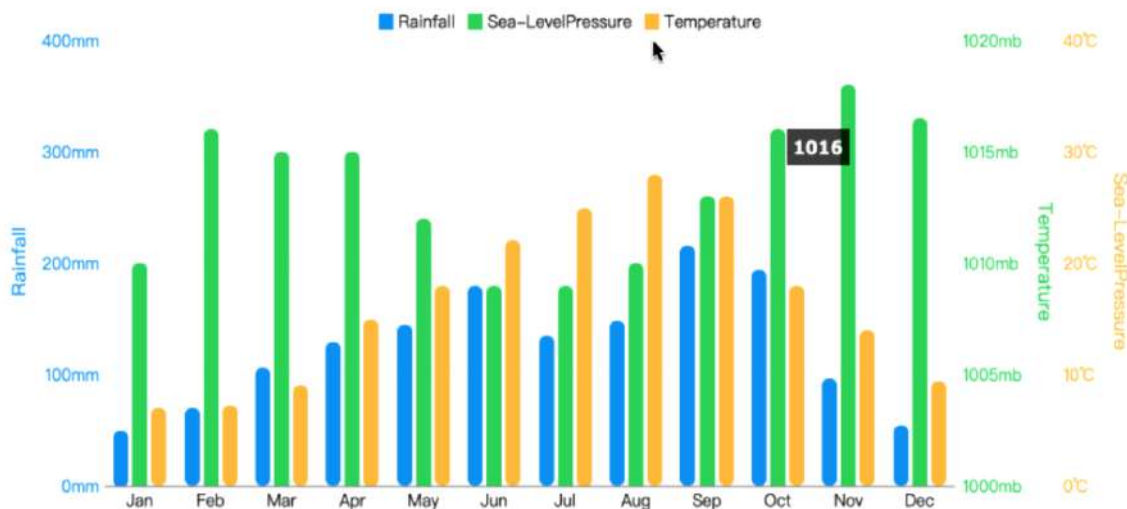
DATA MINING VS. DATA VISUALIZATION

- **Data Visualization** is the process of extracting and visualizing the data in a very clear and understandable way without any form of reading or writing by displaying the results in the form of pie charts, bar graphs, statistical representation and through graphical forms as well.
- In Data Visualization, the primary goal is to convey the information efficiently and clearly without any deviations or complexities in the form of statistical graphs, information graphs, and plots.



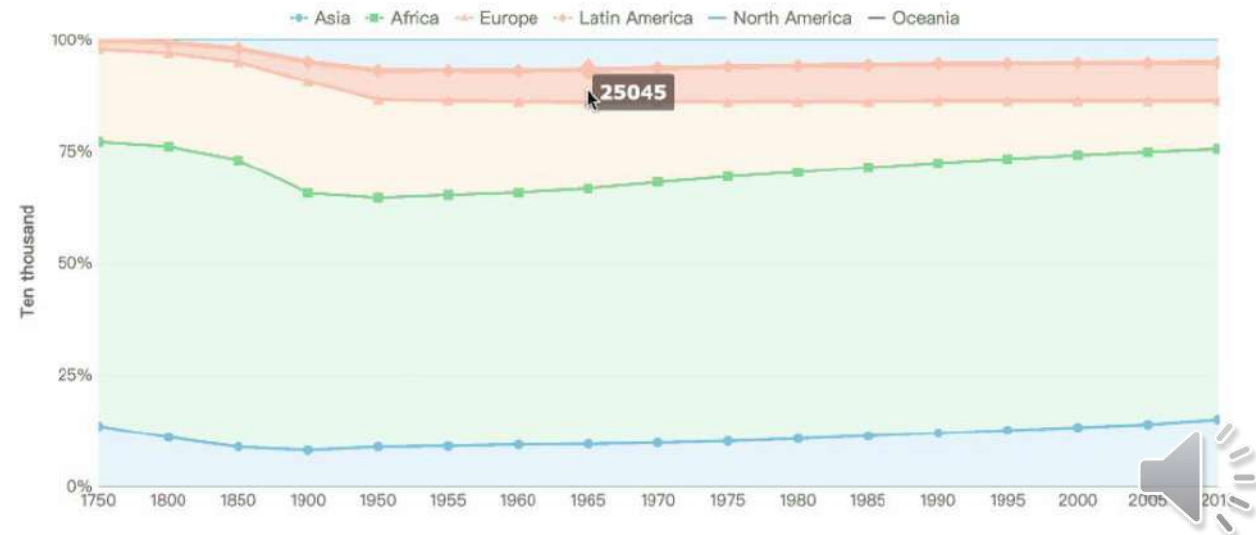
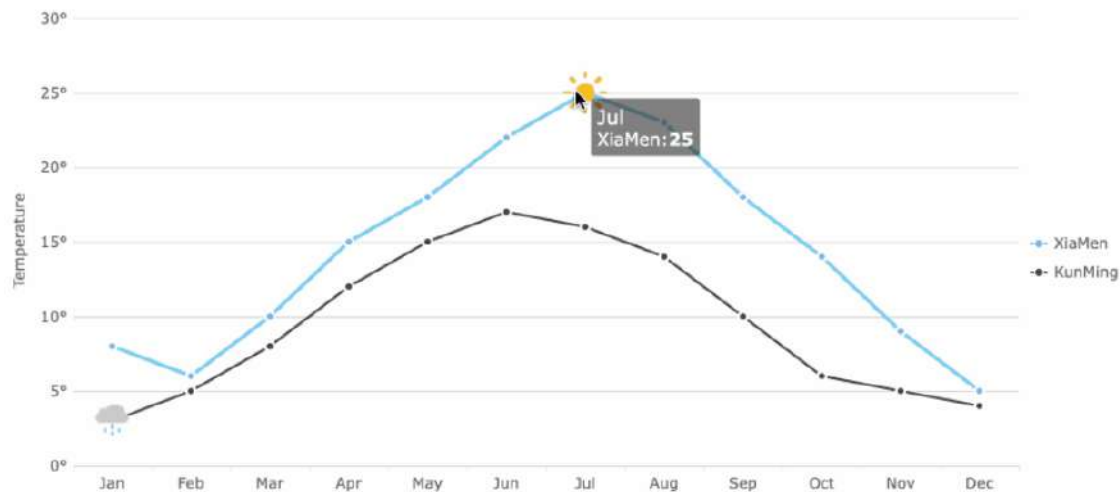
DATA VISUALIZATION TYPES

- **Column Chart** → show numerical comparisons between categories, No. columns should not be too large
- **Bar Chart** → the number of bars can be relatively large.



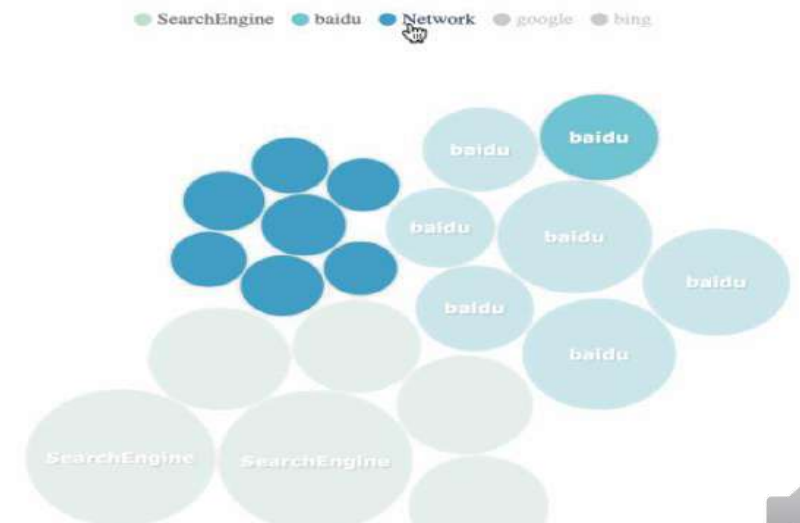
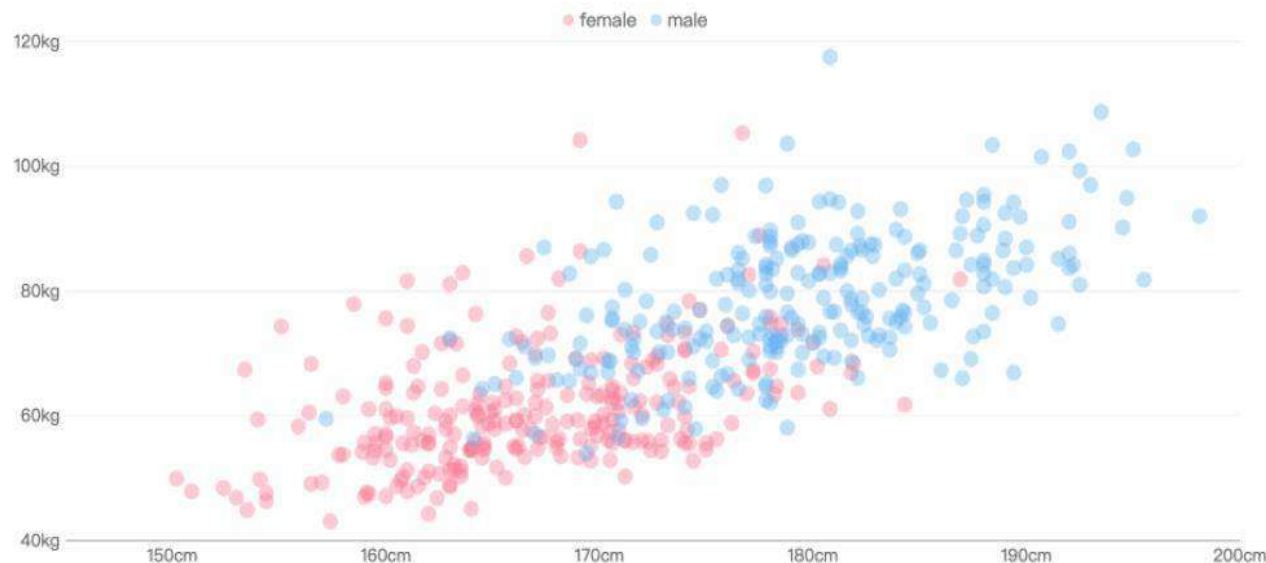
DATA VISUALIZATION TYPES

- **Line Chart** → show the change of data over a continuous time interval or time span.
- **Area Chart** → The filling of the color can better highlight the trend information.



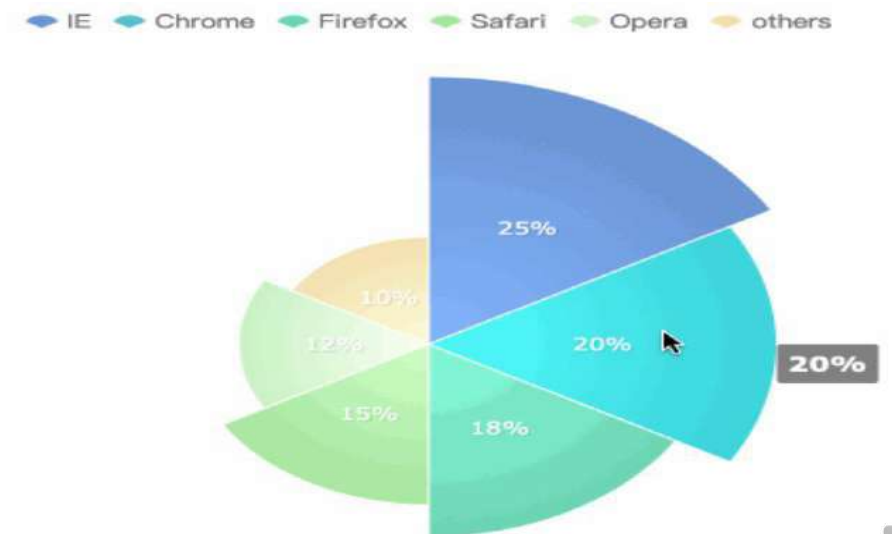
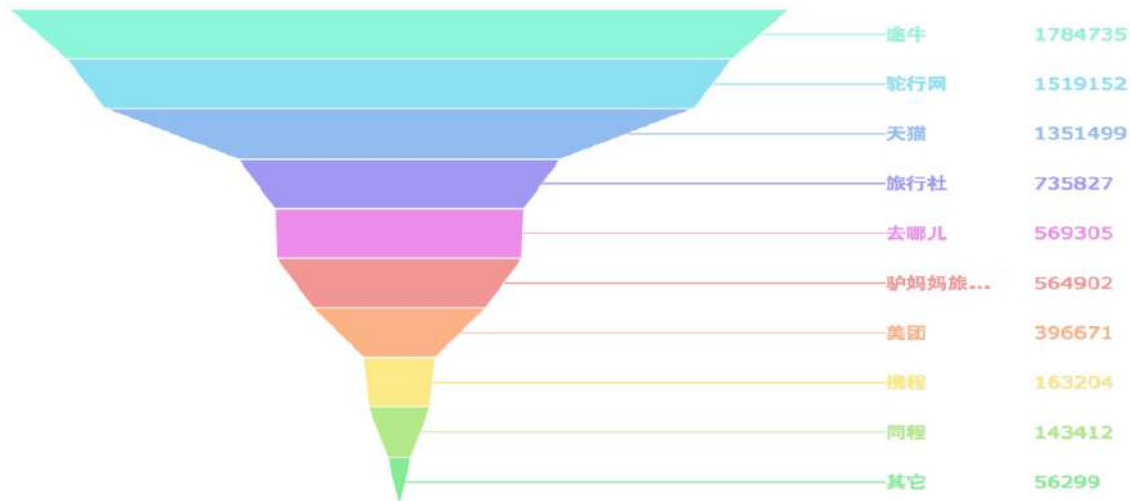
DATA VISUALIZATION TYPES

- **Scatter Plot:** shows two variables in the form of points on a rectangular coordinate system. The position of the point is determined by the value of the variable.
- **Bubble Chart** → is a multivariate chart that is a variant of a scatter plot. Except for the values of the variables represented by the X and Y axes, the area of each bubble represents the third value.



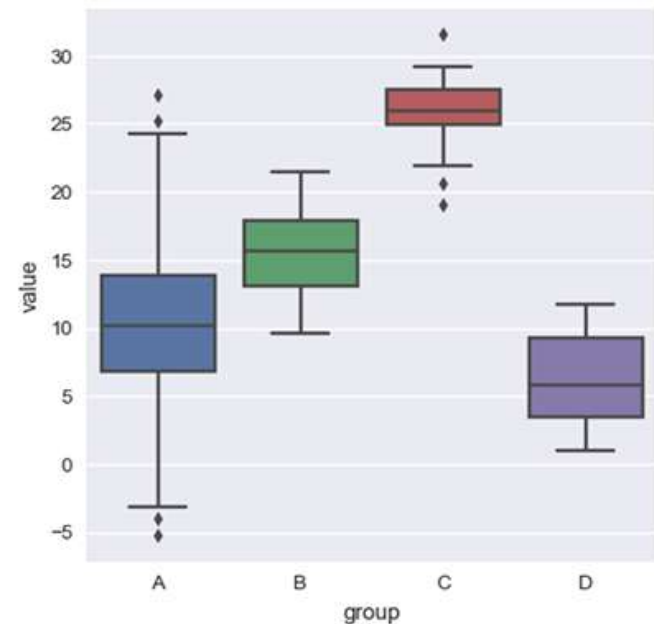
DATA VISUALIZATION TYPES

- **Funnel Chart:** display a series of steps and the completion rate for each step. used to represent how something moves through different stages in a process. A funnel chart displays values as progressively decreasing proportions amounting to 100 percent in total.
- **Pie Chart** → represents one static number, divided into categories that constitute its individual portions.



DATA VISUALIZATION TYPES

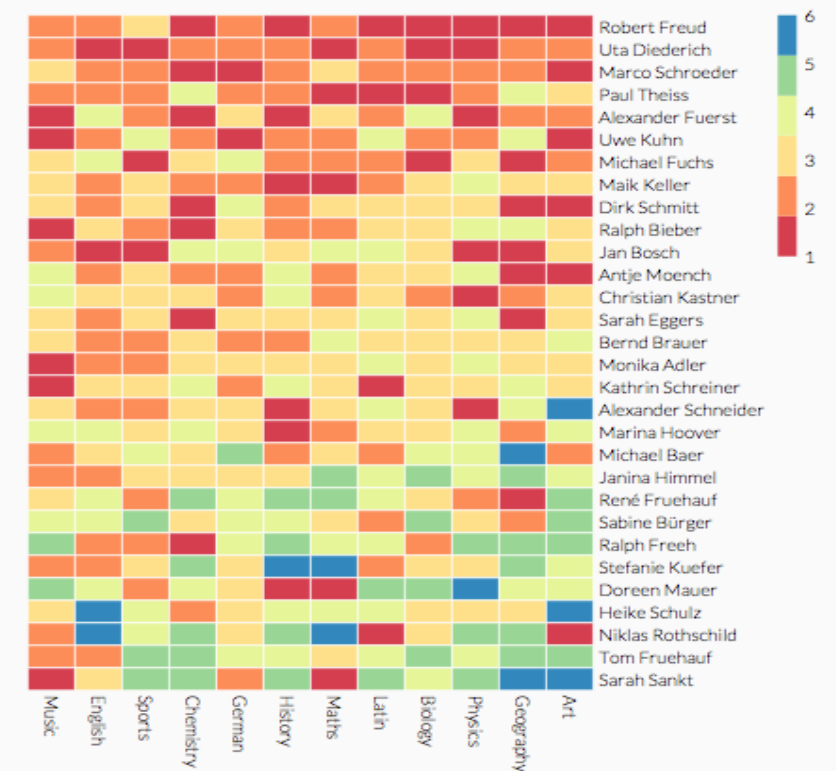
- **Gantt Chart** → shows the timing of the mission
- **Box Plot** → is a visual representation of displaying a distribution of data, usually across groups, based on a five number summary



DATA VISUALIZATION TYPES

- **Heatmap** (choropleth map)
- shows the relationship between two measures and provides rating information. The rating information is displayed using varying colors or saturation and can exhibit ratings such as high to low or bad to awesome, and needs improvement to working well.

Heat map of school grades within a fictional class



Fictional data, names generated with de.fakenamegenerator.com



QUESTION?

